

Element D

Design Selection

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PLTW Capstone

Problem Statement

Medical institutions lack a cost-effective, risk-free method for surgical training and other medical processes. Certified cadavers are the best training material for students, but are in great shortage worldwide, and pose ethical and practical issues. Due to said scarcity, organizations such as government agencies allocate a great proportion of their budget towards acquiring these materials: Medicare spends ~\$180,000 per medical resident each year through subsidies to universities, totaling over \$25 billion.¹ Overall, inefficient distribution and lack of product availability hinder surgical trainees worldwide, thus a more accessible solution is required.

I. Selecting a Design

Various steps were taken to select and finalize a build prototype design of the solution. Based on previously specified criteria and research, basic ideas were brainstormed and incorporated into three base designs. Peers then reviewed these designs, combining them into one final design.

Concept Development

The first design process steps involved multiple stages of brainstorming: individual, peer review, and team concept development. Based on previous research, criteria were set to ensure a product design that satisfies the current market needs and solves the problem effectively. These include, but are not limited to (see Appendix A):

- Scalable restrictive force on fingers
- Accurate position and rotational accuracy of the hand
- Prolonged use features (lightweight, hot-swappable battery,
- Adjustable size / “one-size-fits-all”

From these, an individual concept development stage was executed, where members documented ideas for each specific part of the solution: finger restriction, hand tracking, circuitry, frame, and overall build. These included various forms of haptic-feedback technology, servo integration, and existing VR consoles, piezo-electric and flex sensors, etc. (see Appendix B). With multiple ideas for each component, combinations of promising ideas were merged within a team concept development stage. Each potential solution required:

- Power input
- Use of Arduino or similar boards
- Some form of finger restriction
- Some form of hand/arm tracking
- Flexibility for all sizes

¹ “The Costs of Training Internal Medicine Residents in the United States.” The American Journal of Medicine, U.S. National Library of Medicine, pubmed.ncbi.nlm.nih.gov/25063649/. Accessed 12 Nov. 2024.

- Additional features to improve the solution

Six potential designs were made, and compared in a decision matrix to deem their adherence to criteria and weed out potential repeats. A detailed view of all ideas is listed in Appendix C.

		Weight & Distribution	Profile	Approx. Cost	Force Output	Size Adjustability	Battery Life	Visual Appearance	Feasibility	Ease of Assembly	Software Integration	Total
Designs	Scale Factor:	3	1	2	1	5	2	1	4	3	3	
Flex Sensor Controlled Restriction w/ Palm Vibration		6	5	6	9	5	7	7	4	8	7	151
Vibrational and Pneumatic Glove		5	8	7	3	7	6	8	7	6	5	156
Servo Restriction on Palm w/ Zipper		7	4	7	5	8	6	2	5	4	4	142
Brake Feedback with Forearm Mounted Plates		6	7	6	8	7	7	6	7	6	5	161
Finger and Hand Restriction with Forearm Mounted Sleeve		6	7	6	8	7	8	7	5	7	5	159
Stepper Motor Restriction on Hand and Wrist		5	3	8	5	7	4	2	2	4	7	125

Figure 1.1: A decision matrix comparing six complete potential solutions.

Based on simplified versions of existing specifications, each was rated from one to ten (one being the worst, ten is best). Some criteria were weighed by a scale factor based on their importance. Three designs were chosen from these six (highlighted green in Figure 1.1).

Design #1: Vibrational and Pneumatic Glove

The haptic glove simulates tactile feedback in virtual reality by restricting finger movement and using piezo-electric sensors when an object in VR space is touched. It also uses servo motors, flex sensors, and a pressure plate to provide realistic sensations, helping in low-cost surgical training and skill-based learning. The glove features a one-size-fits design with velcro straps, and the system operates through an Arduino that controls motor movement based on virtual object interactions. This solution is cost-effective, intuitive, and offers easy repairs with off-the-shelf components.

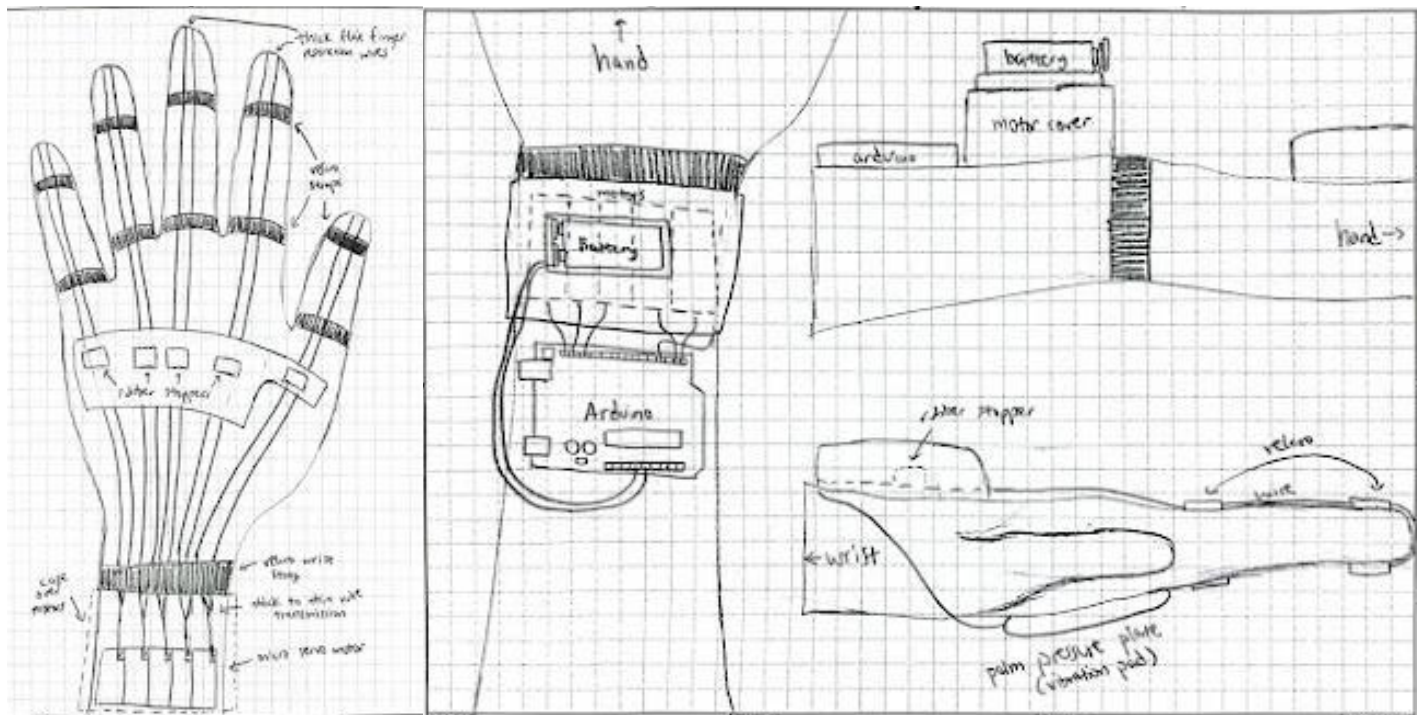


Figure 1.2: Vibrational and Pneumatic Glove design.

Design #2: Brake Feedback with Forearm Mounted Plates

This design features a flexible, zippable fabric for the user to wear, with a tracker (e.g., VIVE Tracker) attached to track hand position and rotation. The glove includes servo motors, flex sensors, and friction-based brakes at the wrist to limit finger movement based on virtual object interactions detected by the Unity-based VR application. Power is supplied by a rechargeable or wireless battery, which also supports the core Arduino system. The modular design allows easy repairs, accommodates a wide range of users with a one-size-fits-all approach, and provides wrist resistance not found in many competitors, improving accuracy and usability.

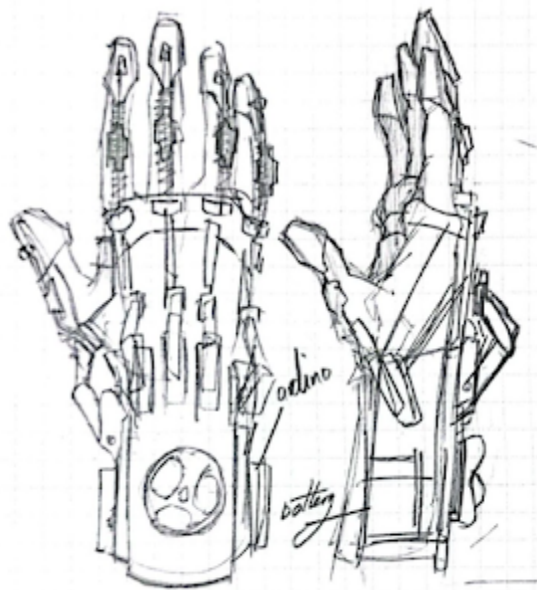


Figure 1.3: Brake Feedback with Forearm Mounted Plates design.

Design #3: Finger and Hand Restriction with Forearm Mounted Sleeve

This design is a glove that mounts to both the hand and the forearm, creating a resistance on the contraction of each finger to mimic the feeling of pressing against an object. The product will enable a user to have the sensation of holding digital objects within a virtual simulation. The object is mounted mostly on the forearm so that the weight of the glove is better distributed which makes the glove easier to use. The glove has wires connected to rings connected to each part of the finger, using basic servo motors to restrict.

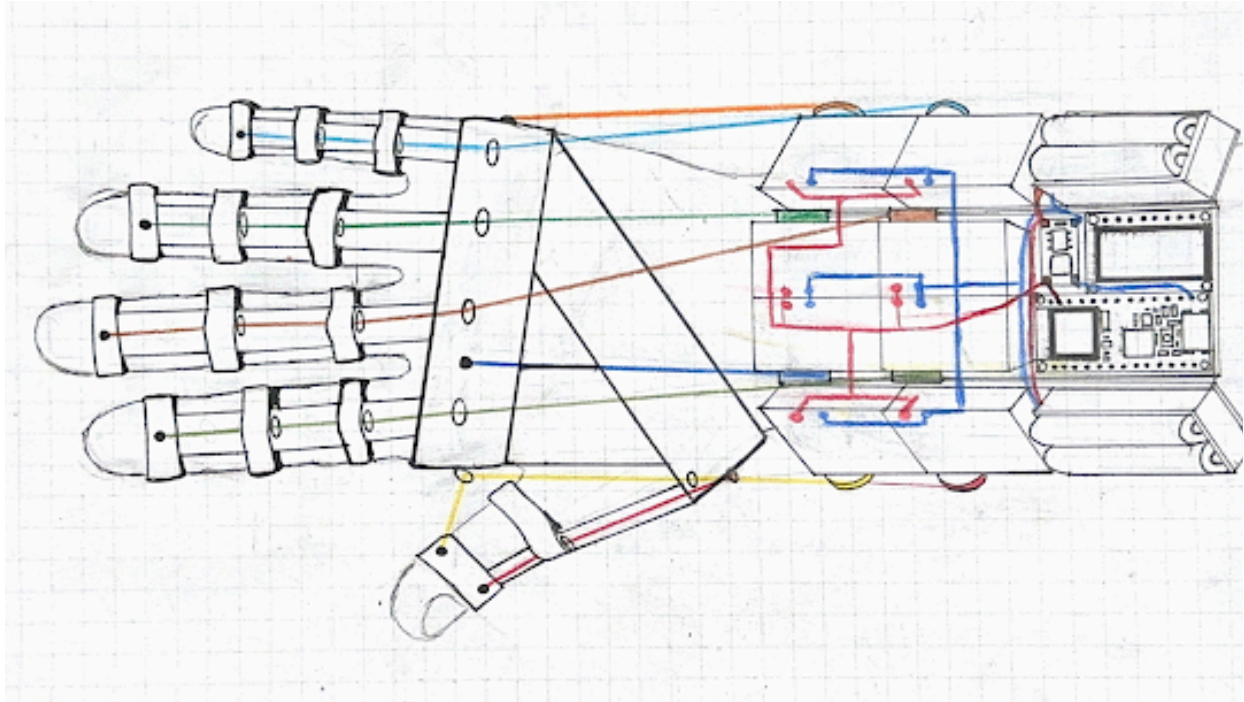


Figure 1.4: Finger and Hand Restriction with Forearm Mounted Sleeve design.

Solution Decision

To decide on the final design, a market survey was conducted to gain peer feedback on the three potential solutions and gather general information on current VR solution opinions (see Appendix D). Users with any VR experience were tasked to rate each design from one to ten (ten meaning the best). From 23 respondents, Design 2 scored an average of 8.13 out of 10, ranking it as the best design out of the three, shown in Figure 2.1, and was chosen as the base for the final design.

Average Design Rating

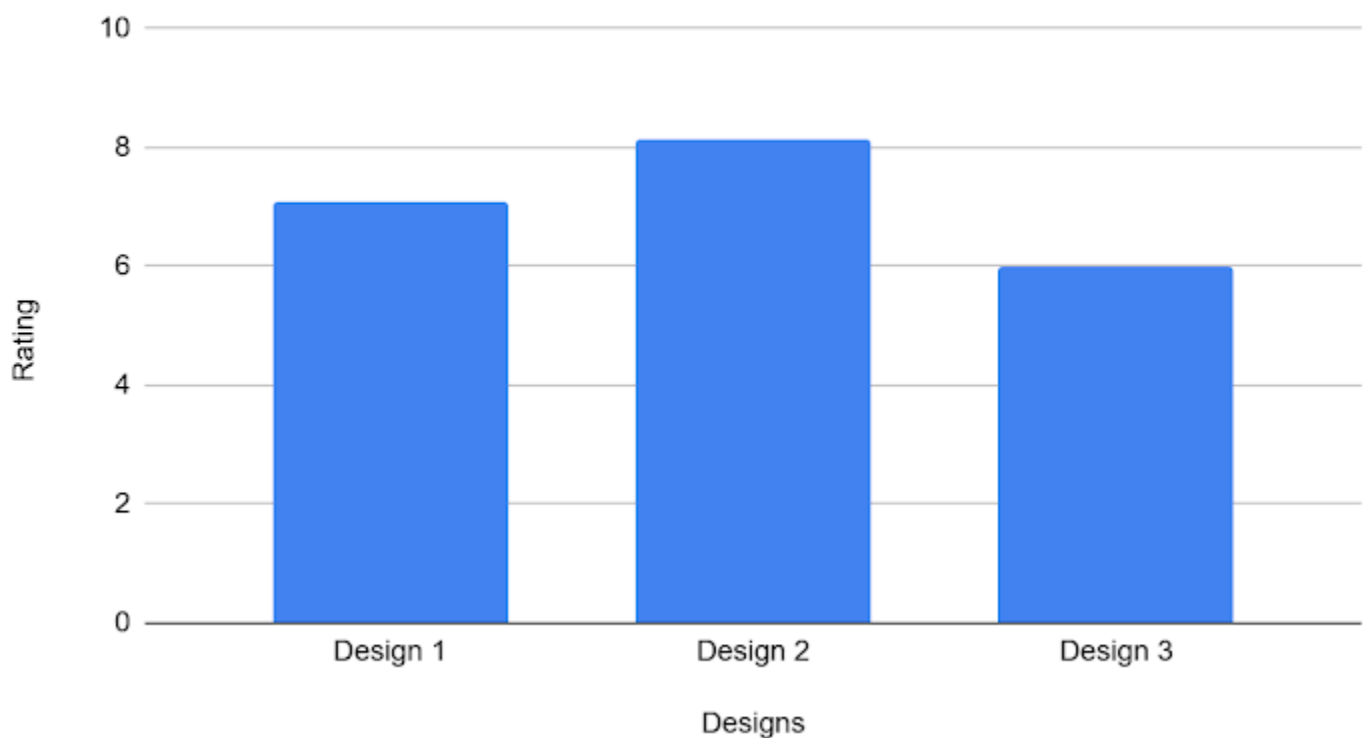


Figure 2.1: Survey results denoting the average ratings of each design.

The next step was to gather all the parts for the design in a bill of materials (BOM) estimate. From these, estimated prices for the build materials would exceed an average budget of \$150. The HTC VIVE tracker, and braking system, for example, were not feasible within an average budget of \$150, as the tracker itself costs that amount. Additionally, a braking system would require additional weight and complication. Thus, the design was switched to using regular MG90S servos in a grid-like frame, compatible with the native VR headset controller the glove would be paired with, whilst still keeping the sleek, plate design it originally had.

II. Final Design

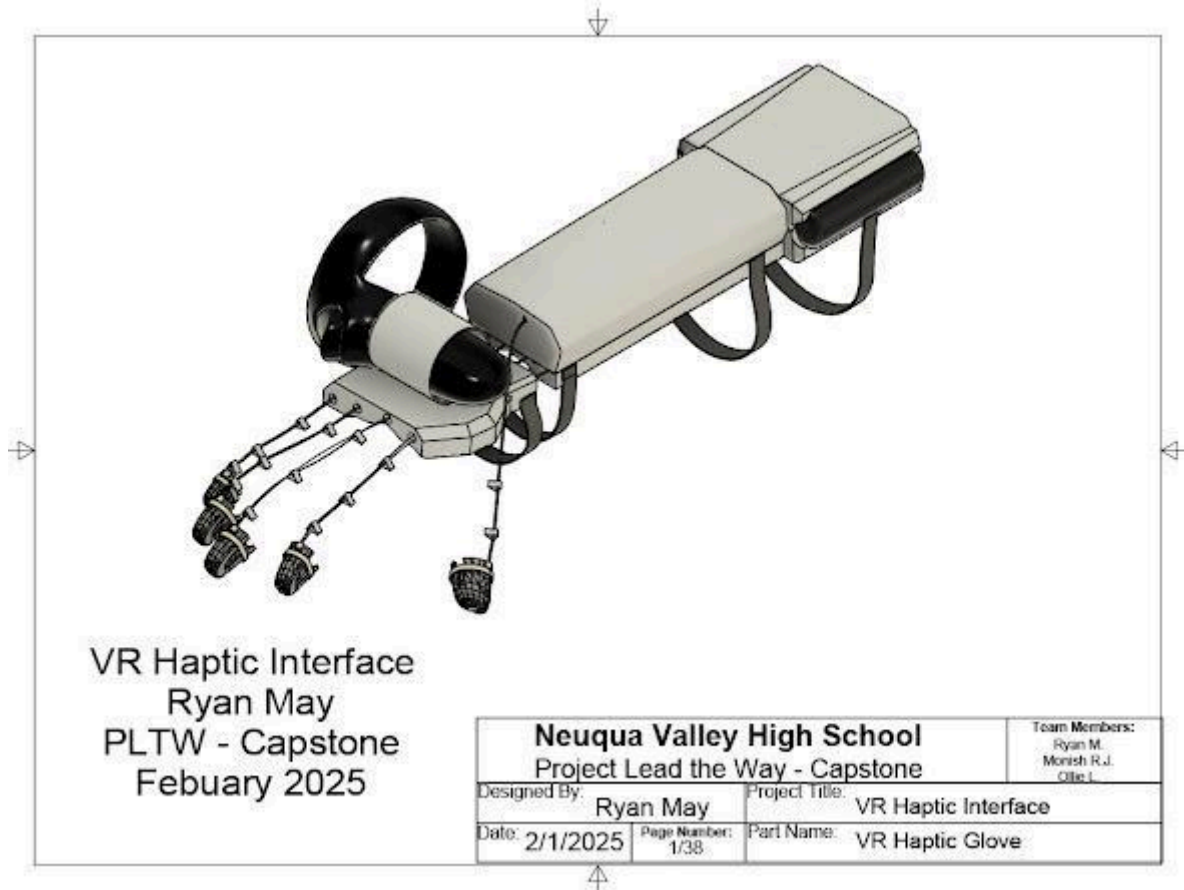


Figure 3.1: Basic isometric view of the final glove.

The solution is a lightweight, modular glove that sends digital information as physical force feedback on a user's hand. It connects to most household headsets, utilizing the default VR controller sensors to track arm movements (in this case it is the Oculus Rift S headset and controller). This glove will be paired with VR applications for any digital usages, relaying object collisions in the digital world to servo restrictions, simulating real touch.

The design basics follow servo laid in a grid-like fashion, threading to flexible finger caps. All components are attached via a spandex glove and fastened with velcro straps for a truly 'one-size-fits-all' capability. The aesthetics are sleek: smooth, 3D-printed bodies housing the main components, with circuitry and other parts distributed evenly across the forearm to minimize weight and keep the center of gravity close to the elbow for maneuvering ease. Detailed diagrams of all parts are located in Appendix F.

Operation

The glove's hardware is powered by two USB-C battery packs connected to an Arduino Nano ESP32. The Arduino controls five MG90S servos housed in a servo mount, connected to potentiometers that measure rotation. A VR controller (in this case, an Oculus Rift S Controller) is inserted into the back palm cylinder to track arm movements. The spandex glove is then put on, and the velcro straps are adjusted to fit the user.

A custom application may be loaded up on the VR headset, allowing the glove to use only the inertial data of the controller, showcasing a digital hand model in the 3D world. The servos are pulled by the fingers when flexed. That rotational data is transposed to the hand model to track the hand accurately.

When the model interacts with a digital object, the collision data is sent to the Arduino, freezing the servos in place via a screw and extension hand connected to the servo axel, restricting the finger from bending further.

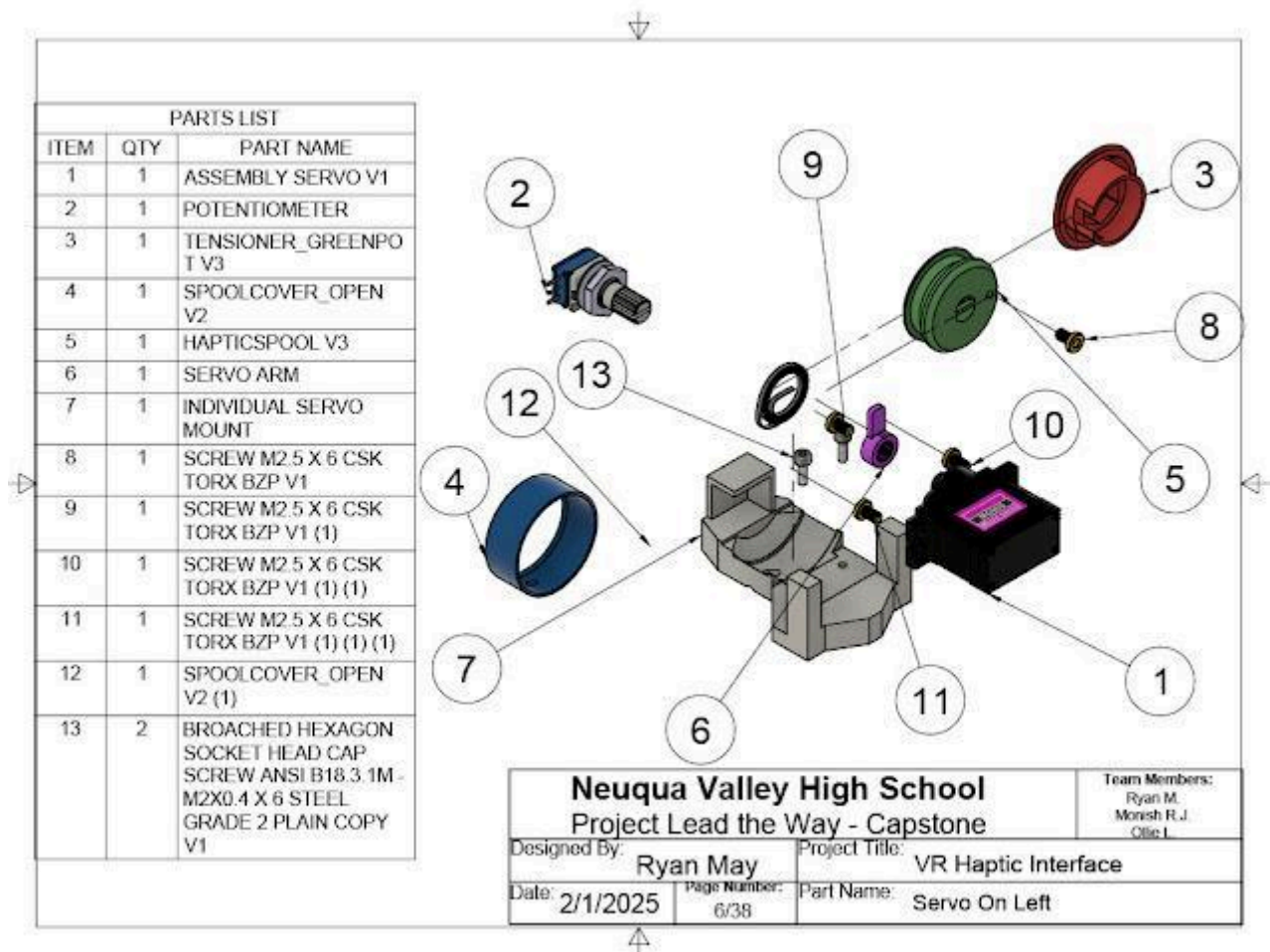


Figure 3.2: Exploded view of the servo mount component.

Justification

The glove is designed mainly to be a cheap, flexible, and reusable solution for VR touch applications compared to other products currently available. Specifically, as the glove is meant to be used in training and surgical institutions, it is designed to fit most hand sizes. Additionally, cheap market VR headsets work the best as the device is designed to make use of what would already be available in these facilities. Both of these qualities allow the glove to be cheaper than other market solutions, and easily integratable with existing infrastructure.

Additionally, the glove fixes great issues with quality of life, especially when it comes to training simulations, being longevity. Long use times contribute to user fatigue and wear of the product itself. This solution would reduce the severity of that issue by having the technology distributed towards the elbow and upwards. This allows for areas where weight naturally rests to be closer to centers of gravity, having less resistance on the arm itself. Additionally, this distribution would allow heat to naturally dissipate out of the main glove body, without components being too stuck together. With swappable batteries, the glove can be perpetually charged with replacement batteries, allowing for indefinite use given a surplus of batteries (easily bought in bulk and stored). All these factors combine for great long-term and short-term use.

With the simplicity and modular design of the glove, it aims to be an easy supplement to any medical course, accessible to most.

Appendices

Appendix A

Design Specifications

1. Restrictive Force on Fingers

- a. Specification: The solution shall exert a scalable force equal and opposite to the force applied by the fingers, with a maximum of 8 lbf.
- b. Rationale: The average human can apply 35N (~7.9lbf) of force with one finger². To replicate physical solid touch, the motors must be able to overpower the fingers within a set range of forces.

2. Position and Rotational Accuracy

- a. Specification: The solution shall track the positions of each finger to a sub 5mm accuracy, and the angle of each finger to sub 5° accuracy.
- b. Rationale: Similar solutions such as the Meta Quest 2's hand tracking have an accuracy of 1.1cm and a rotation accuracy of ~10 degrees³. Medical practices require precision, thus the product shall have less margin of error than comparable market products.

3. Element Resistance and Longevity

- a. Specifications: The solution shall stay below 36 °C in use, and withstand prolonged ambient temperatures, different humidities, and water/sweat precipitation.
- b. Rationale: The solution cannot get too hot to prevent uncomfortable heat for the user. Additionally, corrosion and water resistance should be added to ensure excessive dust or sweat does not compromise the functionality of the solution, helping to prolong product lifespan.⁴

4. Weight

- a. Specifications: The solution shall weigh at most 400 grams.

² Riddle, Michael et al. (2020). Evaluation of individual finger forces during activities of daily living in healthy individuals and those with hand arthritis. *Journal of Hand Therapy*, 33(2), 188-197. [https://www.jhandtherapy.org/article/S0894-1130\(20\)30079-X/abstract](https://www.jhandtherapy.org/article/S0894-1130(20)30079-X/abstract)

³ AbdIkrim, D., Di Luca, M., Aves, P., Maaroufi, M., Yeo, S. H., Miall, R. C., Holland, P., & Galea, J. M. (2024). A methodological framework to assess the accuracy of virtual reality hand-tracking systems: A case study with the Meta Quest 2. *Behavior research methods*, 56(2), 1052–1063. <https://doi.org/10.3758/s13428-022-02051-8>

⁴ Gelston, Jon. *When Should You Waterproof Your Electronics Devices? Here's An Answer.*, www.aimprocessing.com/blog/when-should-you-waterproof-your-electronics-devices-heres-an-answer. Accessed 14 Dec. 2024.

- b. Rationale: Practice surgery sessions can take many hours, and a lightweight solution allows for more natural and unobtrusive usage. Market solutions such as the HaptX Gloves G1 weigh 450g,⁵ hence a comparable yet lighter weight is best.

5. Joint Materials

- a. Specifications: The finger connection points shall be made of a flexible material.
- b. Rationale (Self-Imposed): Flexible materials should be used for the finger mounts and the mounting on the back of the hand to ensure comfortable long-duration sessions.

6. Size Adjustability

- a. Specification: The solution shall be able to fit hand lengths ranging from 5-9 inches.
- b. Rationale: The average male hand length is 7.6 inches and the average female hand size is 6.8 inches⁶. To accommodate this variability, the solution must be able to change length and width to fit most individuals.

7. Battery Life Span

- a. Specification: The solution shall last at least 3 hours of usage time.
- b. Rationale: A standard Meta Quest 3 provides 2-3 hours of battery life on a full charge⁷. To be comparable, the solution shall last the same usage time as the headset.

8. Hot Swappable Battery

- a. Specification: The battery must be replaceable without powering off the solution.
- b. Rationale: A hot-swappable battery provides maximum usage by preventing the need to wait for the product to charge. In addition, only the highest-end VR solutions have this ability, so bringing this to a cheaper size creates a serious market advantage.⁸

9. Hidden Cables

- a. Specifications: All wires should be fully encased and routed throughout the solution yet accessible for proper maintenance.
- b. Rationale (Self-Imposed): Exposed wiring causes the ability to snag and damage the product. To ensure the most seamless usage and the longest product life span, the cables shall be hidden yet accessible for maintenance.

⁵ HaptX. (2024). *HaptX Gloves G1*. HaptX. <https://haptx.com/gloves-g1/>

⁶ Sullivan, D. (2020, July 3). Average hand size: Adults, children, and relation to height. MedicalNewsToday. Retrieved December 12, 2024, from <https://www.medicalnewstoday.com/articles/average-hand-size>

⁷ Carter, R. (2024, May 6). How to Maximize Meta Quest 3 Battery Life: Top Tips. XRToday. <https://www.xrtoday.com/mixed-reality/how-to-maximize-meta-quest-3-battery-life-top-tips/>

⁸ "Weart TouchDIVER Pro Haptic Gloves." *Knoxlabs*, www.knoxlabs.com/products/weart-touchdiver-pro-haptic-gloves?_pos=1&_sid=51b4e4463&_ss=r. Accessed 12 Dec. 2024.

10. Connectivity Types

- a. Specifications: The solution must have wired and wireless compatibility with the headset.
- b. Rationale: Wired connections can simplify troubleshooting issues, but a wireless connection is necessary for typical usage. In the market, Bluetooth and Wi-Fi are standard for connecting VR solutions to VR headsets. Products like bHaptics's TactGlove utilize Bluetooth,⁹ while Weart's TouchDiver pro haptic glove utilizes both Wifi and Bluetooth.¹⁰

11. Manufacturing/Repairs and Part Supply

- a. Specifications: The solution must be made of modular replaceable components.
- b. Rationale (Self-Imposed): To keep manufacturing costs down and reparability viable, the product should be composed of 3D-printed parts and replaceable components. If something breaks the cost of repairing the product is minimized. For example, Velcro is a cheap and durable alternative to traditional zip ties or bolts, with a cycle life of around 20,000 uses.¹¹

⁹ bHaptic. (2024). *bHaptics TactGlove DK1 Wireless Haptic Gloves (Pair) - VR Accessory*. KnoxLabs. Retrieved 12 06, 24, from <https://www.knoxlabs.com/products/bhaptics-tactglove-dk1-haptic-gloves?srsId=AfmBOooZhpYmoK6P69NOkd7TPpjTSU4FyB2ZjaAdarD8ZvzBr73il8YI>

¹⁰ "Weart TouchDIVER Pro Haptic Gloves." *Knoxlabs*, www.knoxlabs.com/products/weart-touchdiver-pro-haptic-gloves?_pos=1&_sid=51b4e4463&_ss=r. Accessed 12 Dec. 2024.

¹¹ Ira, Stephen. "How Long Will It Stick? Understanding the Lifespan of Hook and Loop." *HookandLoop.Com*, 22 Nov. 2023, www.hookandloop.com/blog/how-long-will-it-stick-understanding-the-lifespan-of-hook-and-loop?srsId=AfmBOorzQca6ERDA6PkPlyNbKspVVyYw73iHR_iEGso-OZMVSNLgUUq_. Accessed 14 Dec. 2024.

Brainstorming Ideas

The sketches include the following labels and descriptions:

- Sketch 1 (Top Left):** Labeled "NO GLOVE" and "10".
- Sketch 2 (Top Left):** Labeled "GLA #6" and "12".
- Sketch 3 (Top Right):** Labeled "flexible joint restriction, seize up by then" and "3".
- Sketch 4 (Top Right):** Labeled "caps on top open or closed top" and "2".
- Sketch 5 (Top Right):** Labeled "strapped on or 3D built" and "1".
- Sketch 6 (Middle Left):** Labeled "for better attach point" and "13".
- Sketch 7 (Middle Left):** Labeled "press to menu or even attach own remote" and "14".
- Sketch 8 (Middle Left):** Labeled "etc" and "16".
- Sketch 9 (Middle Left):** Labeled "VR GLOVE MINDMAP" and "9".
- Sketch 10 (Middle Left):** Labeled "TENDON ATTACHMENT" and "10".
- Sketch 11 (Middle Left):** Labeled "FINGER MOUNT" and "11".
- Sketch 12 (Middle Left):** Labeled "plate hinge caps" and "12".
- Sketch 13 (Middle Left):** Labeled "MOTOR" and "13".
- Sketch 14 (Middle Left):** Labeled "flex sensor 2 to M.F. 100" and "14".
- Sketch 15 (Middle Left):** Labeled "potentiometer + sensor" and "15".
- Sketch 16 (Middle Left):** Labeled "inner plate restricts" and "16".
- Sketch 17 (Middle Left):** Labeled "where claws restrict from forearm" and "17".
- Sketch 18 (Middle Left):** Labeled "## but extended to arm" and "18".
- Sketch 19 (Bottom Left):** Labeled "rate for VR capabilities" and "19".

Figure 5.1: Basic brainstorming sketches detailing parts of a potential solution.

Here are some additional ideas on types of finger attachments:

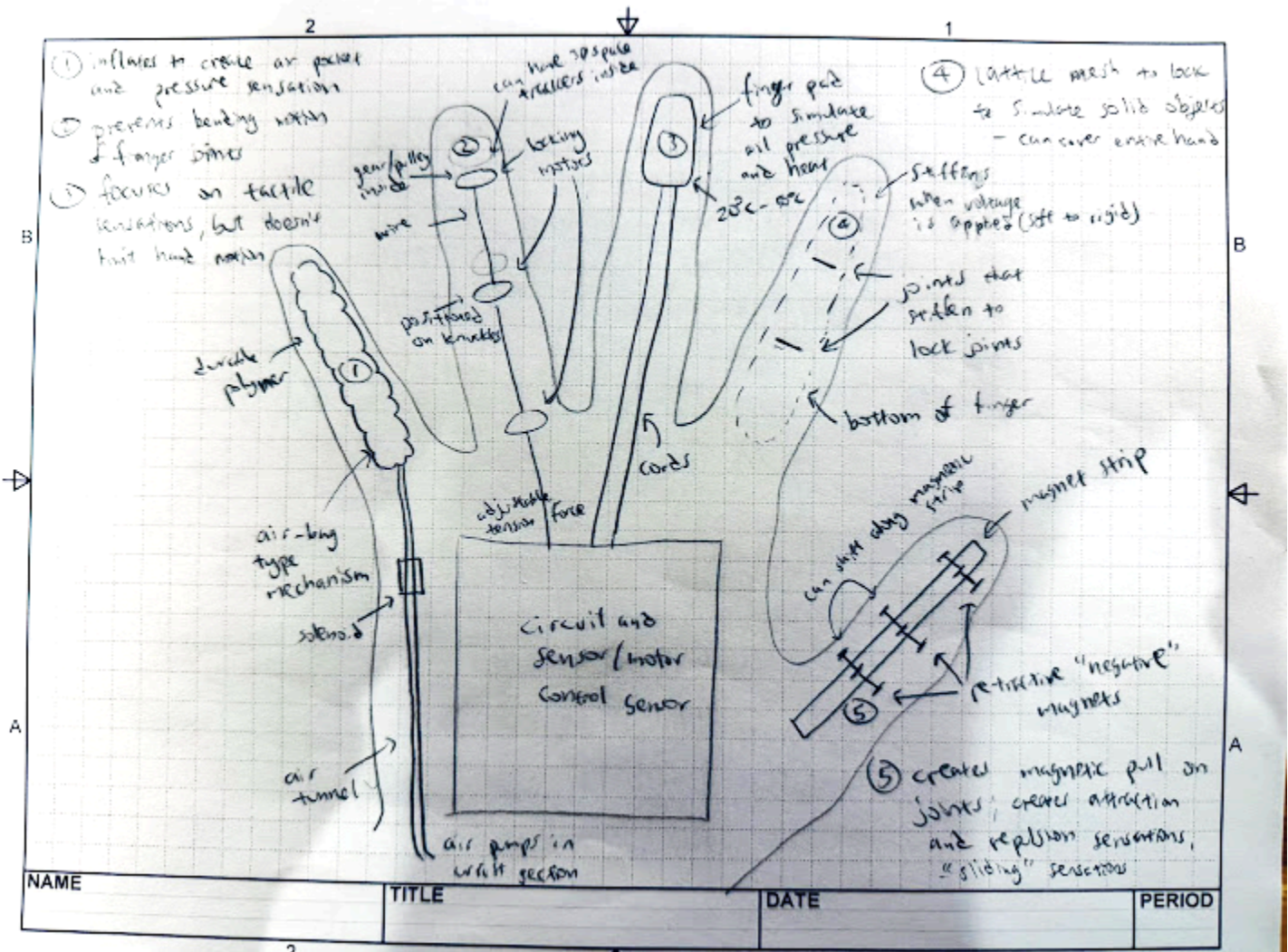


Figure 5.2: Refined individual brainstorming sketches detailing finger attachments for wired solutions.

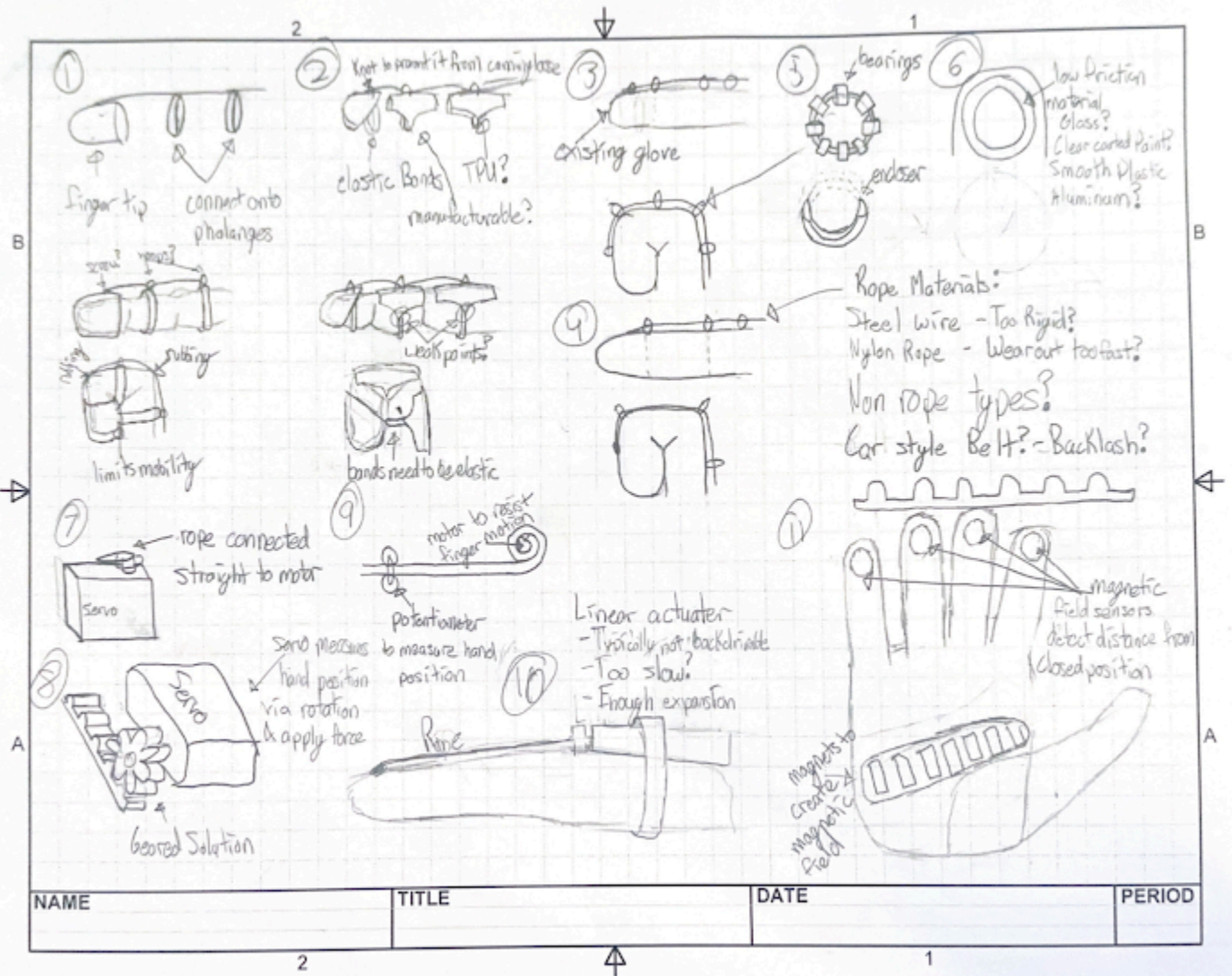


Figure 5.3: Refined brainstorming sketches with added descriptions.

Appendix C

Product Concepts

Product Concept #1

Designer: Ollie Liang

Description

The product is a haptic glove that both restricts finger movement and applies pressure to the palm when an object in VR space is detected being touched. The purpose is to simulate real-life, tactile feedback while in a VR environment to help in low-cost surgical training, hands-on learning in STEM fields, and skill-based training. The appearance of the final design concept will be a one-size-fits glove with several velcro straps, servo motors with wires to control finger movement, a pressure plate on the palm, and many electronic and control components on the back of the hand and forearm.

Operation

The product works by detecting an object touched in VR space, sending a signal to the Arduino which then tells the servo motors to hold or lock the wires in place, restricting finger movement. Depending on how close the palm is to the virtual object, it will activate and place pressure on the palm by vibrating at a certain speed to simulate touching something. It also uses flex sensors to detect and correctly adjust the motor wires to limit finger movement and rotation. All velcro straps should be wrapped at least semi-tightly to ensure that the wire is flush against the glove.

Justification

The product solves the problem by creating easy and repetitive access to hands-on training. While traditional medical training requires cadavers and setup time, the VR glove paired with a virtual environment allows things to be reset quickly while still providing crucial tactile feedback. This concept is more intuitive as it uses mechanical aspects instead of chemical or magnetic qualities to achieve its goal. This design has many off-the-shelf parts for cheap repairs, and it is easy to use since it is a one-size-fits-all and has a hot-swappable battery. One of our main specifications was to have a restrictive force on fingers and position and rotational accuracy, and this glove achieves both with ease and adds on a force feedback component to provide pressure.

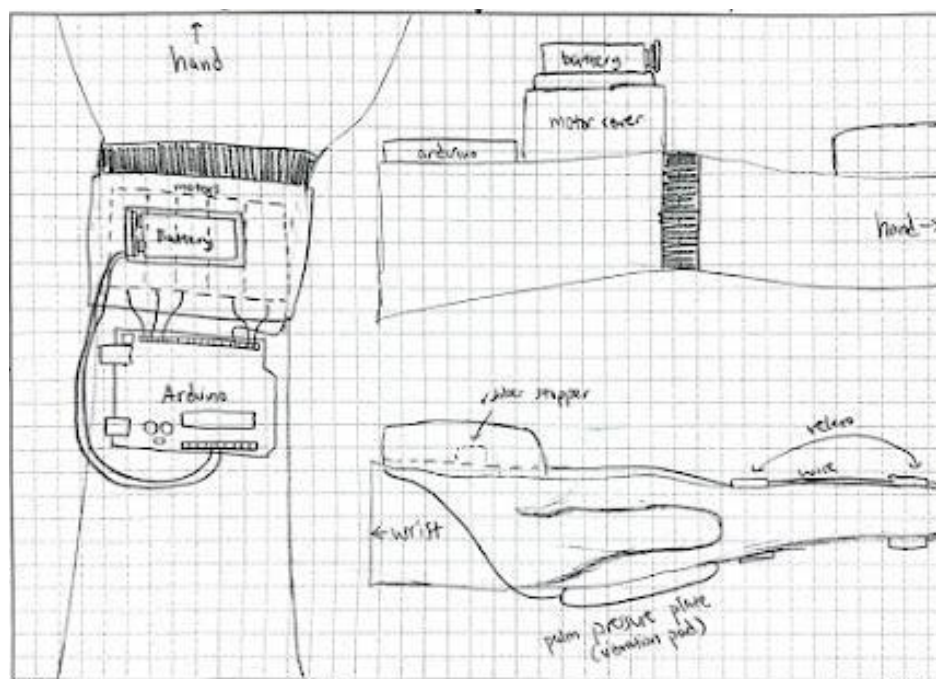
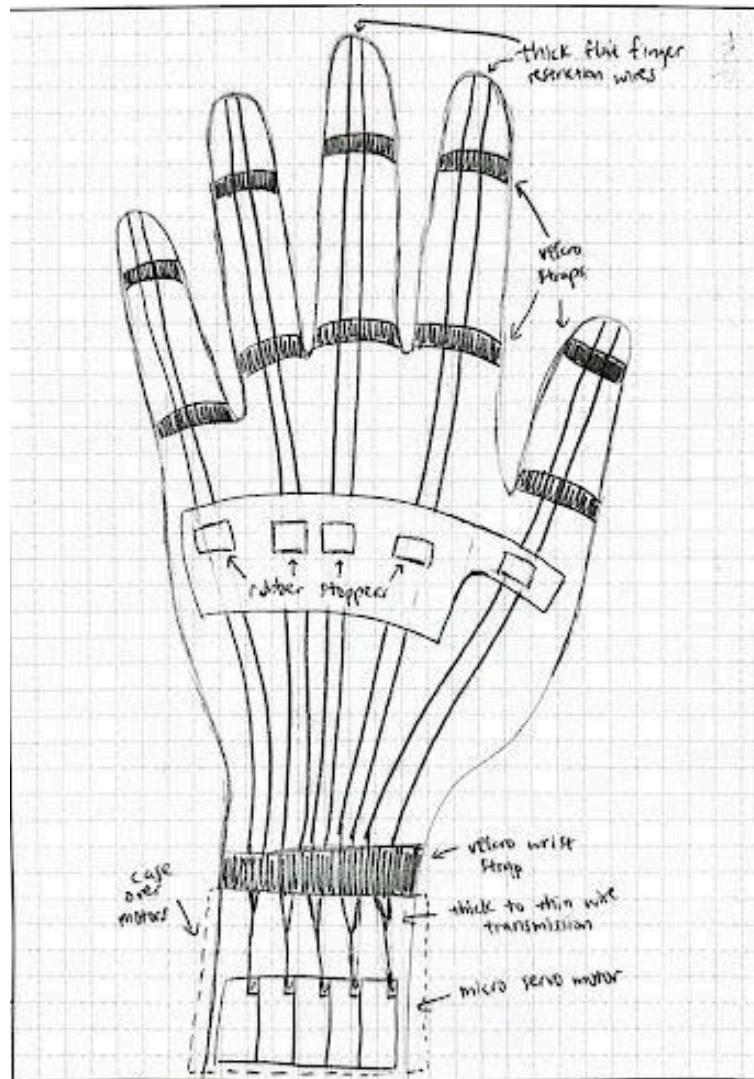


Figure 6.1: Product Concept #1 drawings.

Product Concept #2

Designer: Ollie Liang

Description

The product is a microfluidic haptic glove that uses pressure and vibrations instead of wires to create tactile feedback. The purpose is to create a more unique glove and add more haptics than simply restricting the fingers, which doesn't apply any haptic pressure. The appearance of the final design concept will be a glove that has pneumatic air tubes running across the top of the glove with vibration pads and semi-inflated air tubes on the bottom of the fingers and palm. It will use these air tubes to prevent clenching the hand all the way.

Operation

The product will deliver tactile feedback through pneumatic air pressure and vibrations to create a realistic touch in virtual environments. Using solenoids, a pressure sensor, and air tanks, the glove will provide proportional airflow to the tubes to simulate how hard or soft an object is. There will be LRAs (Linear Resonant Actuators) and piezoelectric actuators, which both help create small vibrations at specific frequencies. An array of them will be placed near the fingertip to simulate different object properties such as roughness or curved edges. The glove will input finger-to-finger distance and object distance in VR using motion sensors and the battery and air tanks must be full for the glove to function properly. Using the semi-inflated air tubes, it will prevent the user from fully closing their fist, and a micro air pump will be used to inflate and deflate the air tubes.

Justification

The product solves the problem by removing the limitations of traditional medical training and practice. By using air pressure and vibration mechanisms, the glove becomes more immersive as it focuses on tactile feedback rather than restrictive forces. This allows the user to experience different textures and shapes more realistically, and since the glove does not need any motors or wires, there is significantly less bulk on the glove itself. This may make the glove more ergonomic and accessible. In addition, the pneumatic system is more cost-effective than microfluidic actuators or electroactive polymers and saves energy to manufacture and test. One of our design specifications was weight and joint materials, and because the air tubes and piezoelectric actuators are flexible and largely lightweight, it allows the user more comfort for long-duration sessions.

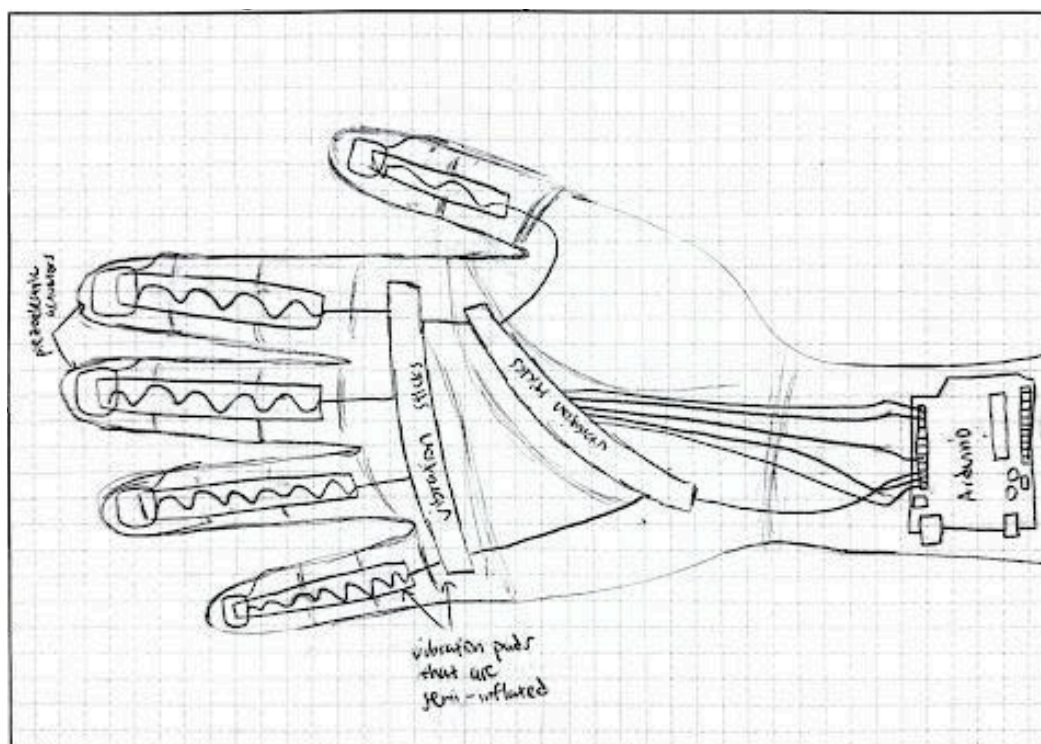
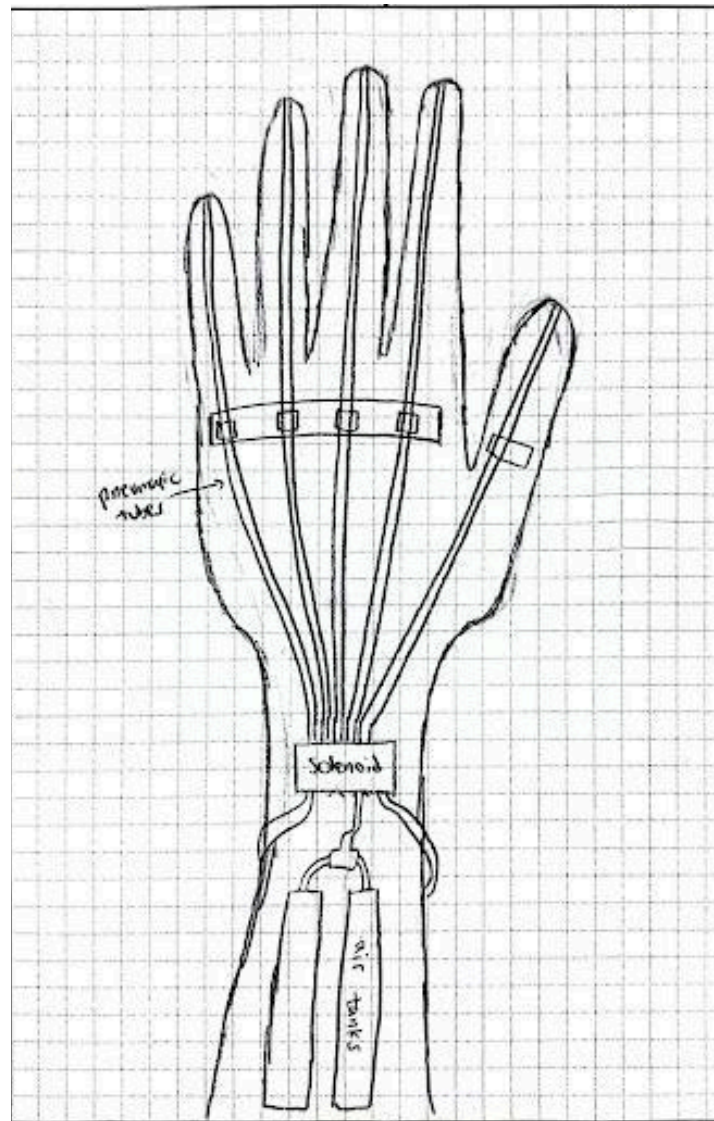


Figure 6.2: Product Concept #2 drawings.

Product Concept #3

Designer: Monish Ramesh Jayakumar

Description

The product is a virtual reality glove paired with a custom VR application to provide a digital alternative to cadaver use and surgical training. The glove can simulate touch using various resistors for position and force feedback. The final glove is made of a flexible, zippable fabric for the user to place their arm into. Users can mount a tracker of choice onto the glove from a preexisting headset.

Operation

The product is first unzipped from the forearm so the user can place their arm into it. They can then zip it back up and turn on the battery, our main input for electricity. This battery (some sort of battery pack) may be wired for charge or wireless for greater mobility. Power is supplied to a hand tracker for position and rotation tracking (for example a VIVE Tracker), servo motors, and through a voltage regulator to the core Arduino and Motor Shield for the device to be on.

Once strapped, the user can flex their fingers. Flex sensors on the back of the hand determine the amount of flex through resistance, translated to a digital hand in VR. The VR application runs through Unity's base VR systems, and waits to detect collisions, then sends those signals to the Arduino. This in turn will activate the servo motors to start resisting motion. Now, each motor has fishing wire threaded through segmented, flattened tubes connecting them to a cap housing a force resistor at the tips of each finger. When resistive force is applied, the cap levels onto the front surface of the finger, where a force resistor. Based on the hardness of the digital object, the force resistors allow for the servos to give the exact output of torque based on the hand motion and rigidity of the digital object.

A Bluetooth receiver connects the Arduino to the headset for all communication. The VR application itself, as a product, would include various medical simulations with open-source code allowing for modifications and allowing users to create their own virtual applications.

Justification

The product is easy to repair with an overall lightweight modular design. It utilizes the whole forearm and gives wrist resistance which many competitors do not have. All swappable pieces are easily reachable, increasing ease of repair. The more open frame of the glove also allows for a wider range of sizes to use the glove, not requiring multiple gloves of various sizes to accommodate a diverse range of users. Those changes improve quality of life and reduce weight and hassle, filling in the criteria of greater accuracy and precision with an accessible design.

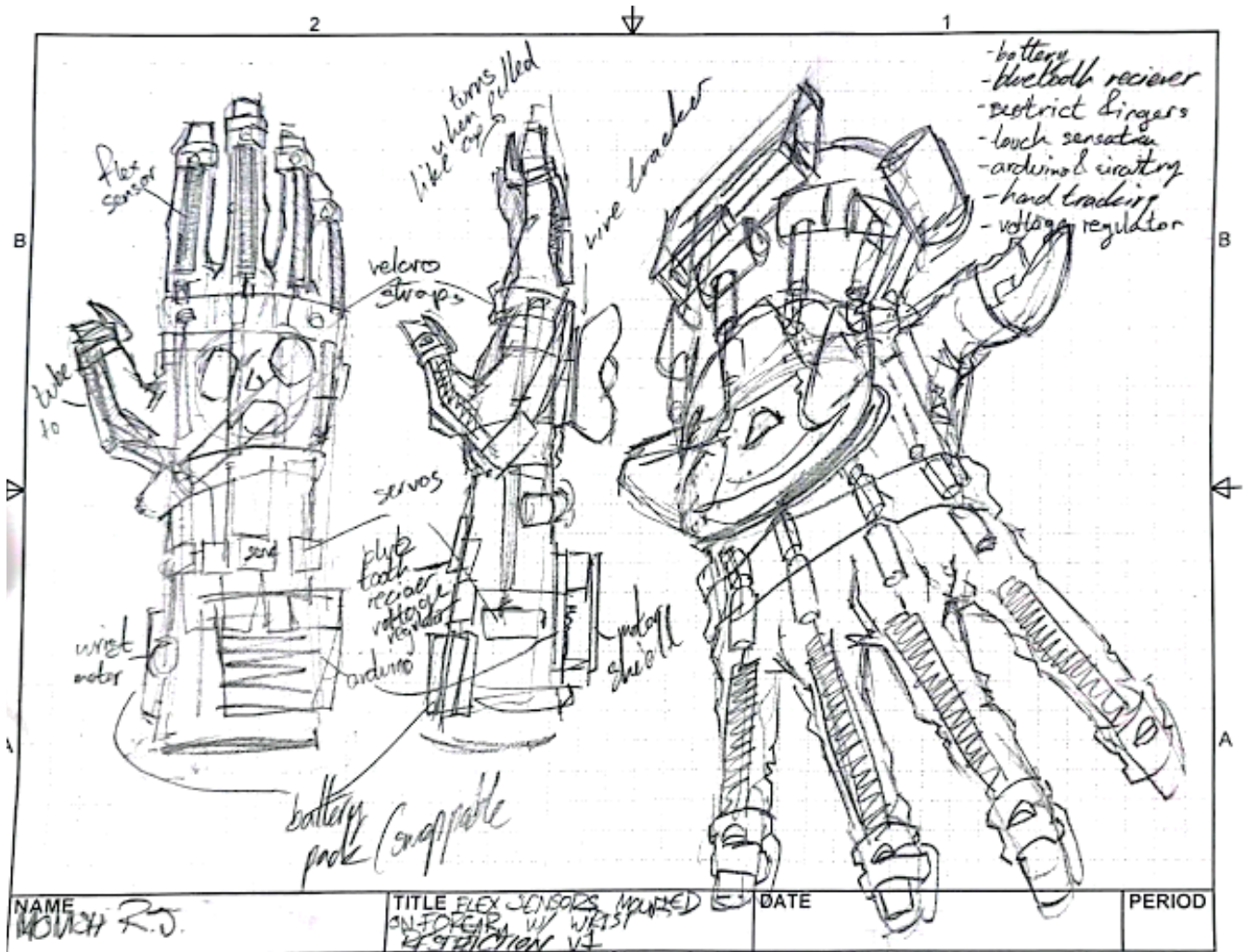


Figure 6.3: Product Concept #3 drawings.

Product Concept #4

Designer: Monish Ramesh Jayakumar

Product Description

The product is an augmented reality glove paired with a custom AR application to provide a digital alternative to cadaver use and surgical training. The glove can simulate touch using various resistors and brakes for position and force feedback integrated into the real world for analytics. Users can mount a tracker of choice onto the glove from a preexisting headset.

Operation

The product is strapped onto the user's hand, aligning the various plates of the glove to the faces of the forearm. A battery is then turned on, our main input for electricity. This battery (some sort of battery pack) may be wired for charge or wireless for greater mobility. Power is supplied to a hand tracker for position and rotation tracking (for example a VIVE Tracker), servo motors, and through a voltage regulator to the core Arduino for the device to be on.

Once strapped, the user can flex their fingers. Flex sensors on the back of the hand determine the amount of flex through resistance. The AR application runs through Unity's base VR systems, and waits to detect collisions to digital objects, then sends those signals to the Arduino. This in turn will activate the brakes located at the wrist of the glove. These brakes will stamp onto the wire, stopping it from extending further through friction. Less braking force equals to the rigidity of the object. Now, each brake base has fishing wire, coiled up for constant tension through a flat spring, threaded through segmented, flattened tubes connecting them to a cap at the tips of each finger. When resistive force is applied, the cap restricts the tips of the fingers.

A Bluetooth receiver connects the Arduino to the headset for all communication. The headset, which shall have AR capability, will run a custom Unity ARCore application allowing it to identify medical objects in the real world and give analytics based on such as an overlay, or simulate cadavers in a real-world environment. This open-source software could be partnered with Anatomage software and greater libraries allowing for a seamless integration of real-world and virtual benefits.

Justification

Although this design would require multiple sizes to accommodate a variety of hands, the benefits outweigh the negatives. The brake system and flat spring allow for constant wire tension and precise calibration, similar to a bike brake. This makes it easy to fix. Additionally, a modular design allows for use with any headsets with AR compatibility and hand tracking, a hot-swappable battery, and accessible circuitry. The AR functionality of the glove also allows for the reusability of current resources in place at medical institutions, enabling seamless integration and enhancing pre-existing curricula.

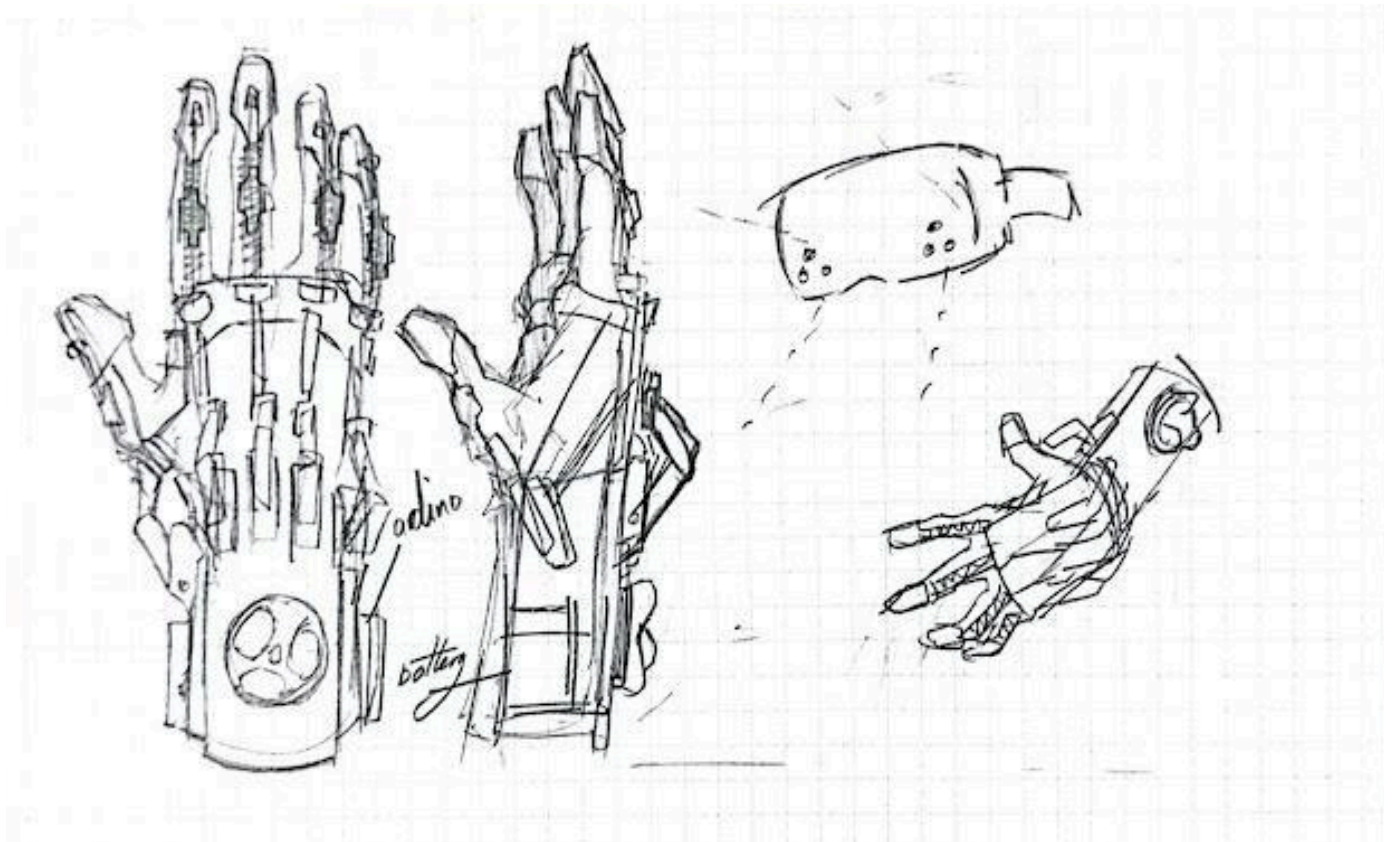


Figure 6.4: Product Concept #4 drawings.

Product Concept #5

Designer: Ryan May

Product Description

The product is a glove that mounts to both the hand and the forearm and creates resistance on the contraction of each finger to mimic the feeling of pressing against an object. The product will enable a user to have the sensation of holding digital objects within a virtual simulation. The object has an open design where components aren't covered so that adjustments are easy to make.

Operation

The logical control of the glove is an Arduino Nano ESP32 which is powered by 4 AAA batteries that output a combined 6v. The 6v is then stepped down to 5v by a buck converter. The 5v out is connected to the microcontroller and each of the 8 MG90S servo motors. Each one of the motors is connected to either one of the fingers or to a part of the hand to restrict the motion of that part of the hand. The glove will track each finger by measuring how much the servo has rotated, and to ensure the cable is extended the correct amount the servo will always be attempting to retract with a small amount of force.

Justification

The product solves the problem by enabling a higher amount of feedback to the user than other products within the market, while at the same time being miles cheaper than the competition. This concept was chosen as one of the final solutions to pursue because the design gives enough space for larger and more powerful servos while still being small enough for the design to not be a hindrance to a user. The multiple batteries as well as a voltage regulator should allow for quick battery swapping which helps save the customer time by reducing time when the glove is unusable. This glove will also save the customer money as it uses mostly off-the-shelf parts so that if a component breaks it is fairly simple and cheap to replace. Although this glove may not be as light as it initially wanted to be, by moving most of the weight farther up the arm the effect of a heavy product will be less noticeable and more comfortable over longer periods of time. This product also improves the quality of life of the user by including more resistance on the hand which enables it to better mimic the sensation of holding an object and feeling the resistance that the object applies on the user.

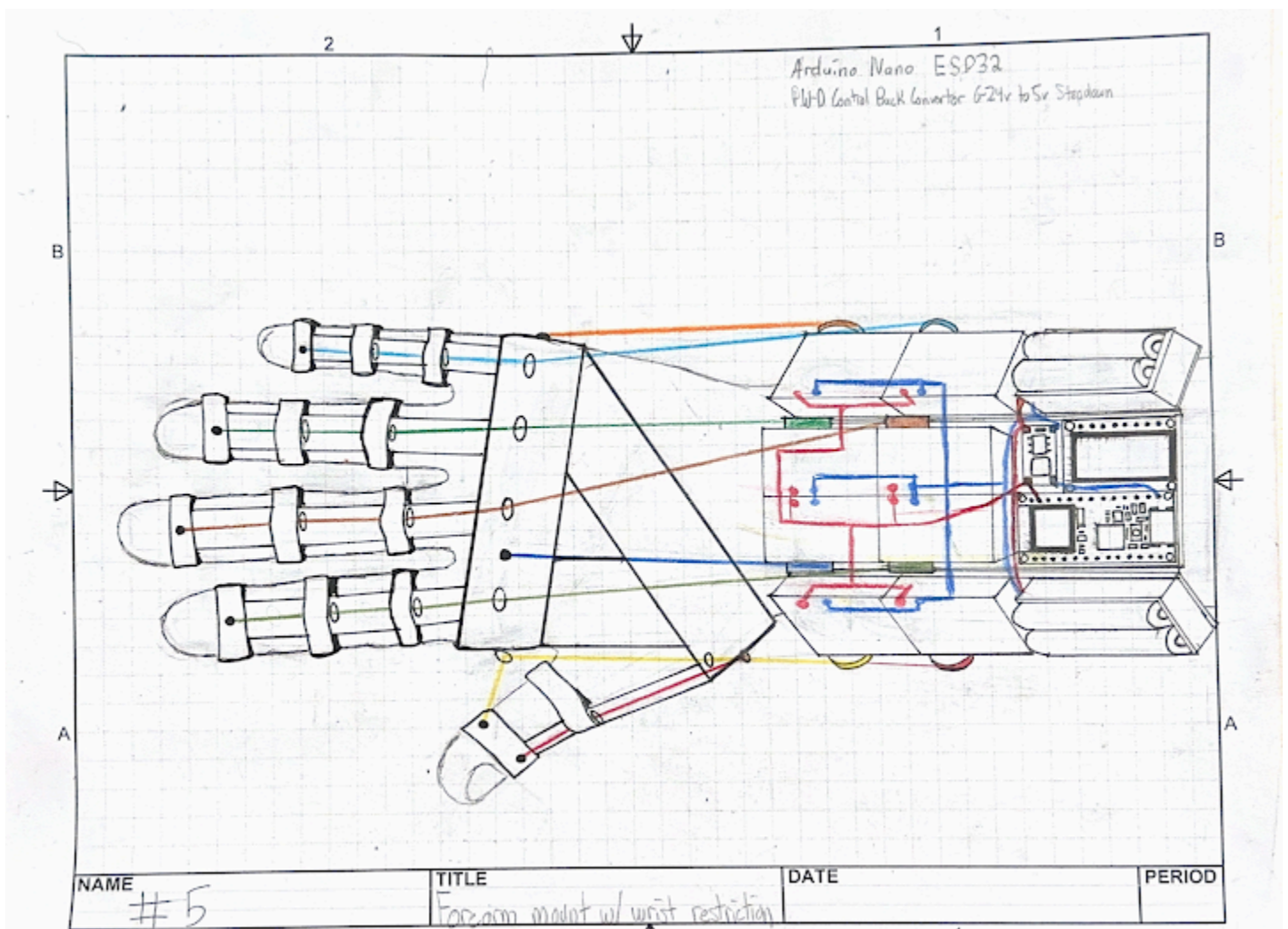


Figure 6.5: Product Concept #5 drawings.

Product Concept #6

Designer: Ryan May

Product Description

The product is a glove that is located mostly on the outside of the hand with a part of the power supply located on the wrists, and it creates a resistance on the contraction of each finger to mimic the feeling of pressing against an object. The product will enable a user to have the sensation of holding digital objects within a virtual simulation. The object replicates the design of a typical glove however on the end of each finger is a cap and an elastic band that connects to the base of the glove. Most of the electronics are exposed.

Operation

The main logic center of the glove is an Arduino Nano ESP32 which is powered by 2x 2 AAA batteries wired in series to output 6v. This is then connected to a step-down buck converter which converts the 6v input to a 5v output, which is connected to the Arduino nano and each of the 5 ULN2003 Driver Test Modules which control the 5 28BYJ-48 stepper motors. Each stepper motor is connected to a spool which houses a wire that connects to a cap on the end of each finger. As each finger contracts the string rotates the stepper motor which records the rotation and calculates the location of each finger. To ensure that the cable is rotated the correct amount the stepper motor will be continuously attempting to rotate with a small amount of force to keep the cable in a small amount of tension. When the software determines that the finger is about to or is colliding with a digital object it will send a signal to the microcontroller which tells each stepper motor how much to retract to replicate the sensation of an object in 3d space. The software connects to the microcontroller via Bluetooth which is built into the Arduino Nano.

Justification

The product solves the problem by reducing the cost of the ability to have this level of feedback and by reducing the weight, which improves the quality of life of the user when using the product. The product manages to reduce the weight of existing solutions by using a smaller and more compact microcontroller, as well as smaller stepper motors, which still have enough force to create a realistic sensation. This design in particular was considered as one of the final designs to pursue because of its small size, with all the parts compacted into a small space. As well as its highly simplified design which maximizes the ratio between weight and ability by removing potentially unnecessary motors. Additionally, the glove saves the consumer money by creating an all-in-one design that can change the size to fit any and every possible user.

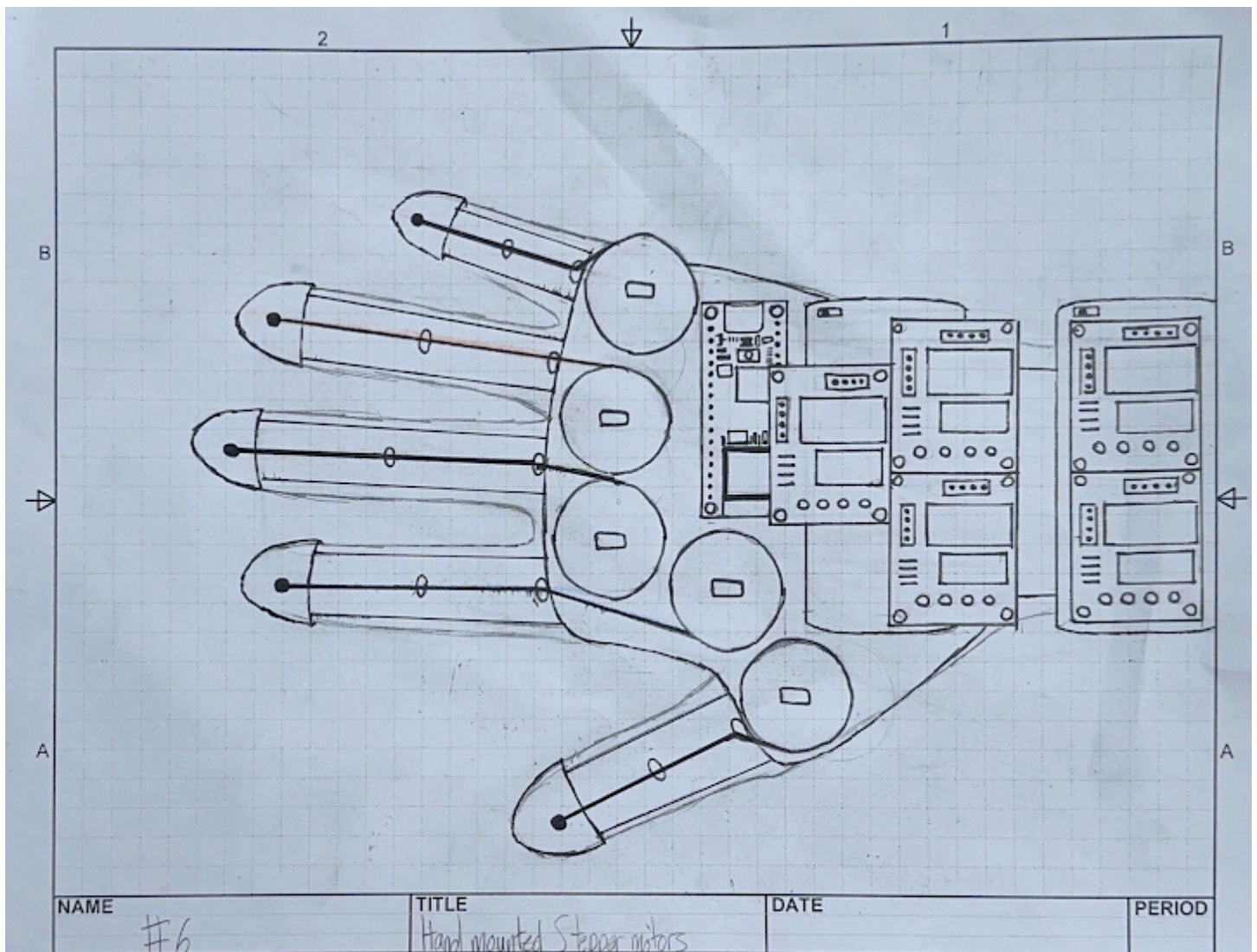


Figure 6.6: Product Concept #6 drawings.

Appendix D

Consumer Feedback

A user survey was conducted to determine the quality of life of current VR products to aid in determining human benchmarks for the potential solution. Users with experience in VR were also asked to rate three potential designs and give additional feedback.

Have you used virtual reality (VR) technology before?

32 responses

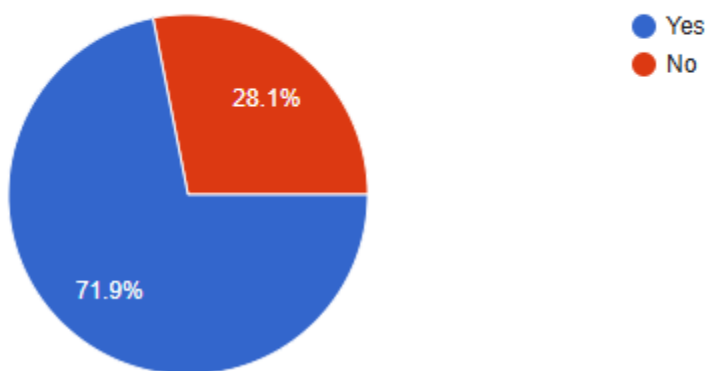


Figure 7.1: Pie chart of percentage of people who have used VR technology before

Through the 72% of people who have used VR before, they provided useful information on the frequency of their VR uses, comfortability, and immersion, which will help determine which aspects of the glove we will prioritize to meet consumer expectations.

How long is a typical VR Session (30 minute increments)?

23 responses

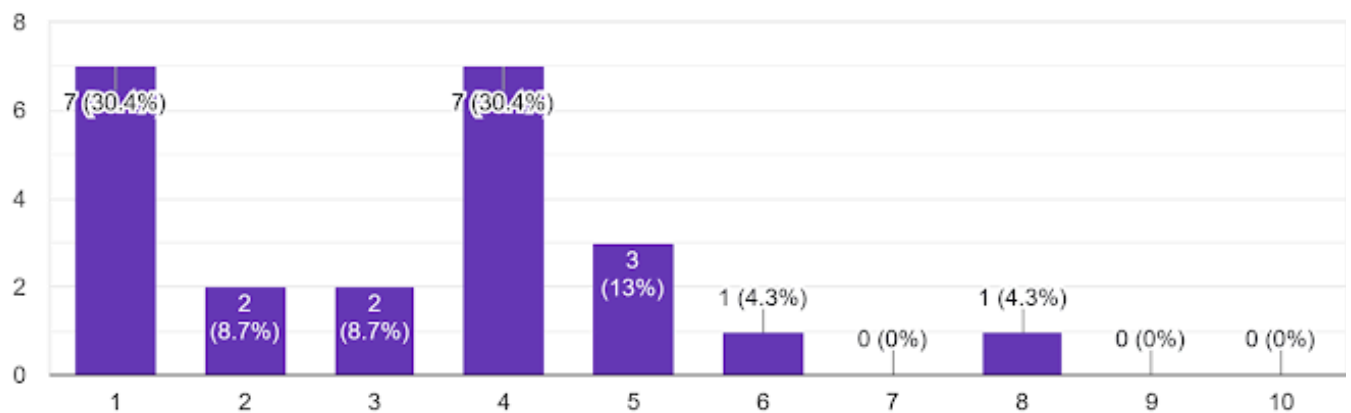


Figure 7.2: Bar graph of average lengths of sessions for VR users

Do you think buttons take away from the immersion of VR?

23 responses

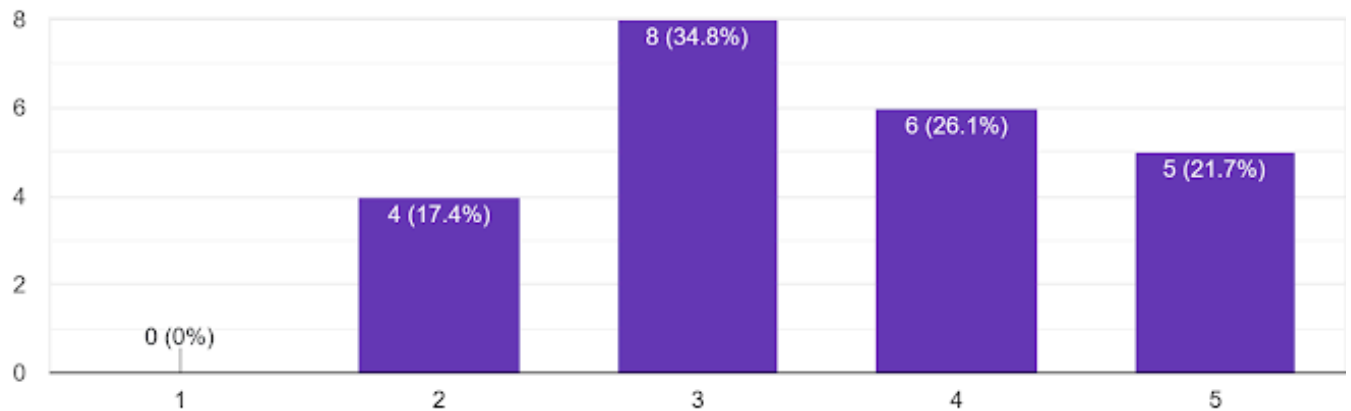


Figure 7.3: Bar graph with data on button impact on VR immersion

How much do you sweat while using VR?

23 responses

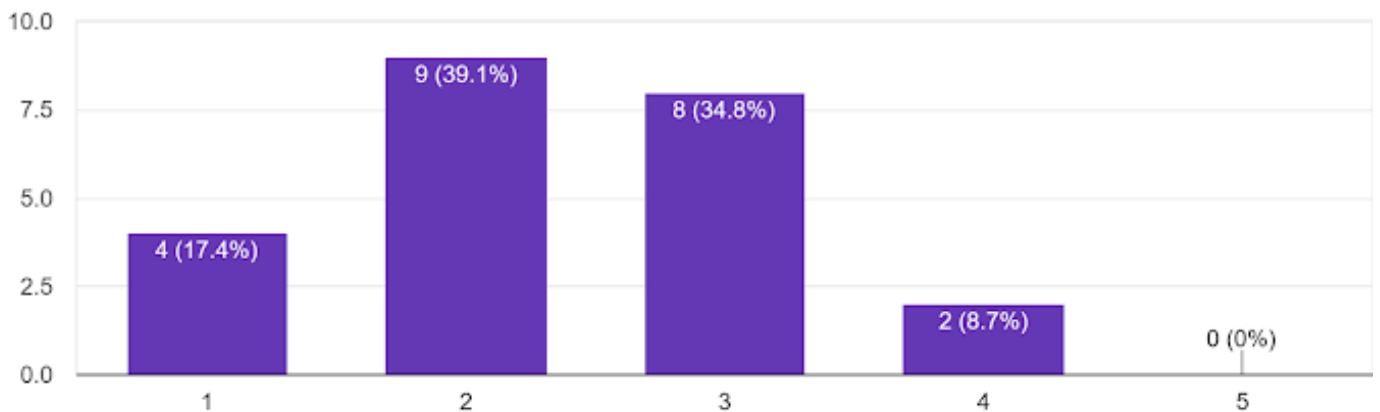


Figure 7.4: Bar graph detailing amount of sweat when using VR

In addition, we gave all survey takers the opportunity to leave some feedback. Here are some useful responses:

- The glove would increase sweat unless it had breathable materials or even a cooling function
- Concept 2 is significantly more advanced because it uses brakes instead of motors, which uses less power and is much less likely to break. For haptics you should use magnetic vibrators because they don't require waiting for a motor to spin up, and response time is very important in VR.
- The weight of the arm is a huge problem because if anything is too heavy no one would wanna use it or buy it. Like servos being used to control each finger may be heavy.
- There's a company that is making a VR glove that transmits a touch feeling to your hand using electric signals. I would consider learning from them(forgot name).

By collecting all this data, we can ensure that the final haptic glove can be built to accommodate weight and immersion more.

Appendix E

Decision Matrix

		Weight & Distribution	Profile	Approx. Cost	Force Output	Size Adjustability	Battery Life	Visual Appearance	Feasibility	Ease of Assembly	Software Integration	Total
Designs	Scale Factor:	3	1	2	1	5	2	1	4	3	3	
Flex Sensor Controlled Restriction w/ Palm Vibration		6	5	6	9	5	7	7	4	8	7	151
Vibrational and Pneumatic Glove		5	8	7	3	7	6	8	7	6	5	156
Servo Restriction on Palm w/ Zipper		7	4	7	5	8	6	2	5	4	4	142
Brake Feedback with Forearm Mounted Plates		6	7	6	8	7	7	6	7	6	5	161
Finger and Hand Restriction with Forearm Mounted Sleeve		6	7	6	8	7	8	7	5	7	5	159
Stepper Motor Restriction on Hand and Wrist		5	3	8	5	7	4	2	2	4	7	125

Figure 8.1: Decision matrix for six refined concept designs to determine top three potential solutions.

Appendix F

Final Design Drawings

Appendix F contains all sketches of the glove including exploded views, part list tables, individual components, and fully dimensioned drawings.

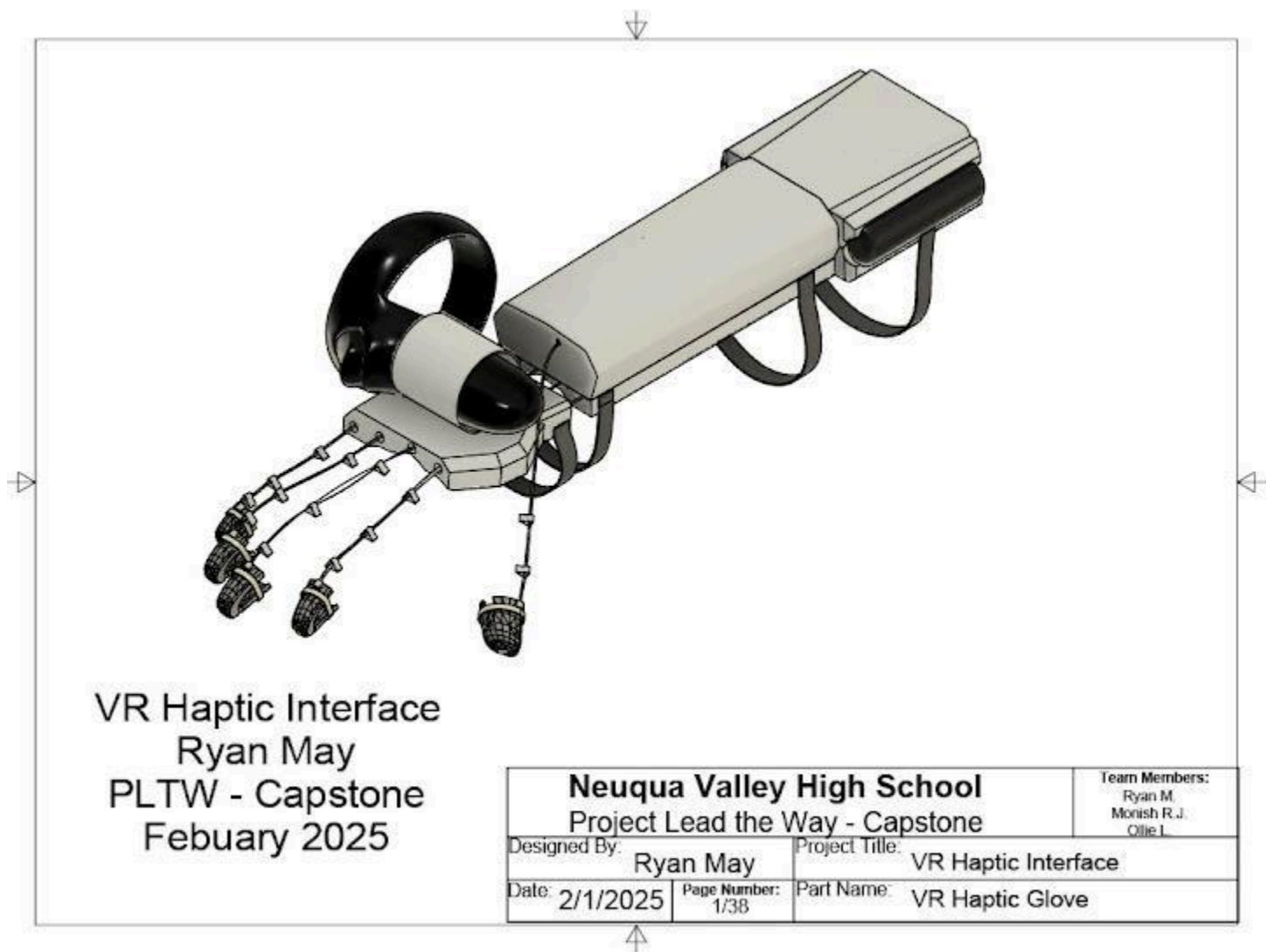


Figure 8.1: Standard View of Assembled Glove

Figure 8.1 & 8.2 shows the glove in its fully assembled state from various angles.

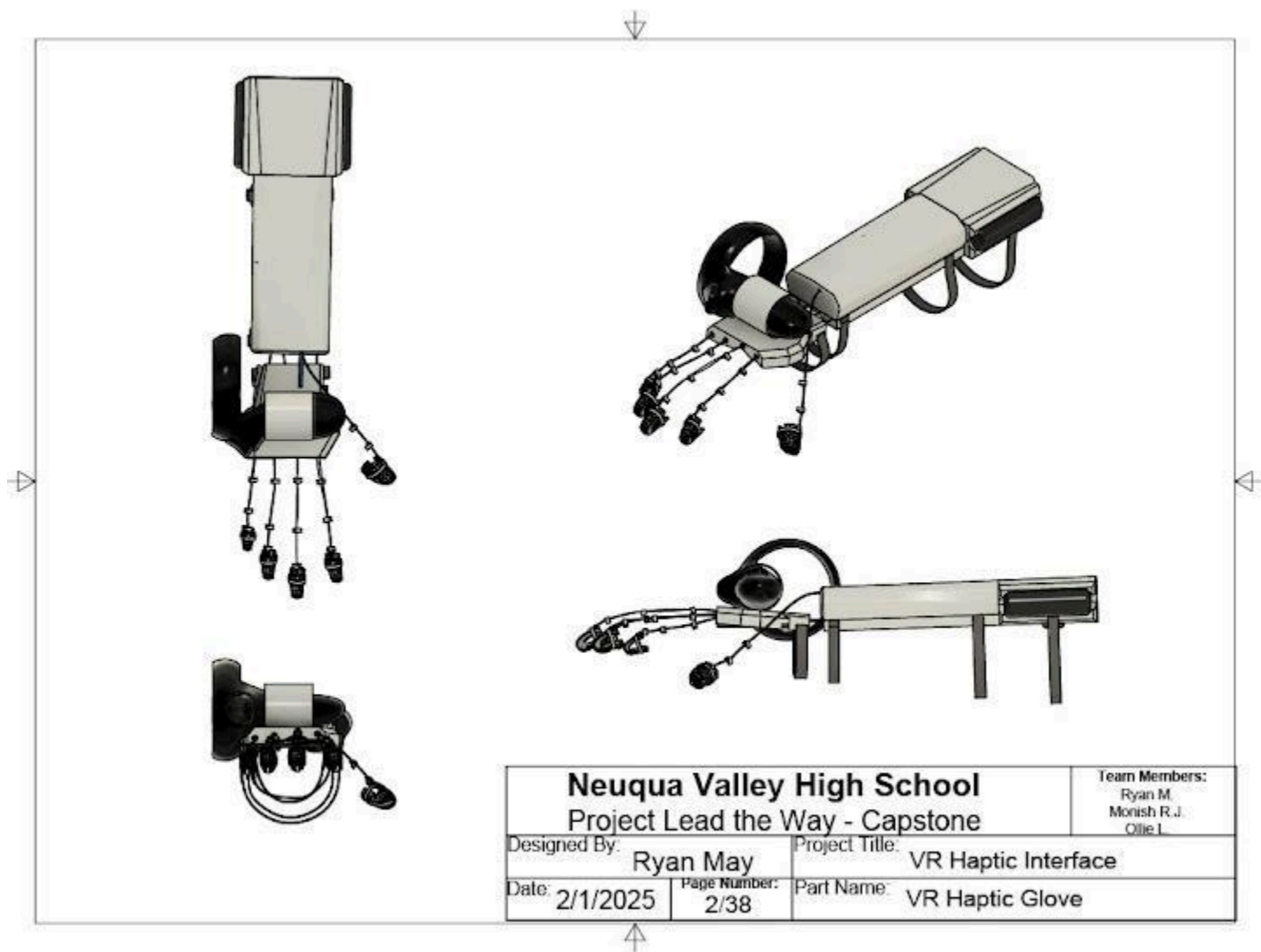


Figure 8.2: Top, Front, Side, and Standard views of the fully assembled glove.

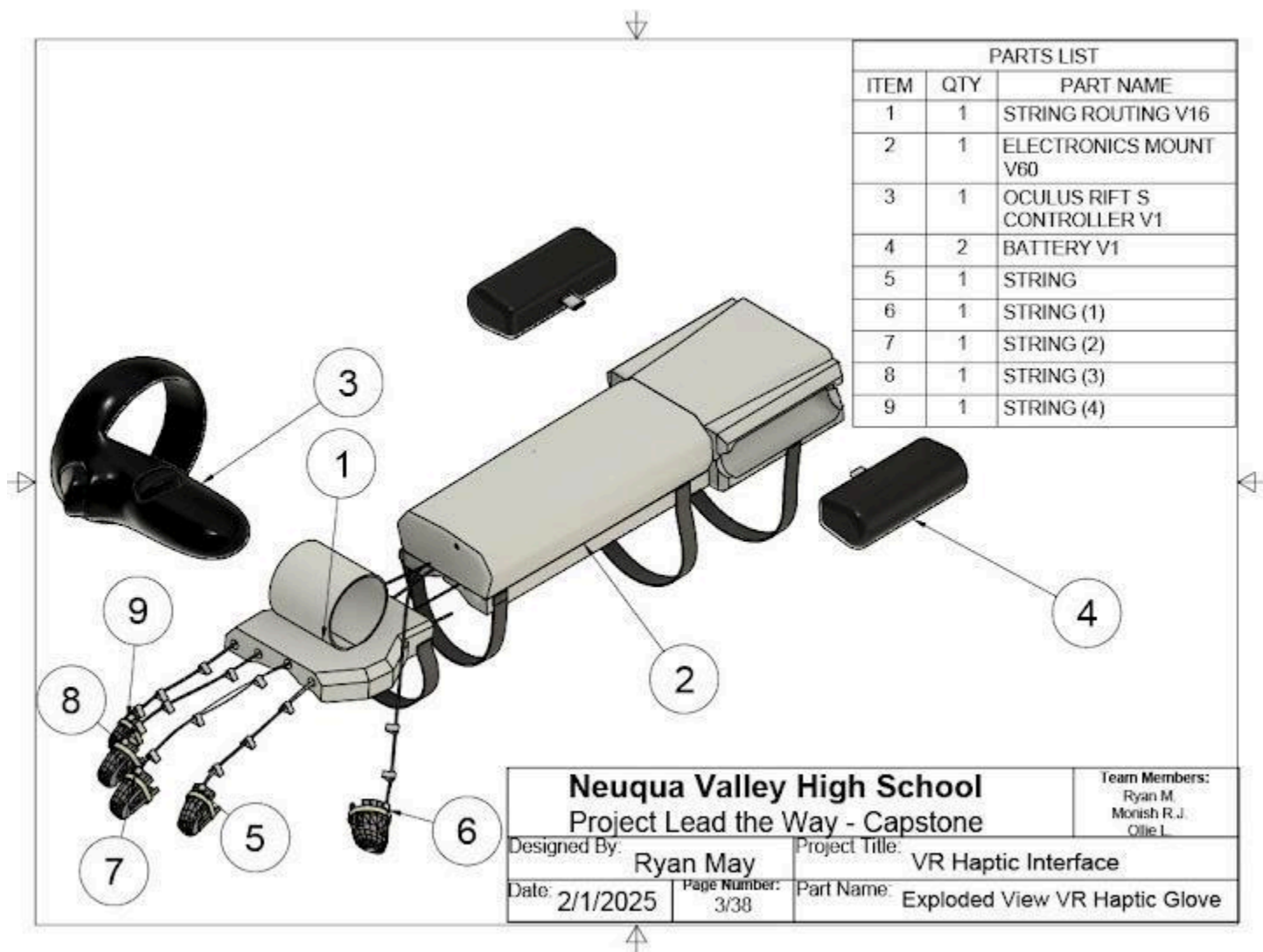


Figure 8.3: Exploded View of Assembled Glove with Parts List (“Electronics Mount” and “String Routing” are considered unique parts and have their own parts lists.)

Figure 8.3 shows each part labeled as well as a parts list for each component/assembly within the fully assembled glove.

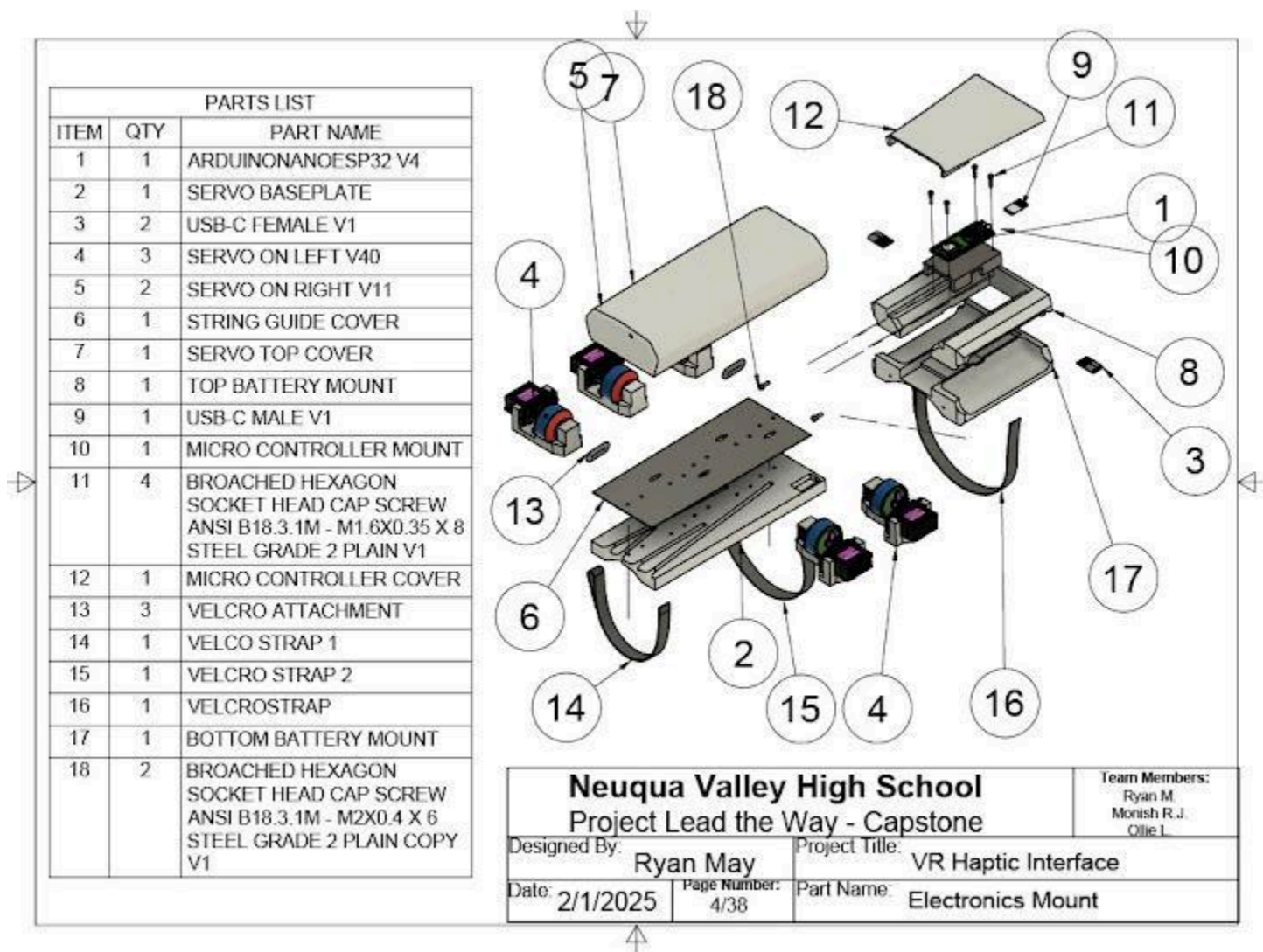


Figure 8.4: Exploded View of “Electronics Mount” from Figure 8.3, including a parts list. (“Servo on left” and “Servo on right” are considered unique parts and have their own parts lists.)

Figure 8.4 shows each part labeled as well as a parts list for each component within the “Electronics Mount”.

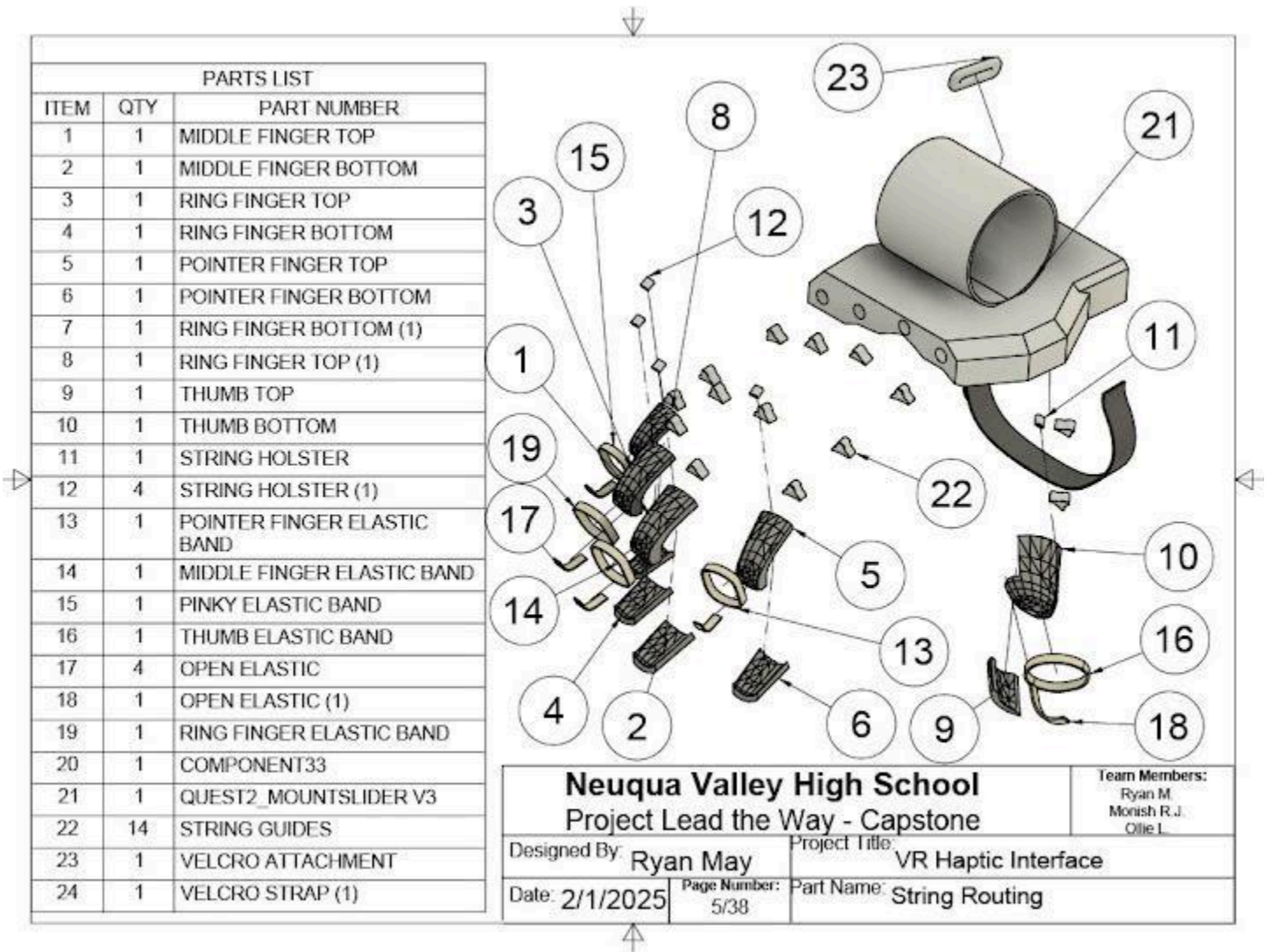


Figure 8.5: Exploded View of the “String Routing” component from Figure 8.1.

Figure 8.5 shows each part labeled as well as a parts list for each component within “String Routing”.

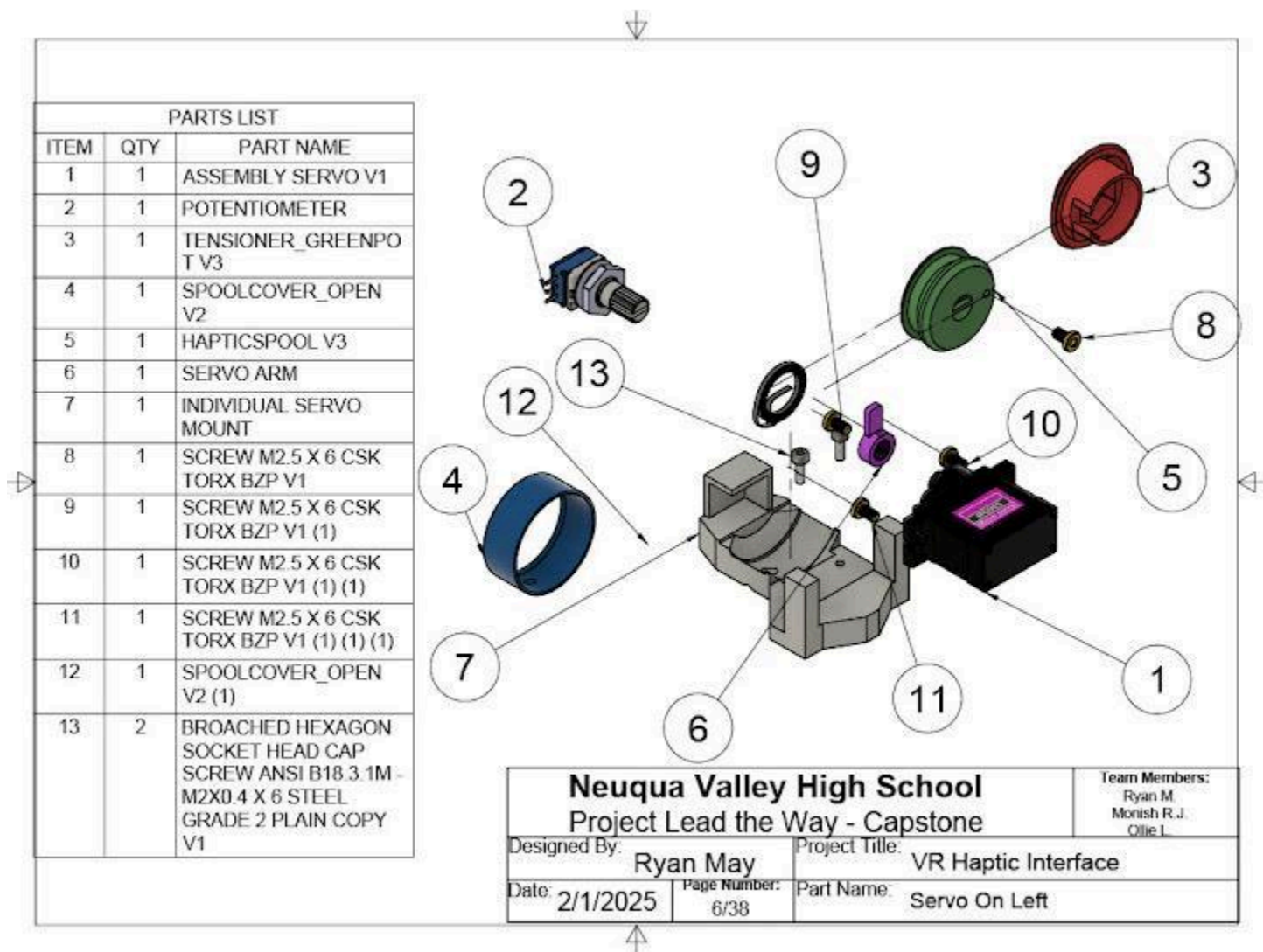


Figure 8.6: Exploded View of the “Servo on Left” component from Figure 8.4.

Figure 8.6 shows each part labeled as well as a parts list for each component within “Servo on Left”.

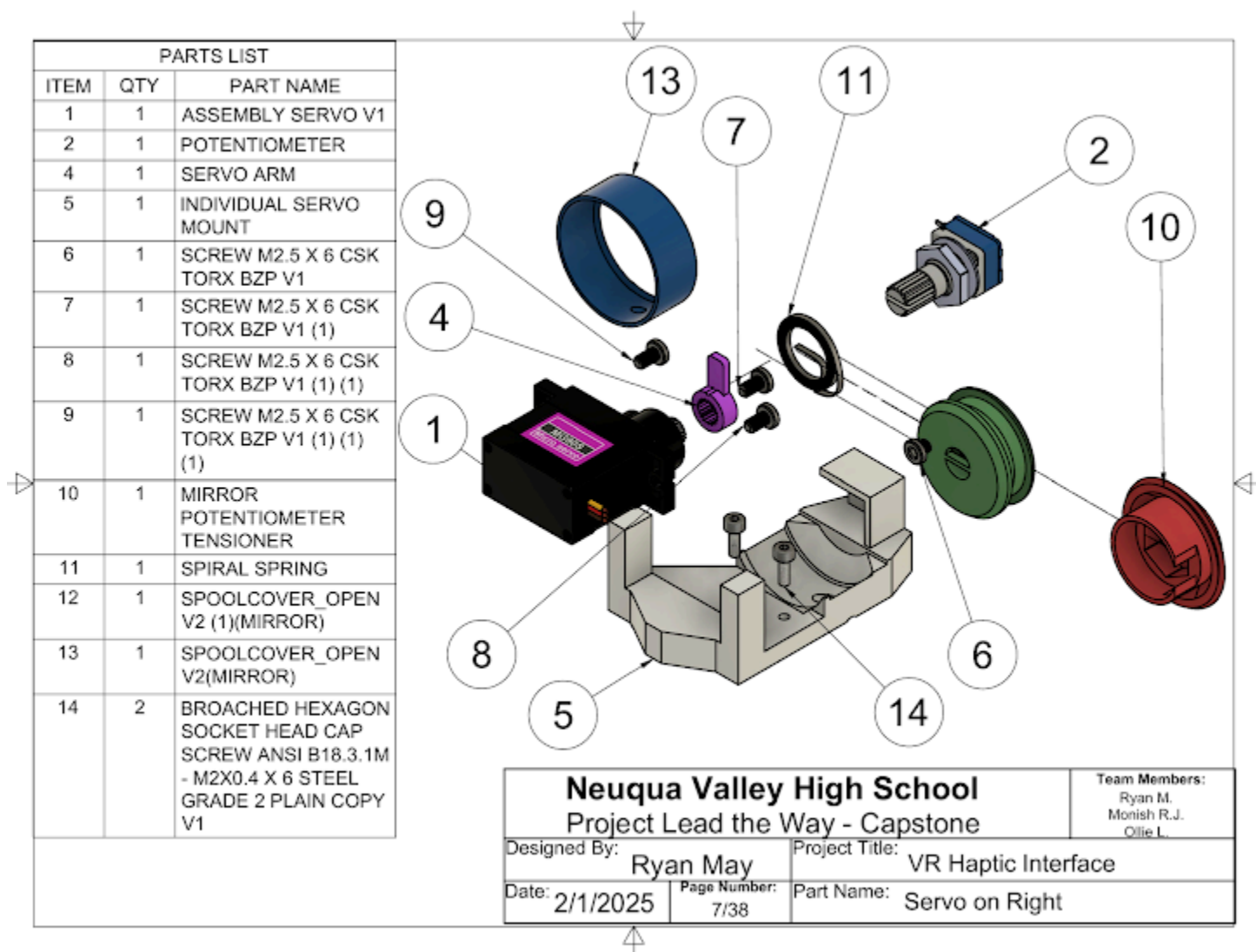


Figure 8.7: Exploded View of the “Servo on Left” component from Figure 8.5.

Figure 8.7 shows each part labeled as well as a parts list for each component within “Servo on Right”.

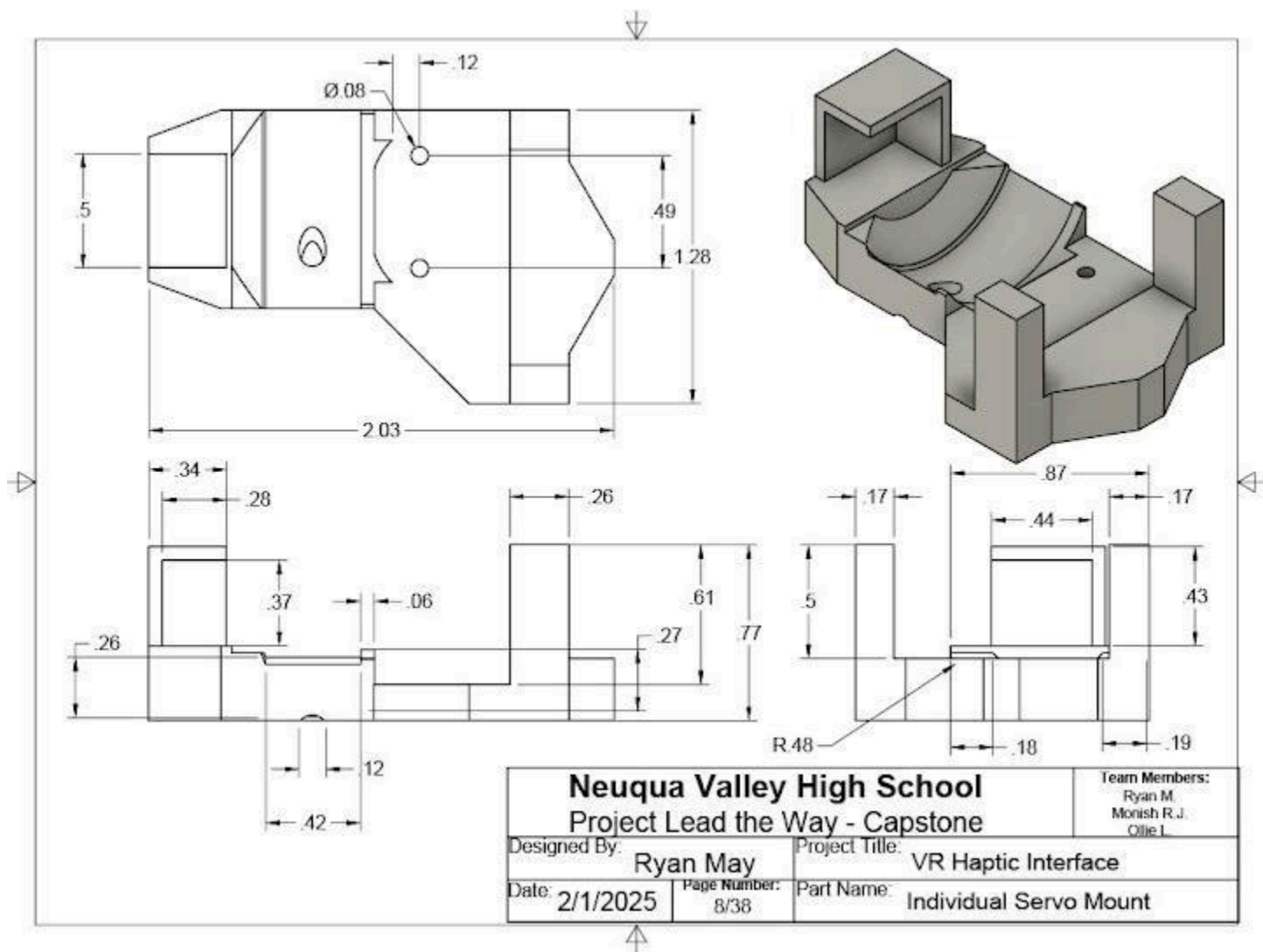


Figure 8.8: Dimensioned Sketch of the “Individual Servo Mount” used in “Servo on Left” and “Servo on Right”.

This mount in Figure 8.8 houses all parts used in “Servo on Left” and “Servo on Right” and can be assembled separately from the rest of the glove to reduce the difficulty of assembly.

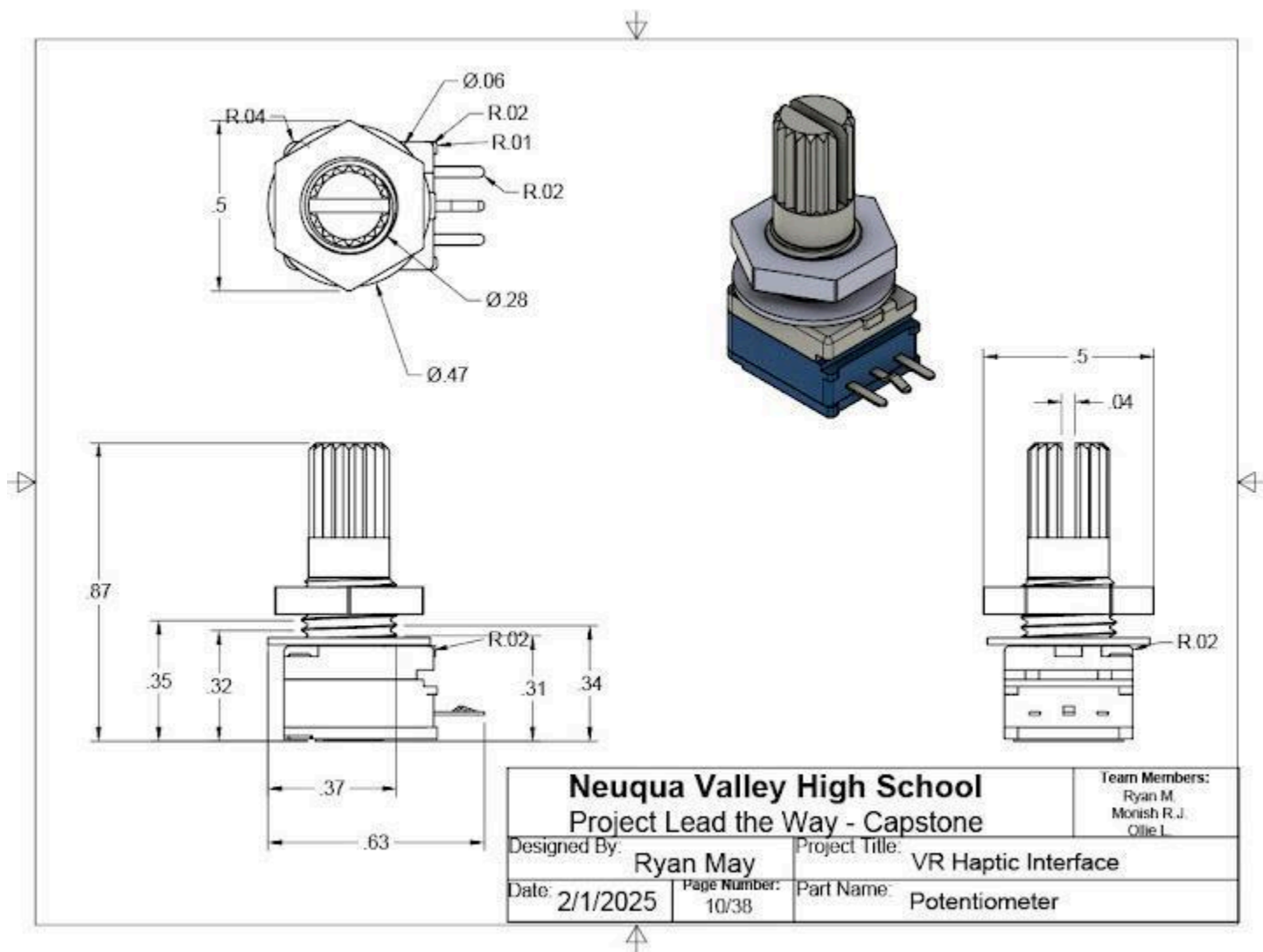


Figure 8.10: Dimensioned Sketch of the potentiometers used in both “Servo on Left” and “Servo on Right”.

The voltage outputted by the potentiometer shown in Figure 8.10 is read by the microcontroller and calculates how expanded or contracted that particular finger is.

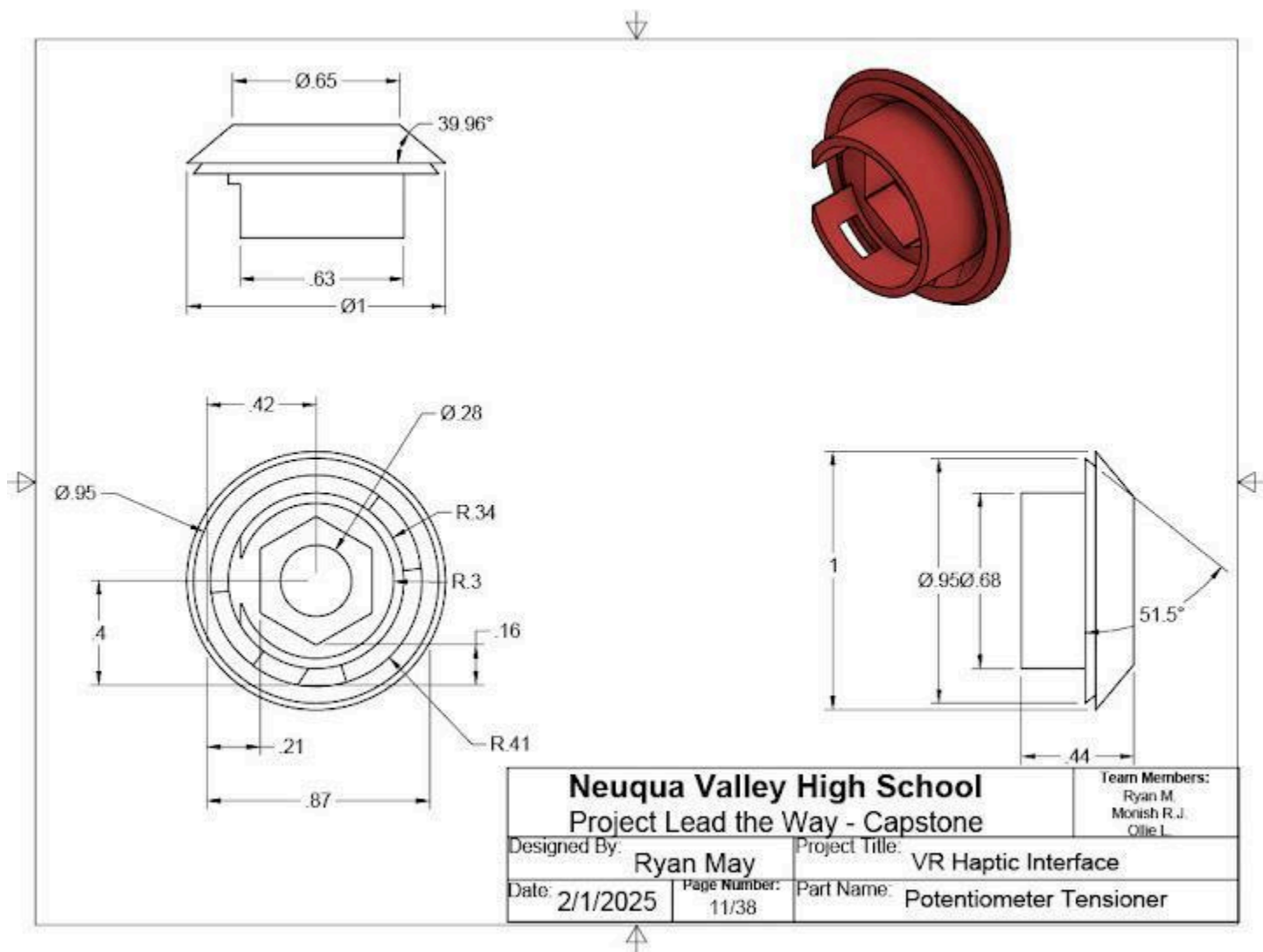


Figure 8.11: Dimensioned Sketch of” Potentiometer Tensioner” in “Servo on Left”

“Potentiometer Tensioner” shown in Figure 8.11 houses a spring that connects a spring from the glove to the potentiometer which keeps it under tension.

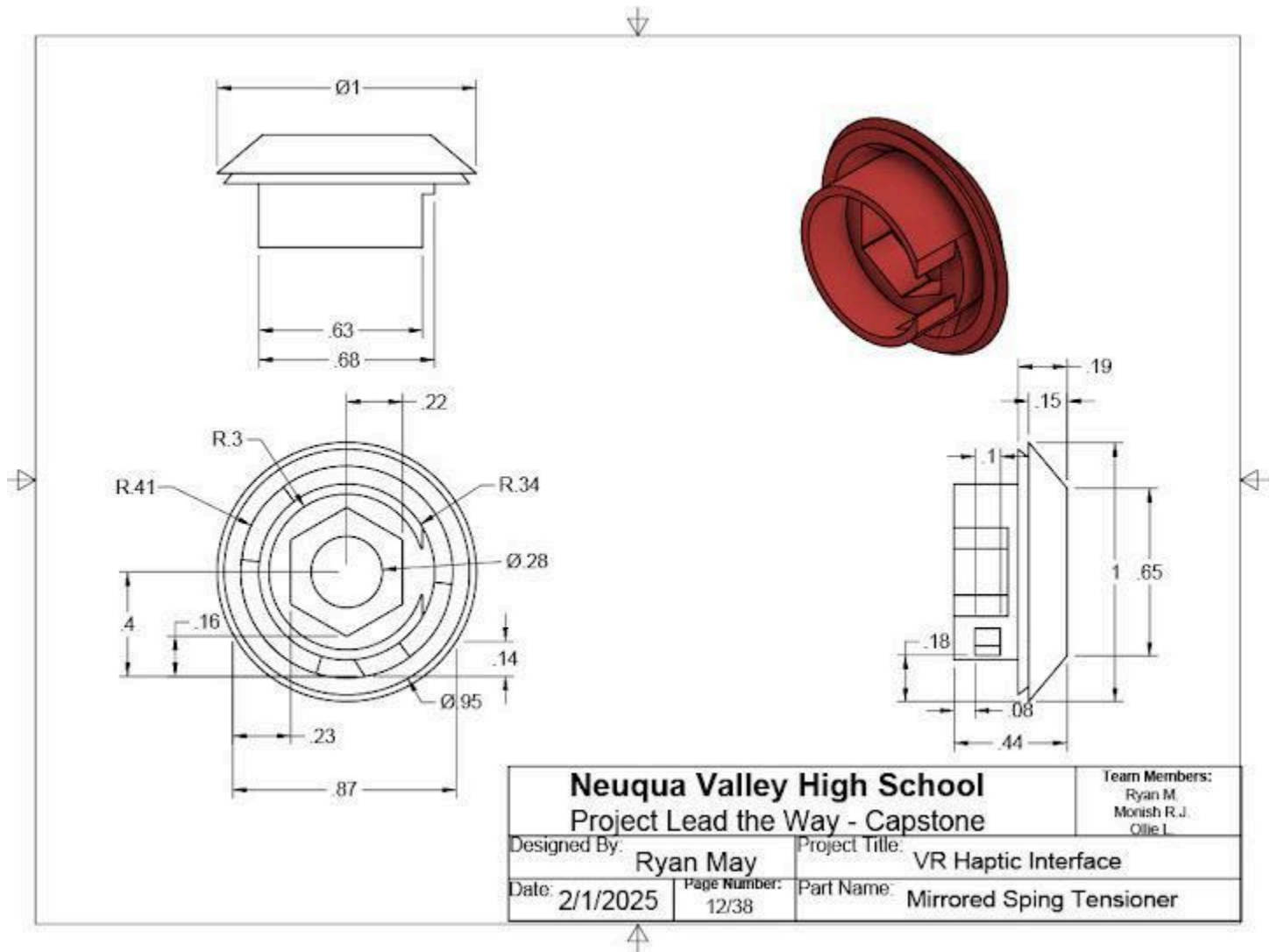


Figure 8.12: Dimensioned Sketch of "Potentiometer Tensioner" in "Servo on Right".

Figure 8.12 depicts the mirrored version of "Potentiometer Tensioner" from Figure 8.11.

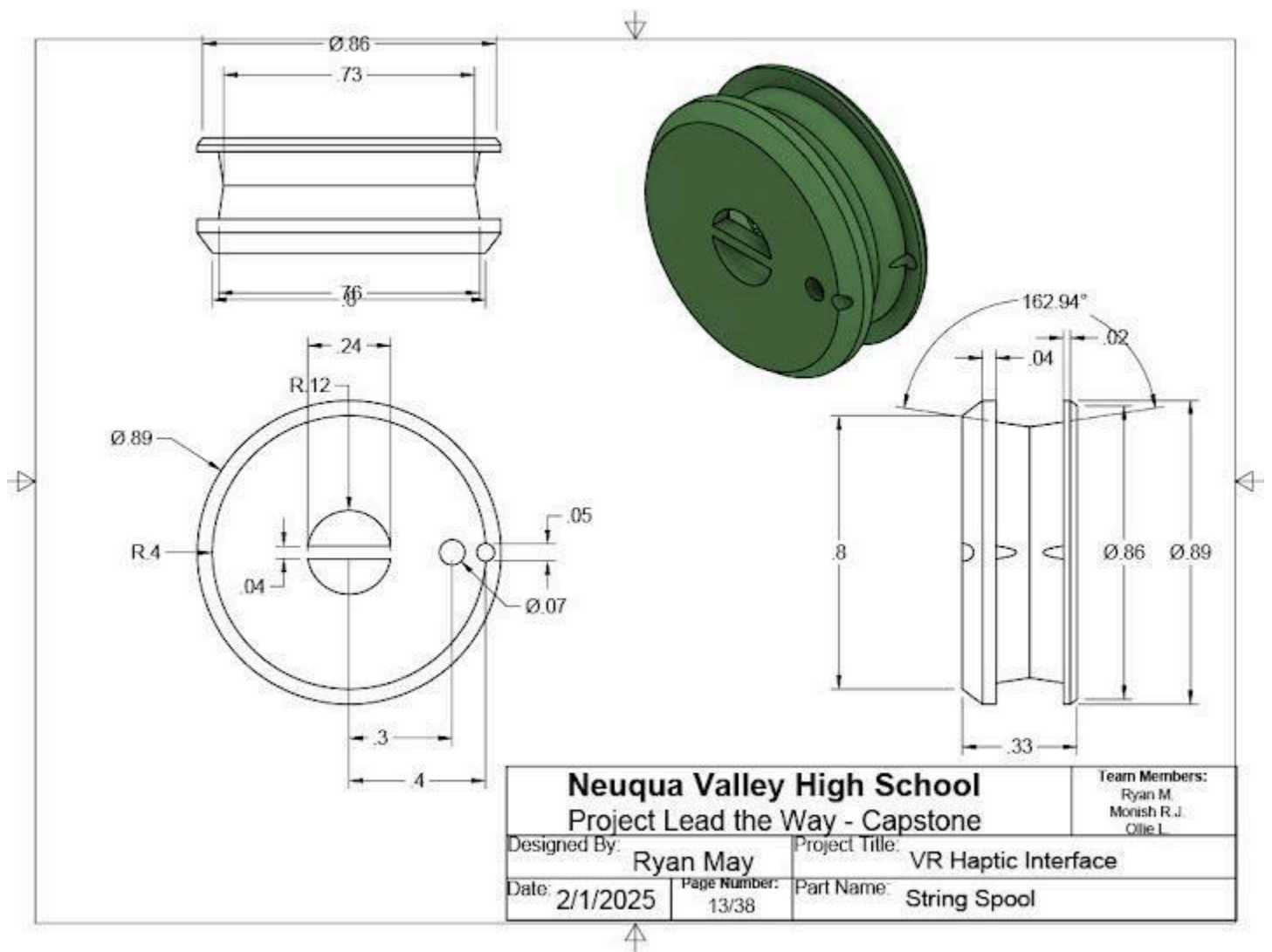


Figure 8.13: Dimensioned Sketch of “String Spool” used in both “Servo on Left” and “Servo on Right”.

“String Spool” in Figure 8.13 connects the Potentiometer to a string which changes the rotation of the potentiometer depending on the position of the fingers.

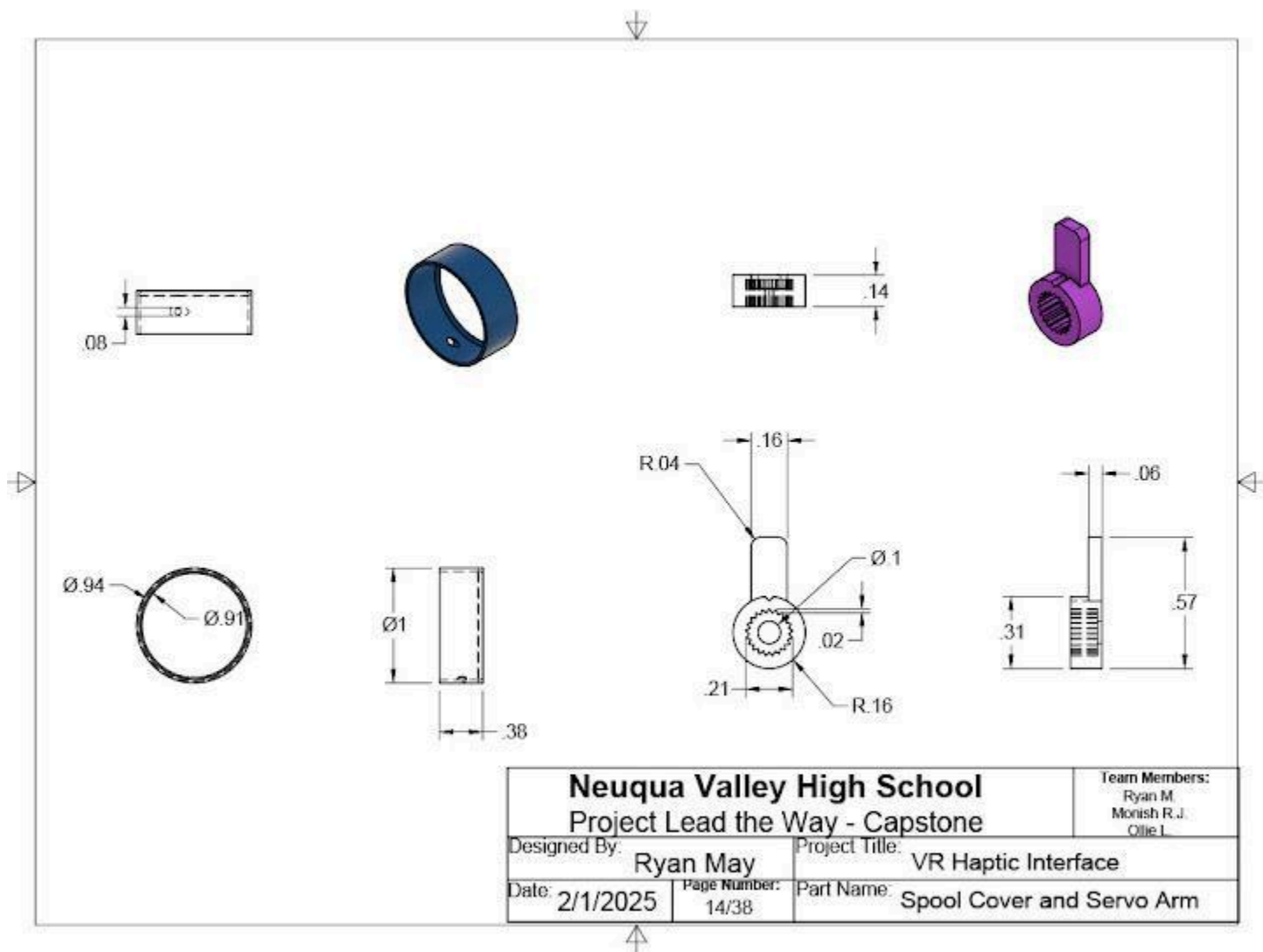


Figure 8.14: Dimensioned Sketch of “Spool Cover” and “Servo Arm” used in both “Servo on Left” and “Servo on Right”.

The Spool Cover in Figure 8.14 guides the string out of the spool and into each of its routing tunnels. The Servo Arm in the Figure connects to the servo and allows the servo to push against the Spool to limit the rotation spool of spring and thus also the fingers.

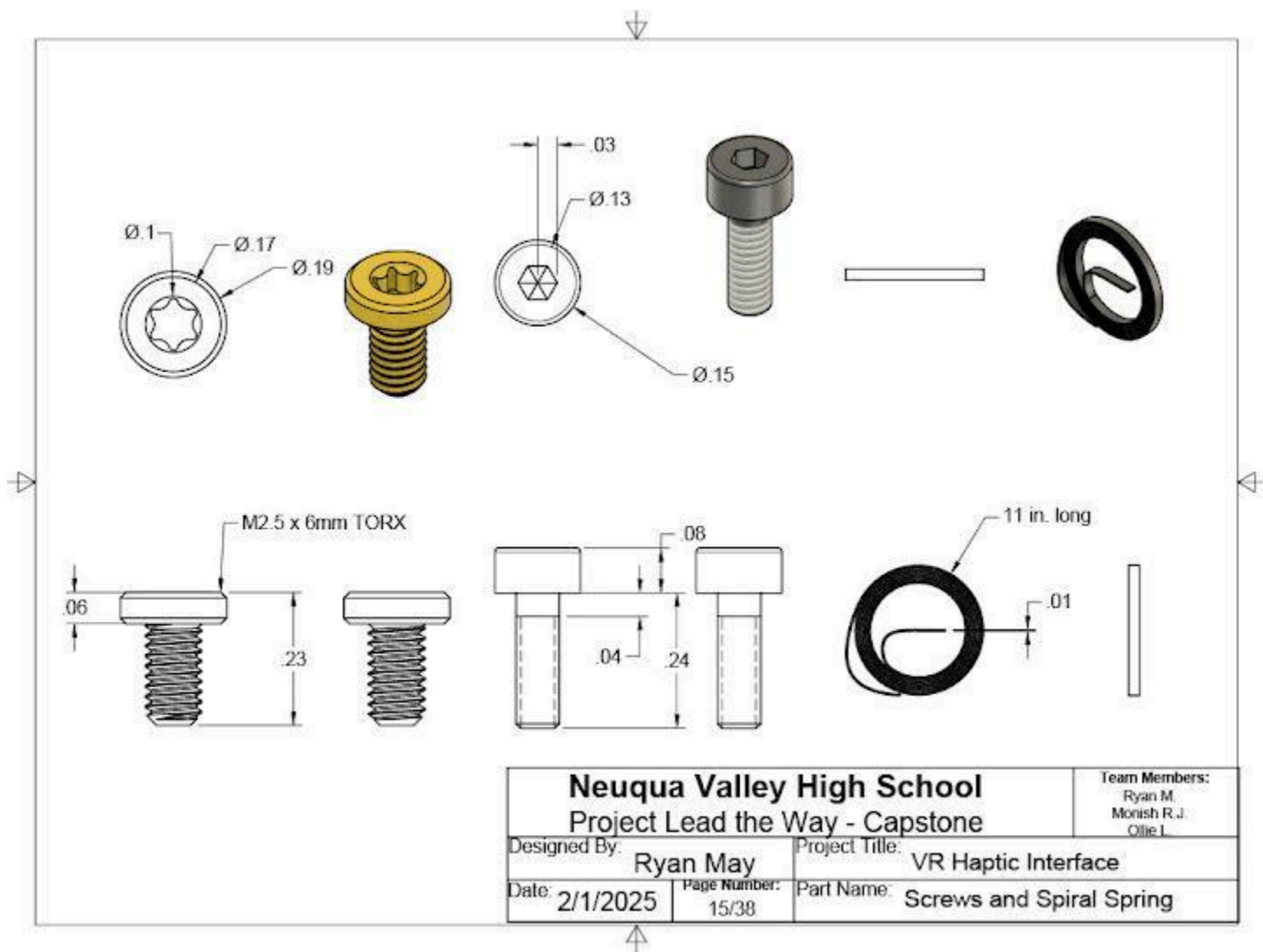


Figure 8.15: Dimensioned Sketch of the m2.5 x 6mm Torx, m2 x 6mm, and the Spiral Spring used in both “Servo on Left” and “Servo on Right”.

The M2.5 x 6mm Torx Screw depicted on the left side of Figure 8.15 connects the servo to the “Individual Servo Mount” as well as acts as the protrusion from the String Spool which the “Servo Arm” pushes against to create resistance.

The M2 x 6mm screw shown in the middle of Figure 8.15 connects the “Individual Servo Mount” to the “Servo Baseplate”.

The Spiral Spring on the right side of Figure 8.15 connects the “Spring Tensioner” to the Potentiometer and provides a small yet constant force that retracts the spring when the fingers open.

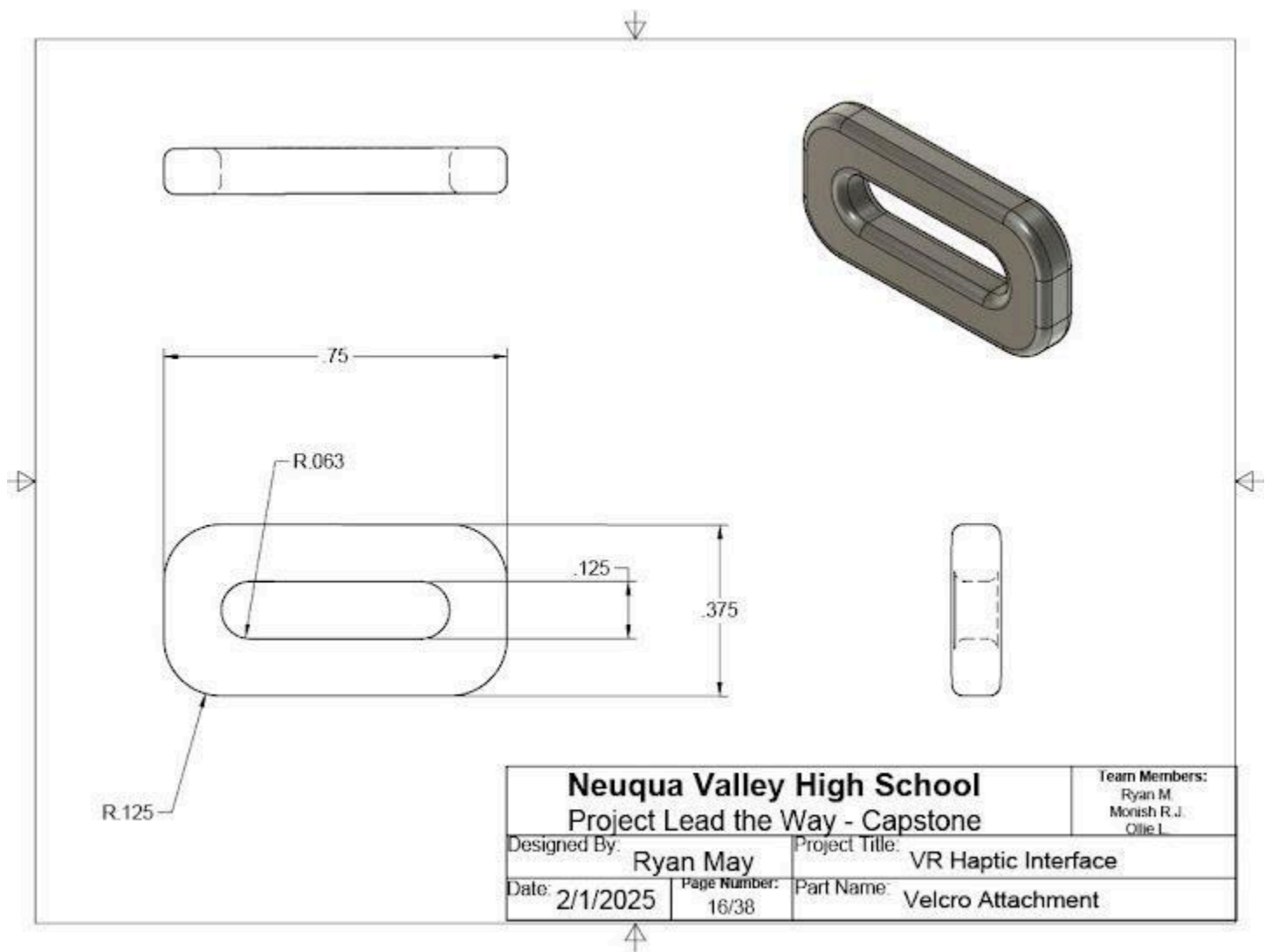


Figure 8.16: Dimensioned Sketch of “Velcro Attachment” used in both “Electronics Mount” and “String Routing”.

“Velcro Attachment” in Figure 8.16 connects the Velcro to the Glove which enables the Glove to mount on to a wide variety of arm sizes.

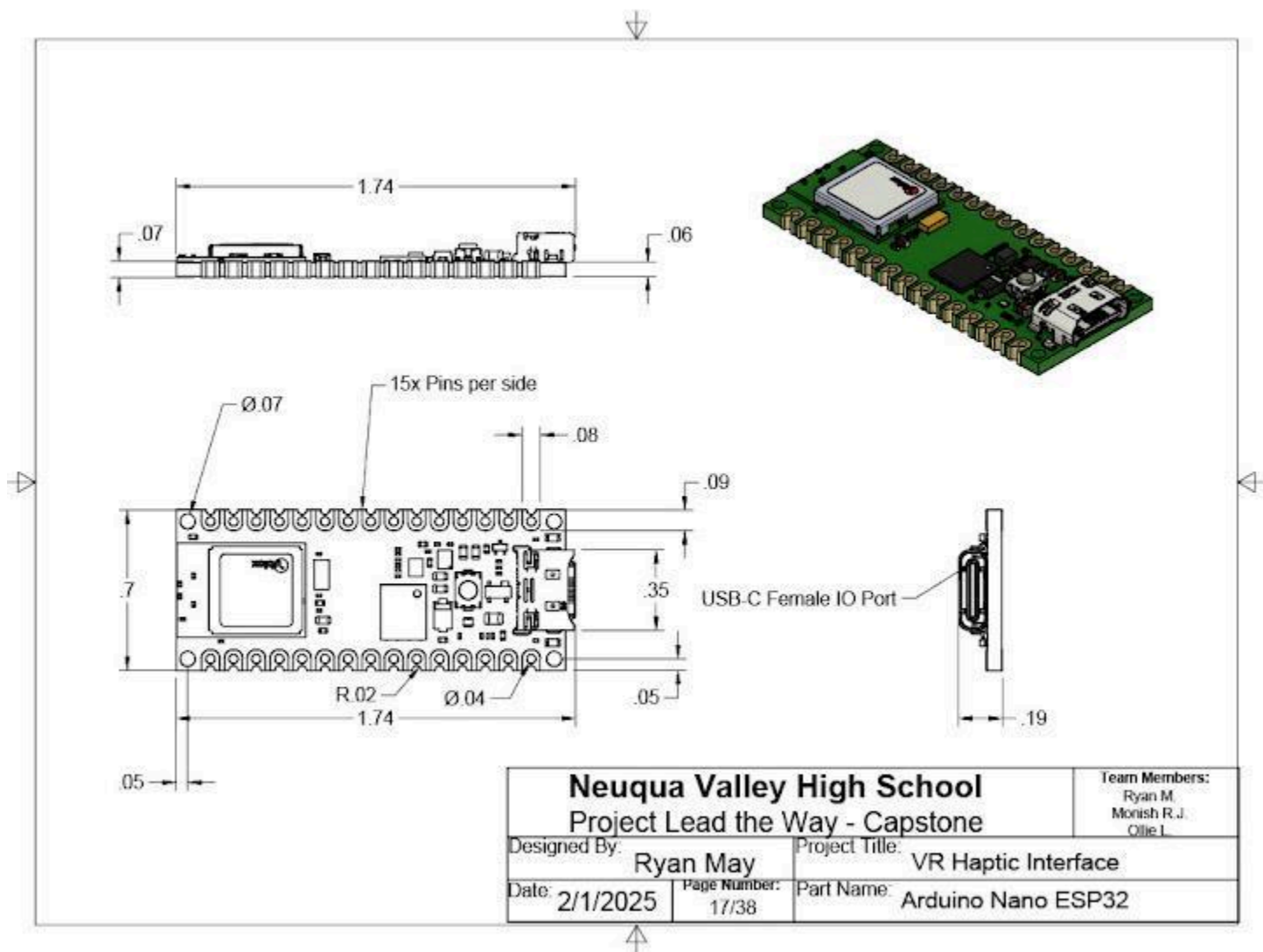


Figure 8.17: Dimensioned Sketch of the microcontroller used in “Electronics Mount”.

The microcontroller (Arduino Nano ESP32) in Figure 8.17 connects to the VR headset via Bluetooth and is mounted on the glove via the microcontroller mount.

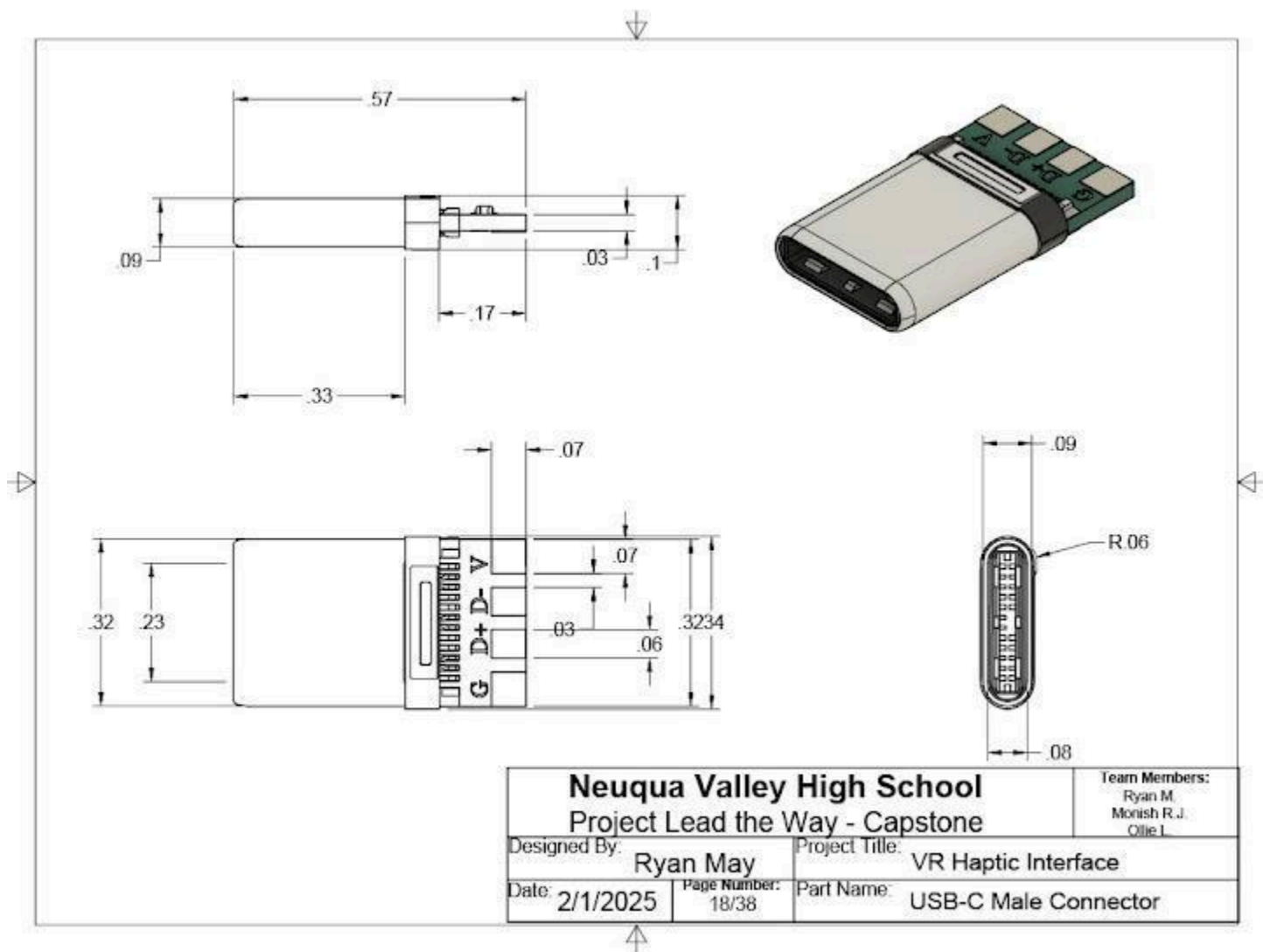


Figure 8.18: Dimensioned Sketch of the USB-C Male used in “Electronics Mount”.

The USB-C Male connector in Figure 8.18 is a breakout board that allows the microcontroller to receive power from the batteries.

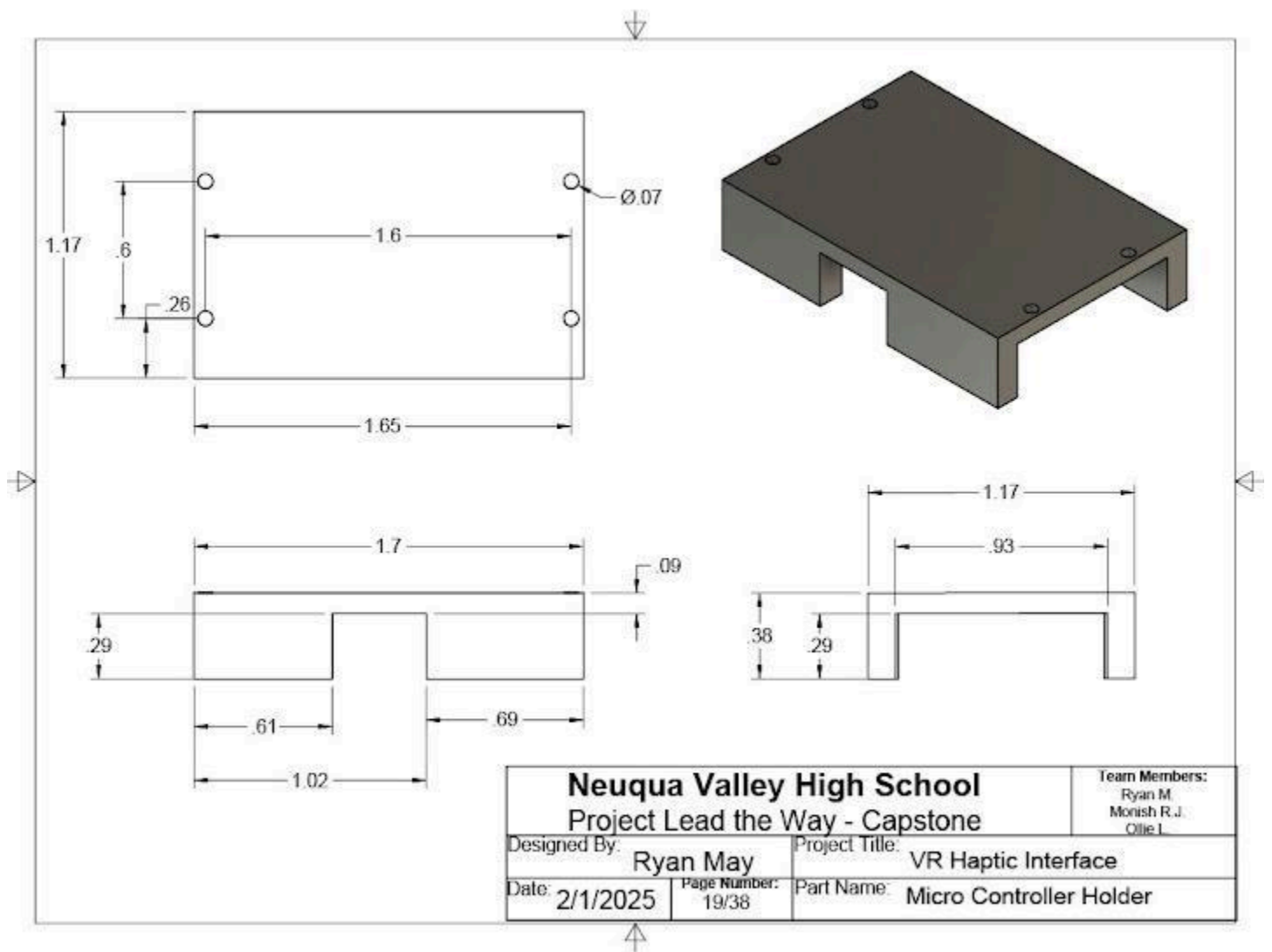


Figure 8.19: Dimensioned Sketch of the “Micro Controller Holder” used in “Electronics Mount”.

The “Micro Controller Holder” in Figure 8.19 connects the microcontroller to the Battery Holders.

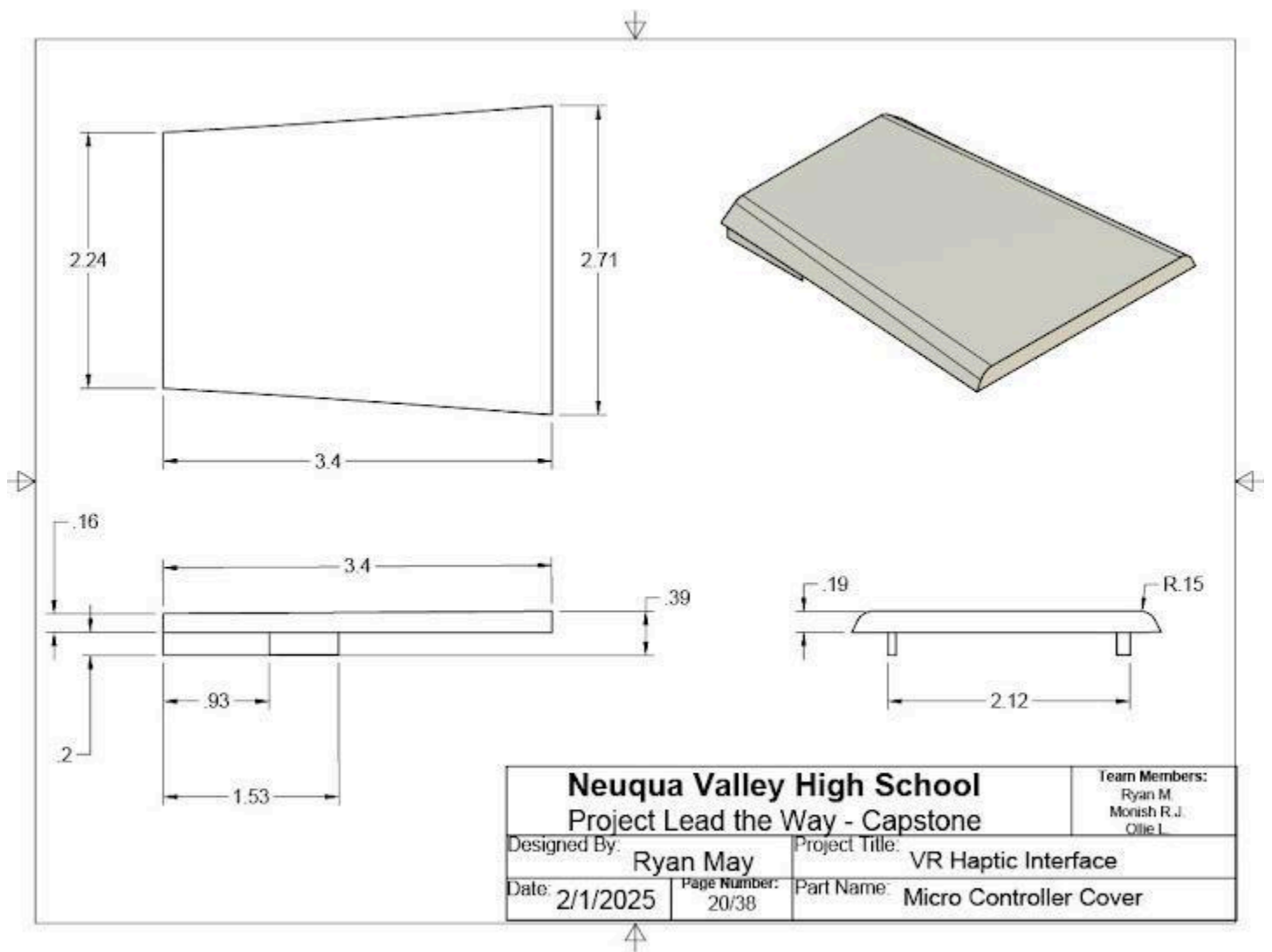


Figure 8.20: Dimensioned Sketch of the “Micro Controller Cover” used in “Electronics Mount”.

Figure 8.20 depicts the “Micro Controller Cover” which allows for easy access to the electronics within the glove while not leaving any wires exposed.

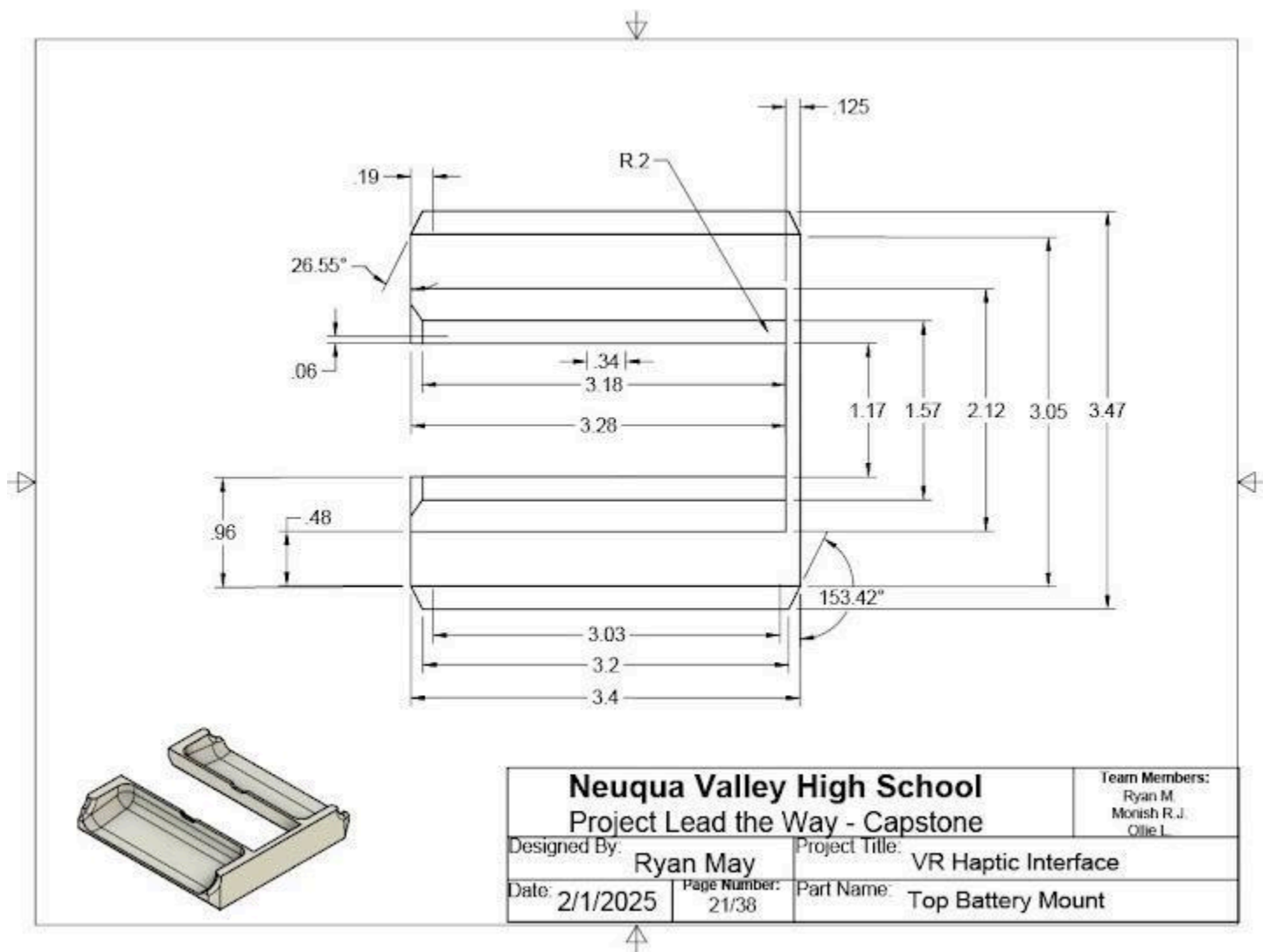


Figure 8.21: 1st Dimensioned Sketch of “Top Battery Mount” used in “Electronics Mount”.

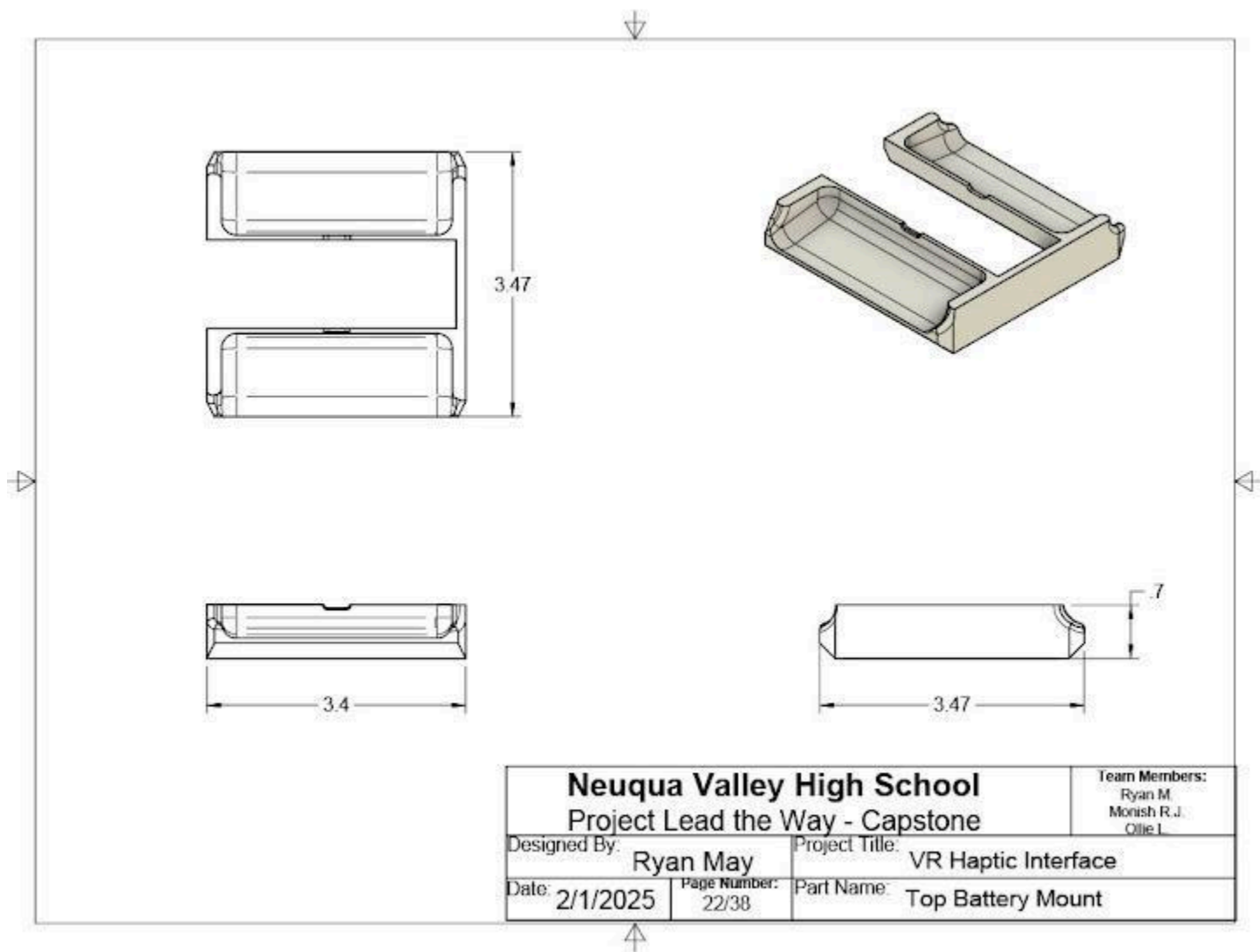


Figure 8.22: 2nd Dimensioned Sketch of “Top Battery Mount” used in “Electronics Mount”.

The “Top Battery Mount”, depicted in Figure 8.21 and Figure 8.22, is the top half of the battery mount which allows the microcontroller to connect to the rest of the glove and keeps the batteries in one place in the glove.

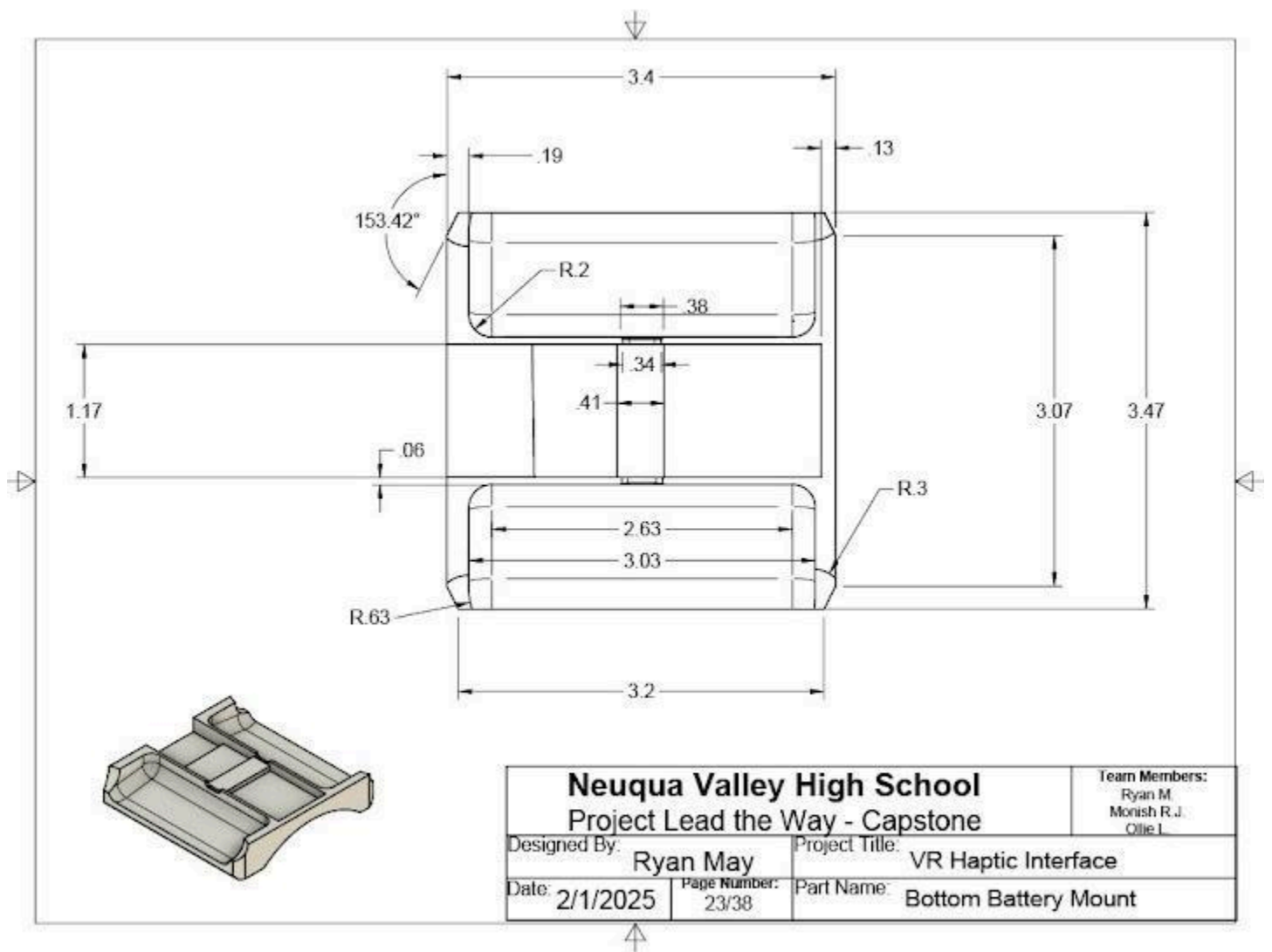


Figure 8.23: 1st Dimensioned Sketch of "Bottom Battery Mount" used in "Electronics Mount".

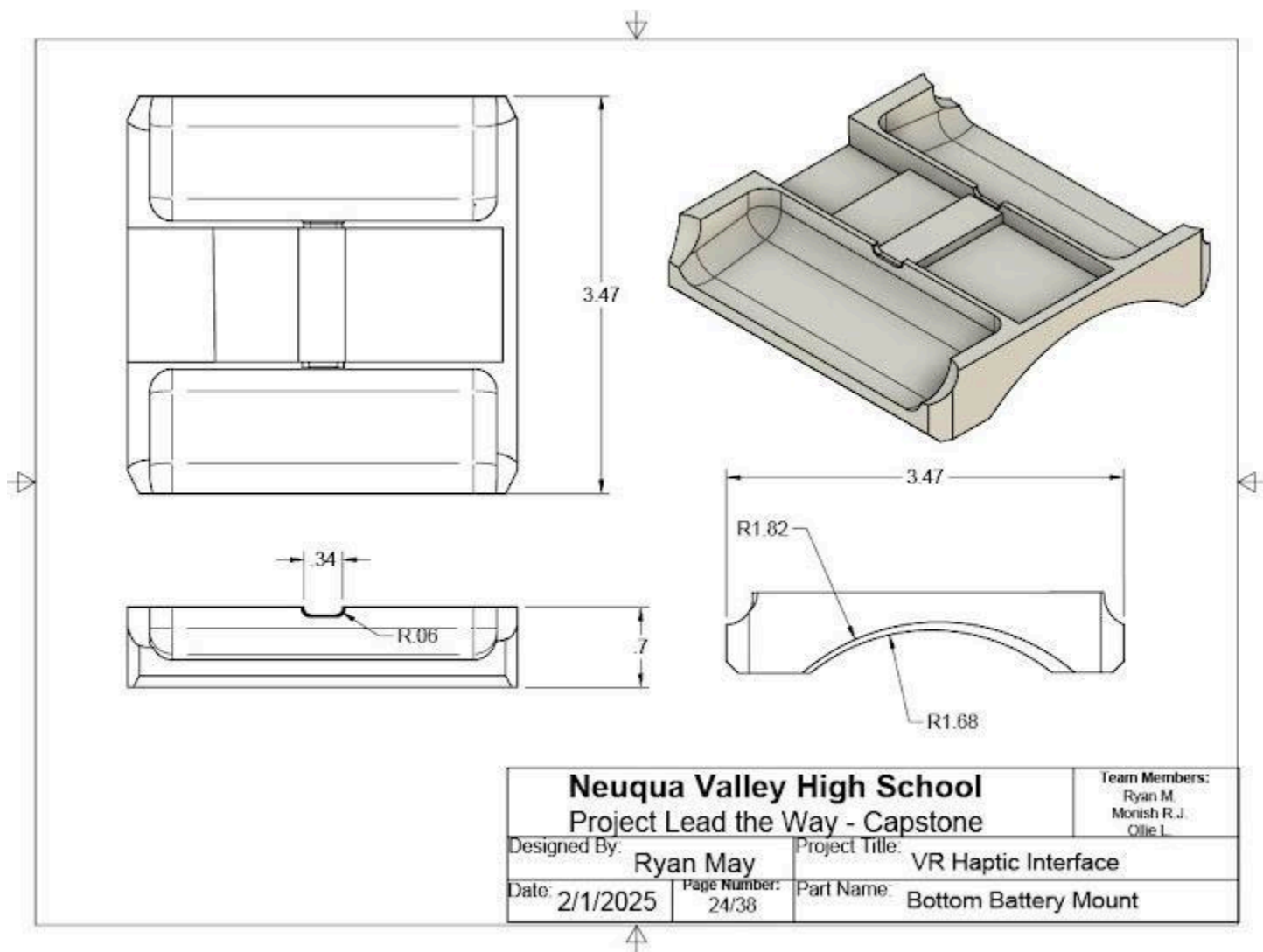


Figure 8.24: 2nd Dimensioned Sketch of “Bottom Battery Mount” used in “Electronics Mount”.

Figures 8.23 and 8.24 depict the “Bottom Battery Mount” which is the bottom half of the battery mount that connects both parts of the battery mount and the microcontroller to the “Servo Base Plate” and keeps the batteries in one place in the glove.

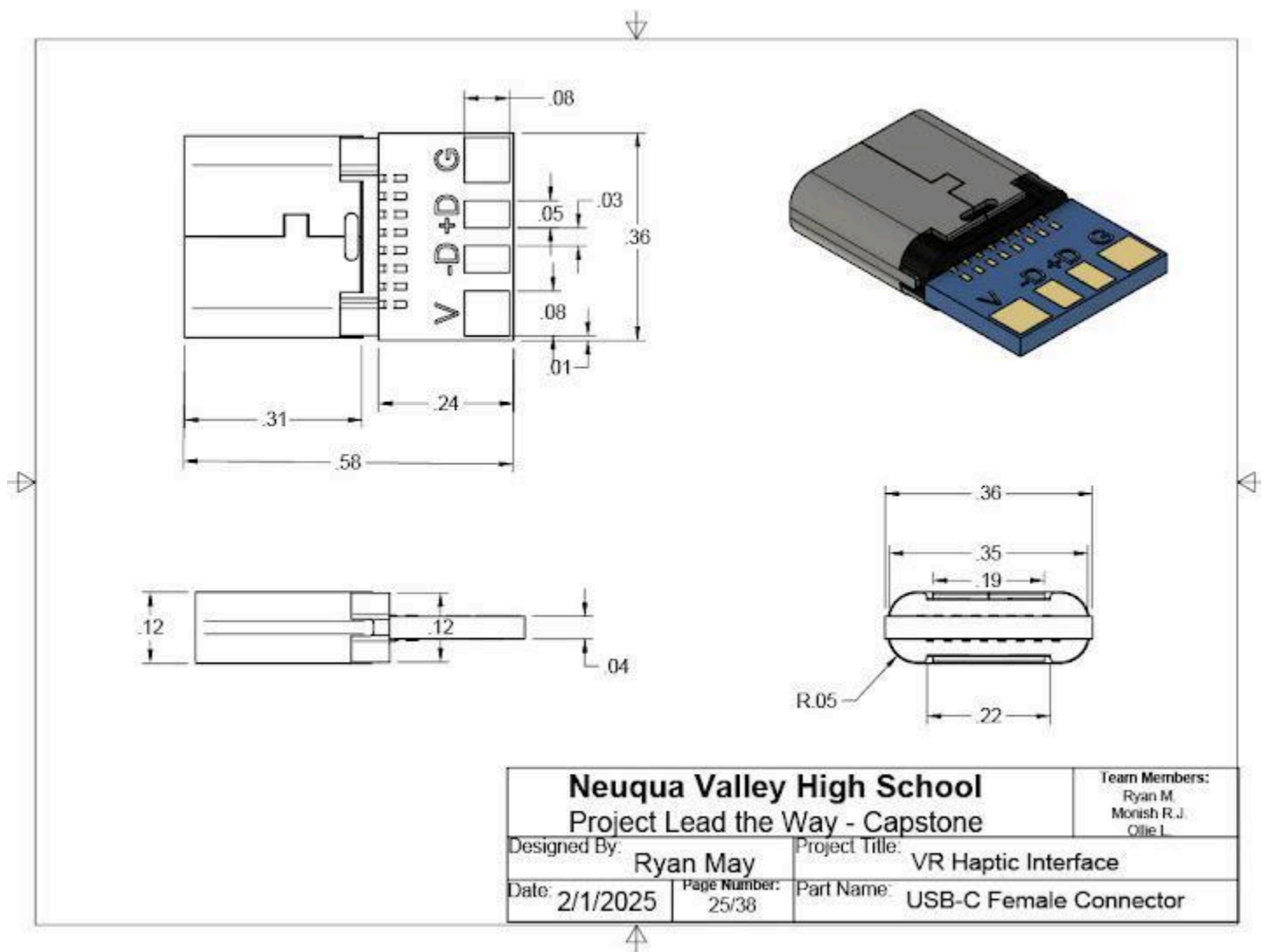


Figure 8.25: Dimensioned Sketch of “USB-C Female Connector” used in “Electronics Mount”.

“USB-C Female Connector” in Figure 8.25 is a breakout board that the batteries plug into and gives a 5v out and Ground connection to the Servos and microcontroller.

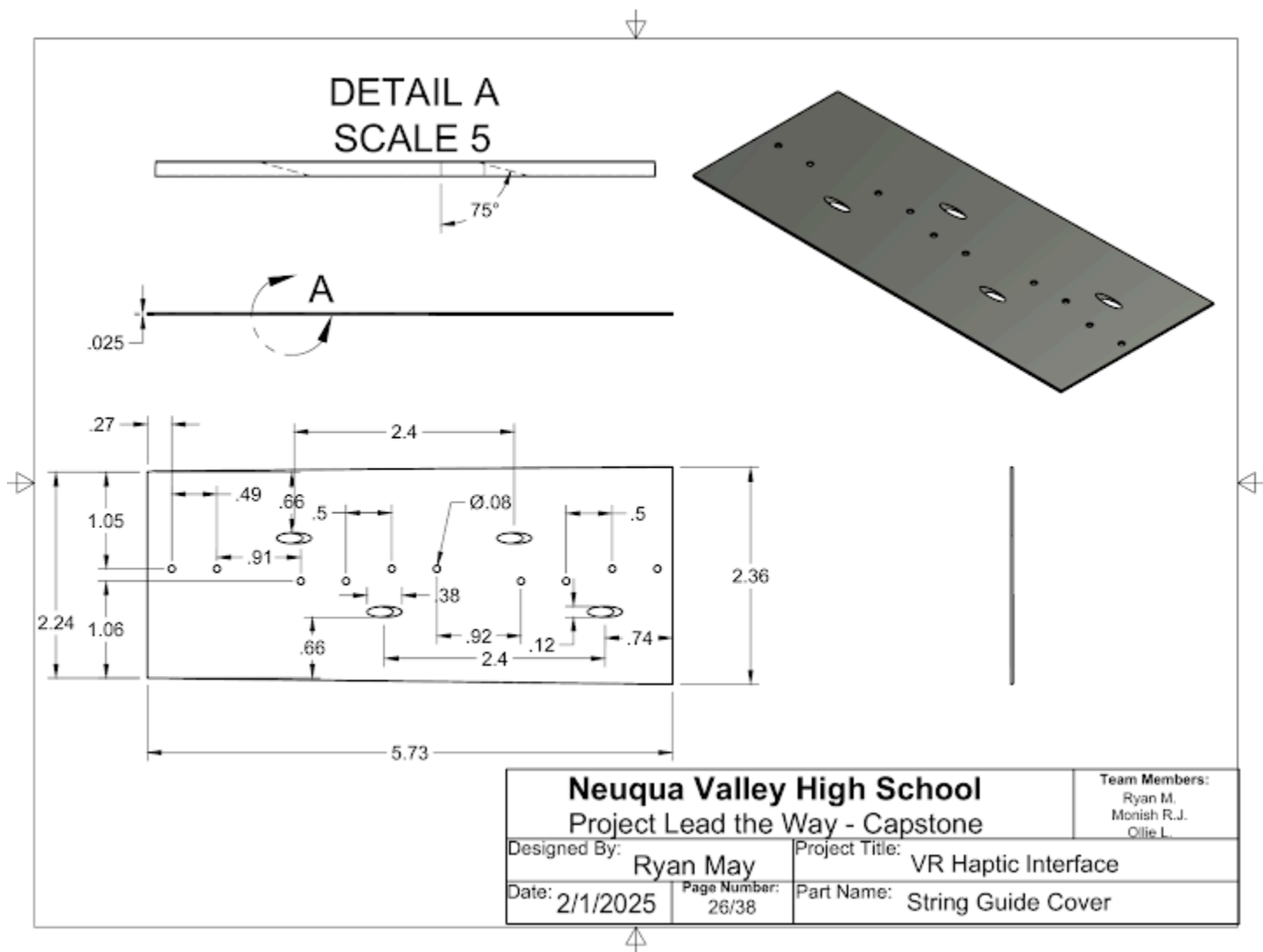


Figure 8.26: Dimensioned Sketch of “String Guide Cover” used in both “Electronics Mount”.

Figure 8.26 depicts the “String Guide Cover” which goes on top of the Servo Base Plate and covers the strings to prevent additional wear to them.

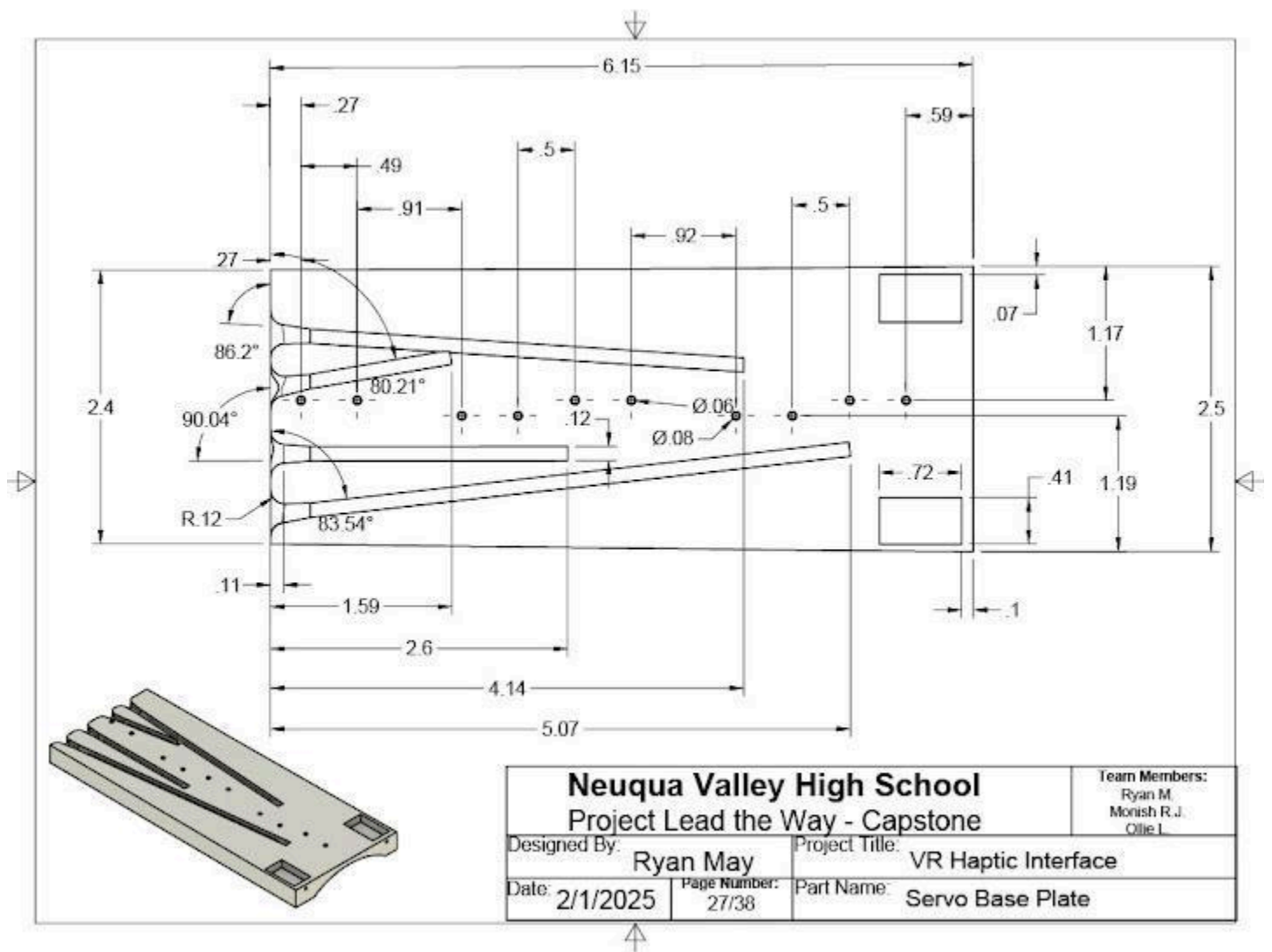


Figure 8.27: 1st Dimensioned Sketch of “Servo Base Plate” used in “Electronics Mount”.

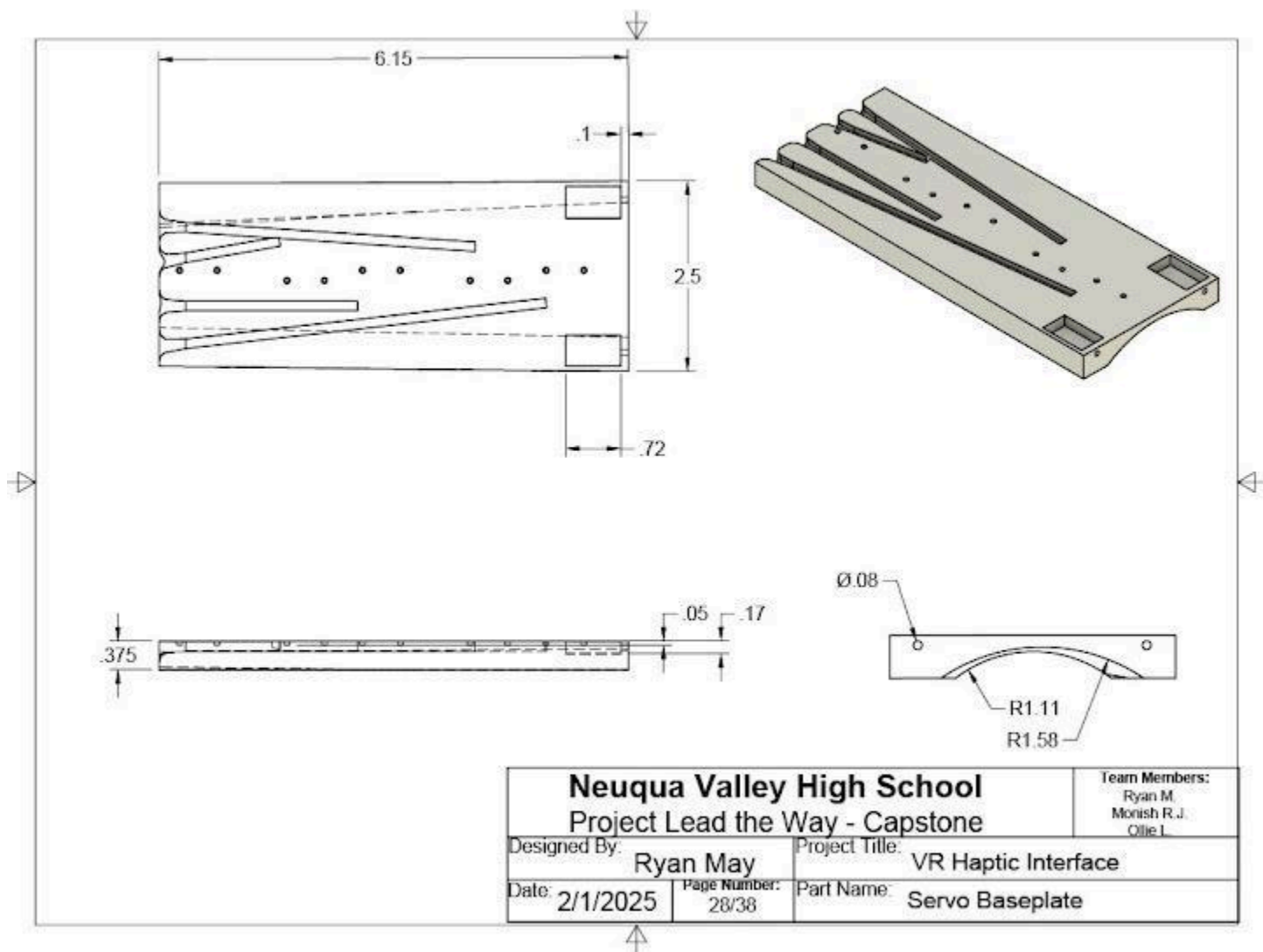


Figure 8.28: 2nd Dimensioned Sketch of “Servo Base Plate” used in “Electronics Mount”.

Figures 8.27 and 8.28 show the “Servo Base Plate” which is the main base of the glove and connects most components together. It houses the strings and tunnels them to the front of the glove. It also connects all of the “Individual Servo Mount” together as well as the “String Guide Cover” and the “Servo Top Cover”

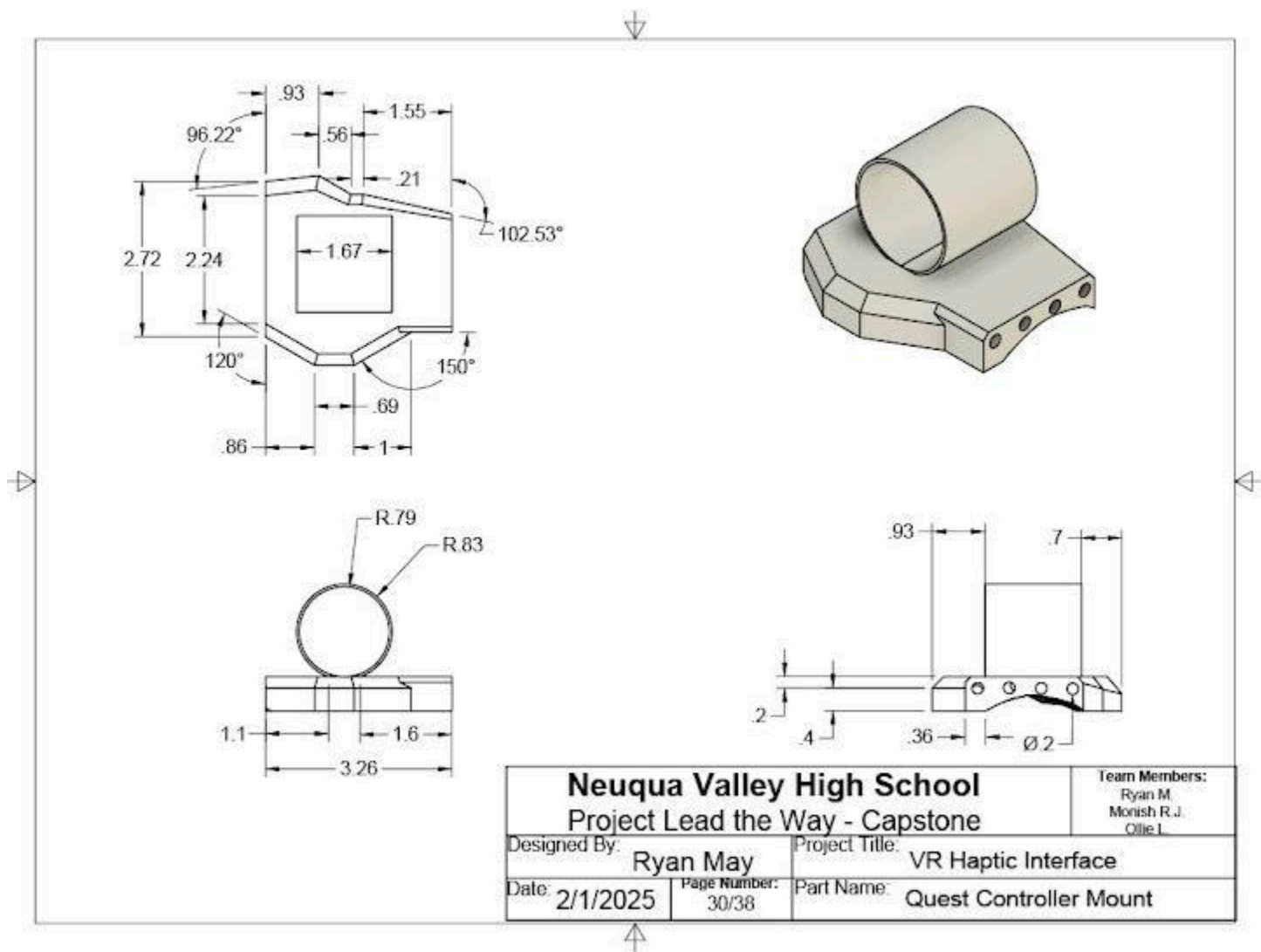


Figure 8.30: Dimensioned Sketch of “Quest Controller Mount” used in “String Routing”.

“Quest Controller Mount” in Figure 8.30 is the main component in “String Routing”. It connects to the user's hand and houses an Oculus Quest Controller which tracks the user's movements.

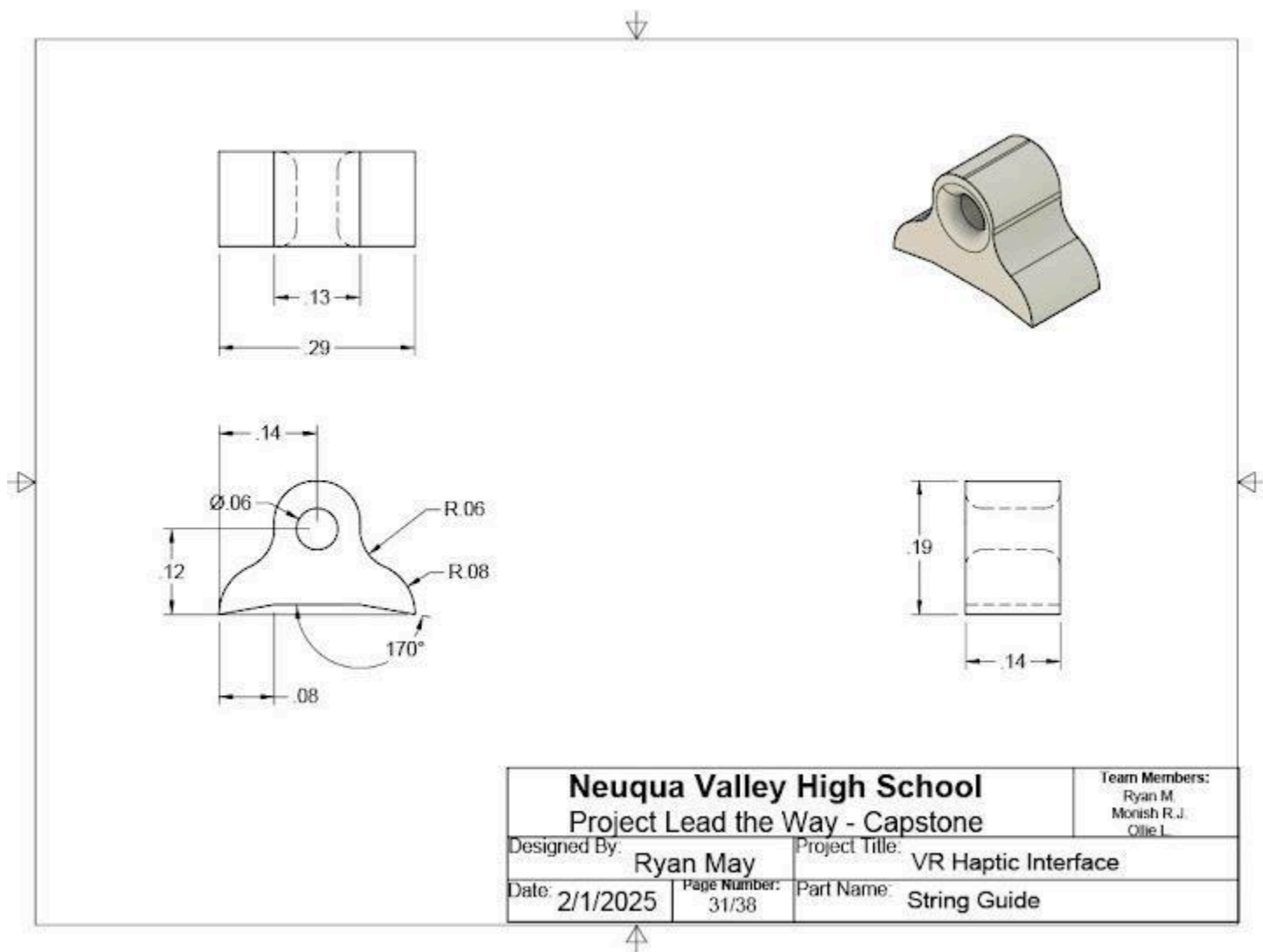


Figure 8.31: Dimensioned Sketch of the “String Guide” used in “String Routing”.

“String Guide” in Figure 8.31 goes on the user's knuckles and each bend in the user's fingers to prevent the string from rubbing against the user's fingers.

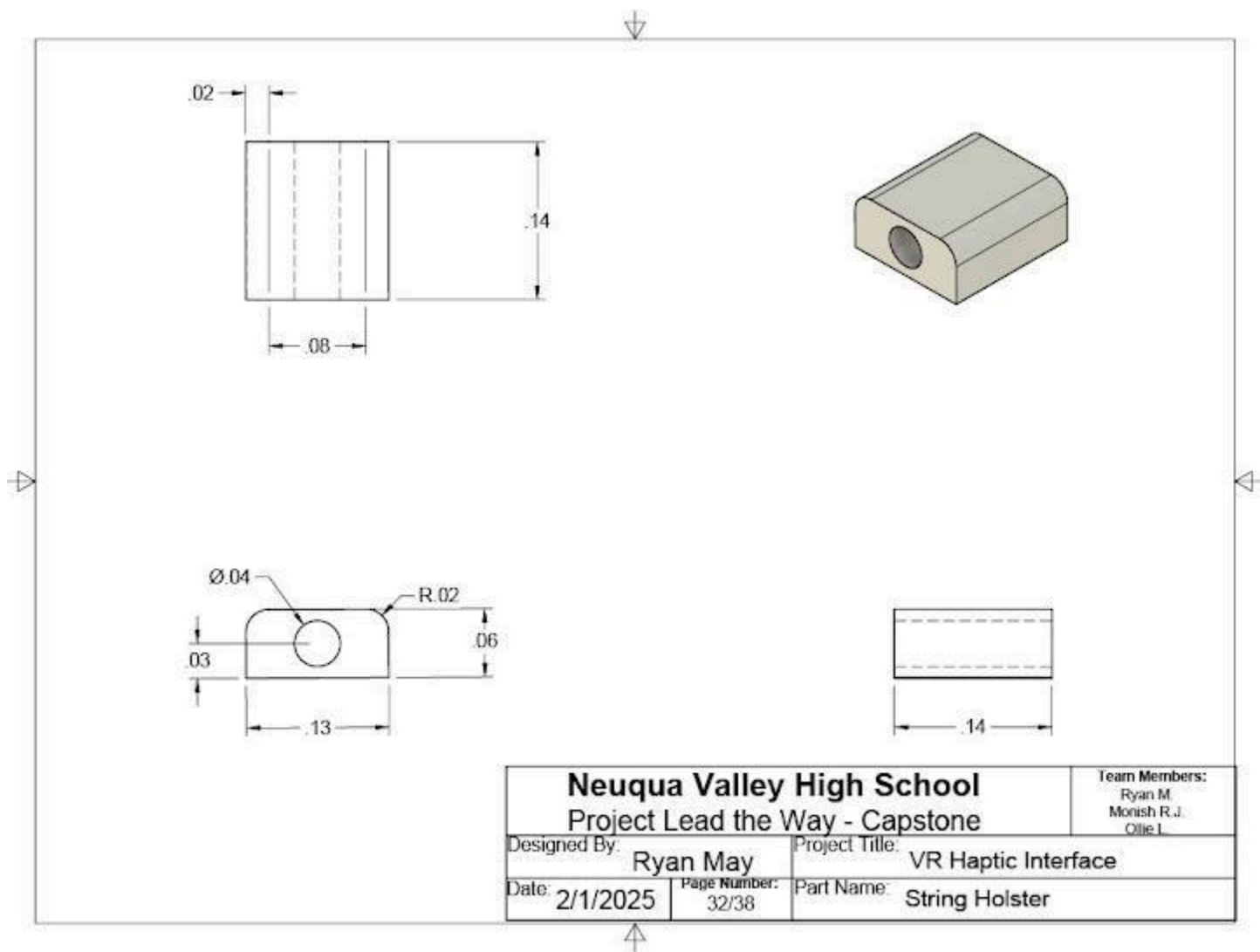


Figure 8.32: Dimensioned Sketch of the String Holster used in “String Routing”.

“String Holster” in Figure 8.32 connects to the top part of each finger and keeps the string connected to the fingers.

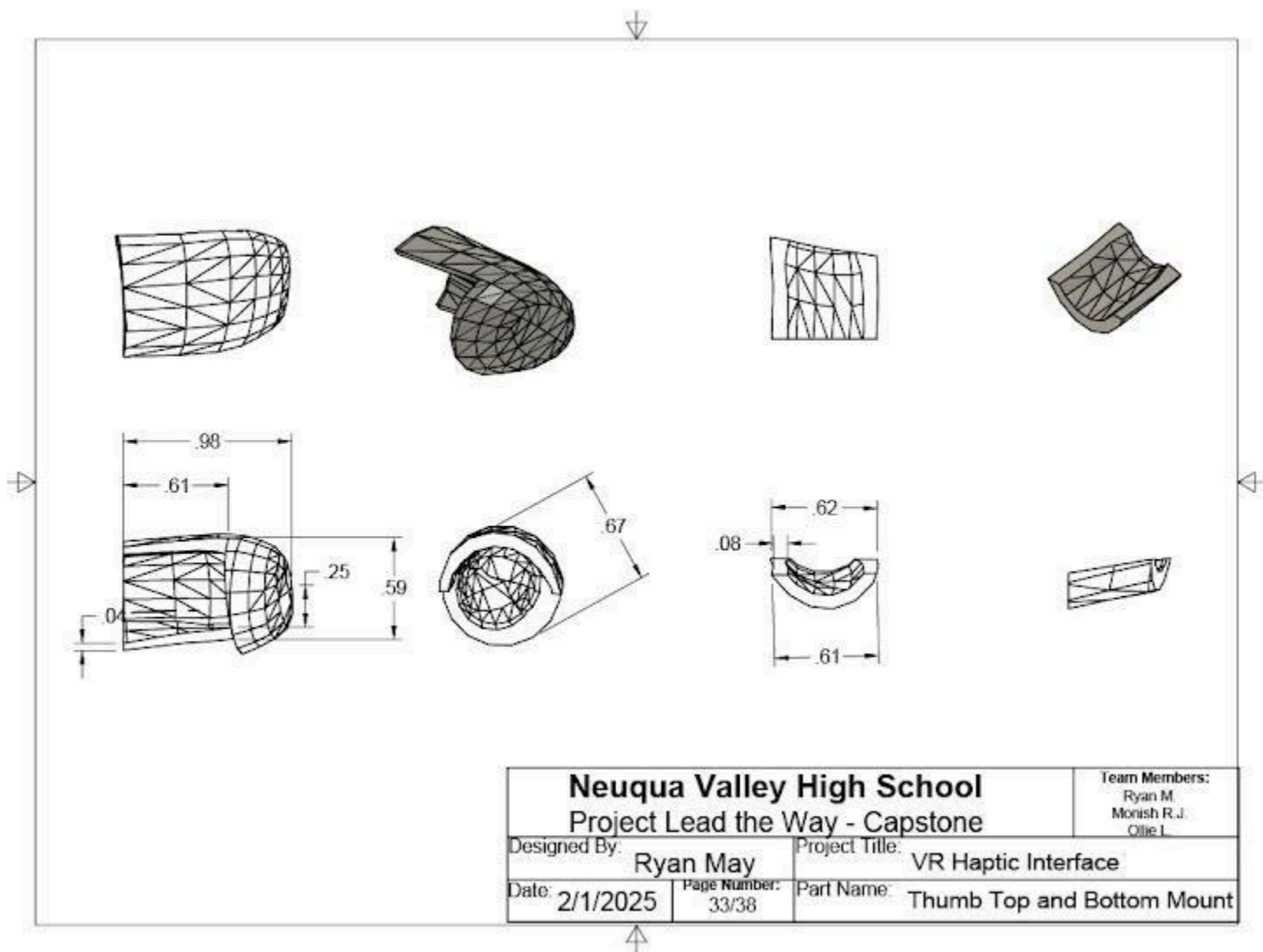


Figure 8.33: Dimensioned Sketch of “Thumb Top” and “Thumb Bottom” used in “String Routing”.

“Thumb Top” and “Thumb Bottom” in Figure 8.33 are the top and bottom parts of the adjustable cap which goes on the end of the thumb.

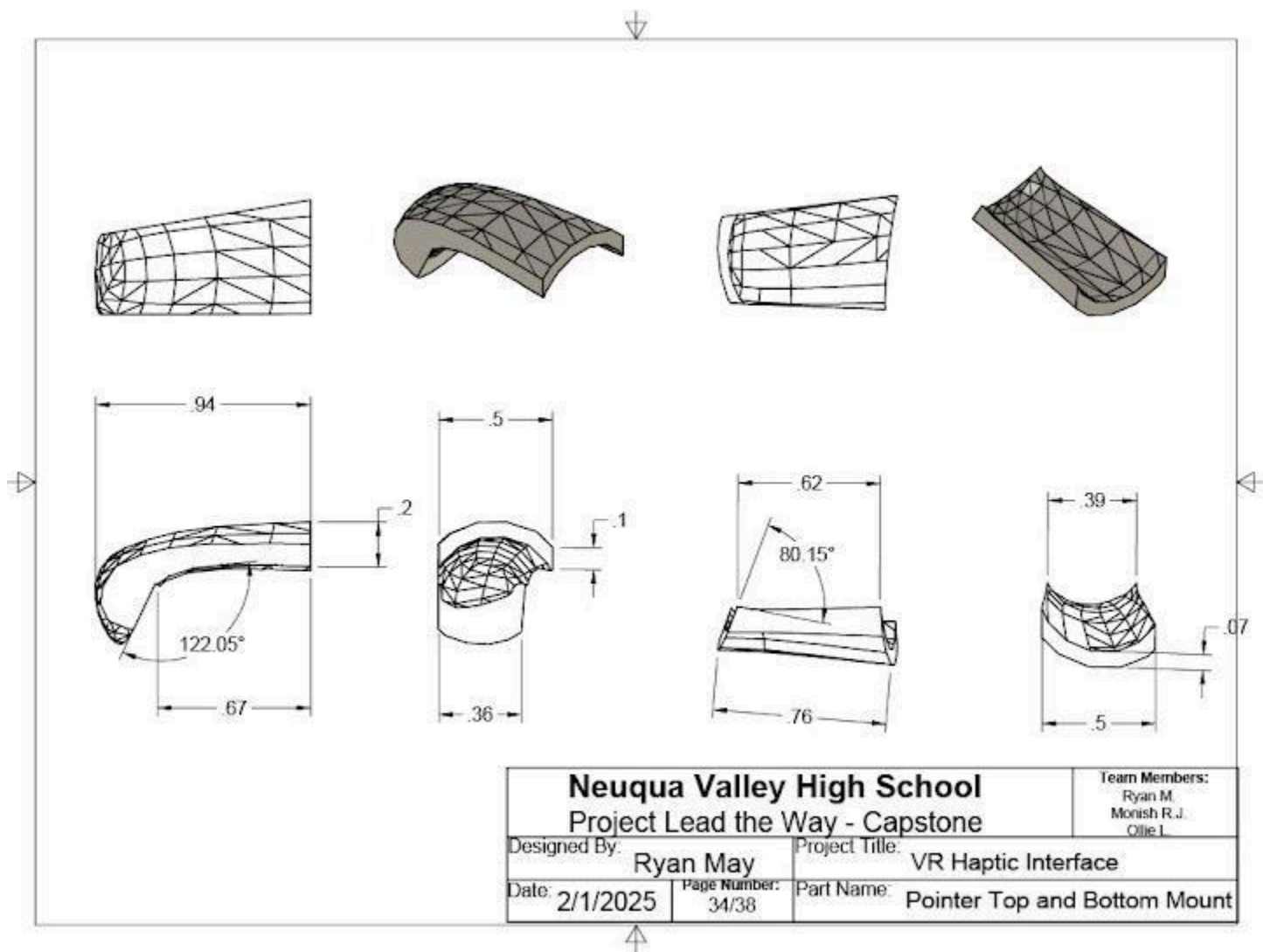


Figure 8.34 Dimensioned Sketch of the “Pointer Top” and “Pointer Bottom” used in “String Routing”.

“Pointer Top” and “Pointer Bottom” in Figure 8.34 are the top and bottom parts of the adjustable cap which goes on the end of the pointer finger.

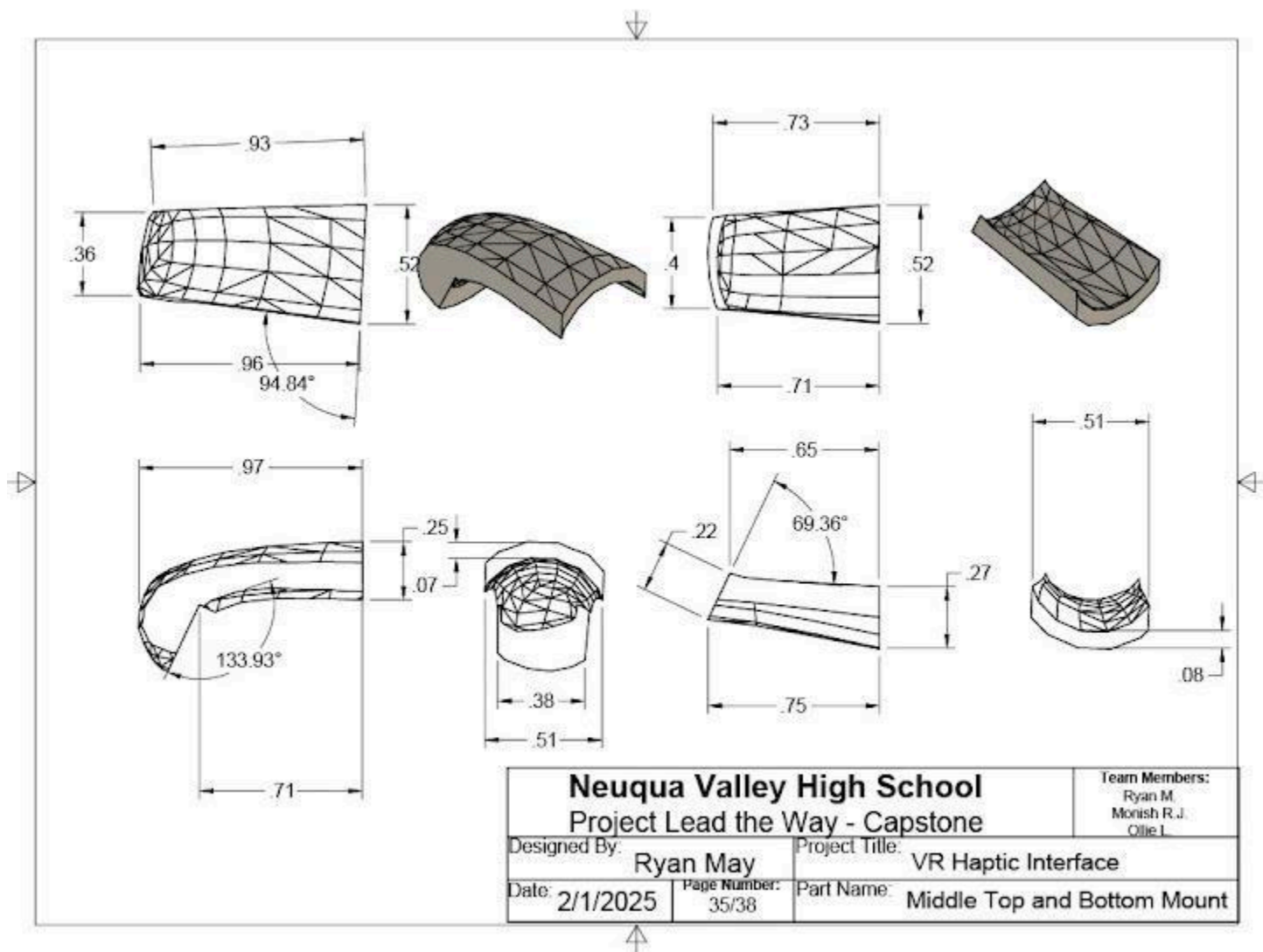


Figure 8.35: Dimensioned Sketch of the “Middle Top” and “Middle Bottom” used in “String Routing”.

“Middle Top” and “Middle Bottom” in Figure 8.35 are the top and bottom parts of the adjustable cap that go on the end of the middle finger.

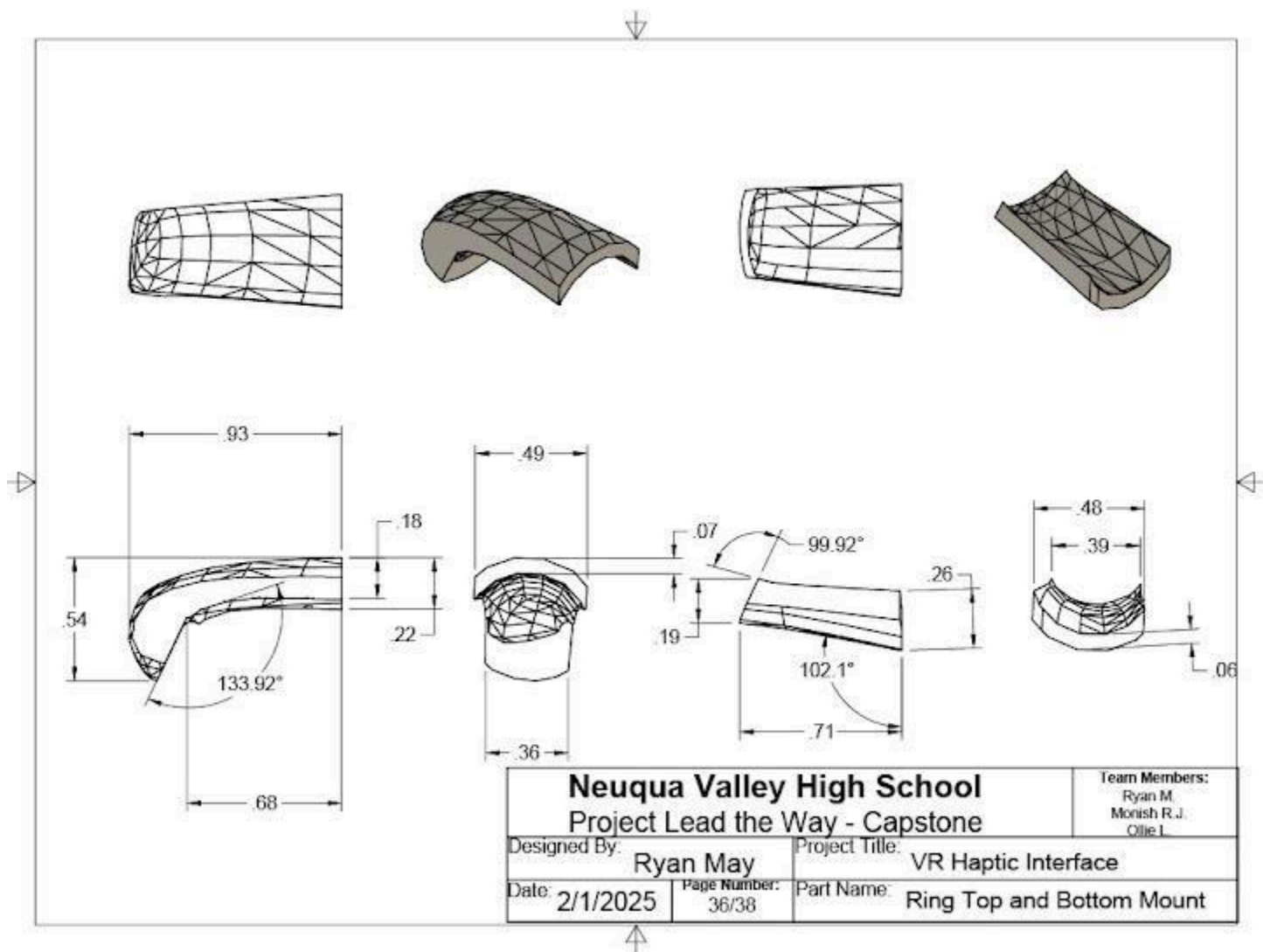


Figure 8.36: Dimensioned Sketch of “Ring Top” and “Ring Bottom” used in “String Routing”.

“Ring Top” and “Ring Bottom” in Figure 8.36 are the top and bottom parts of the adjustable cap that goes on the end of the ring finger.

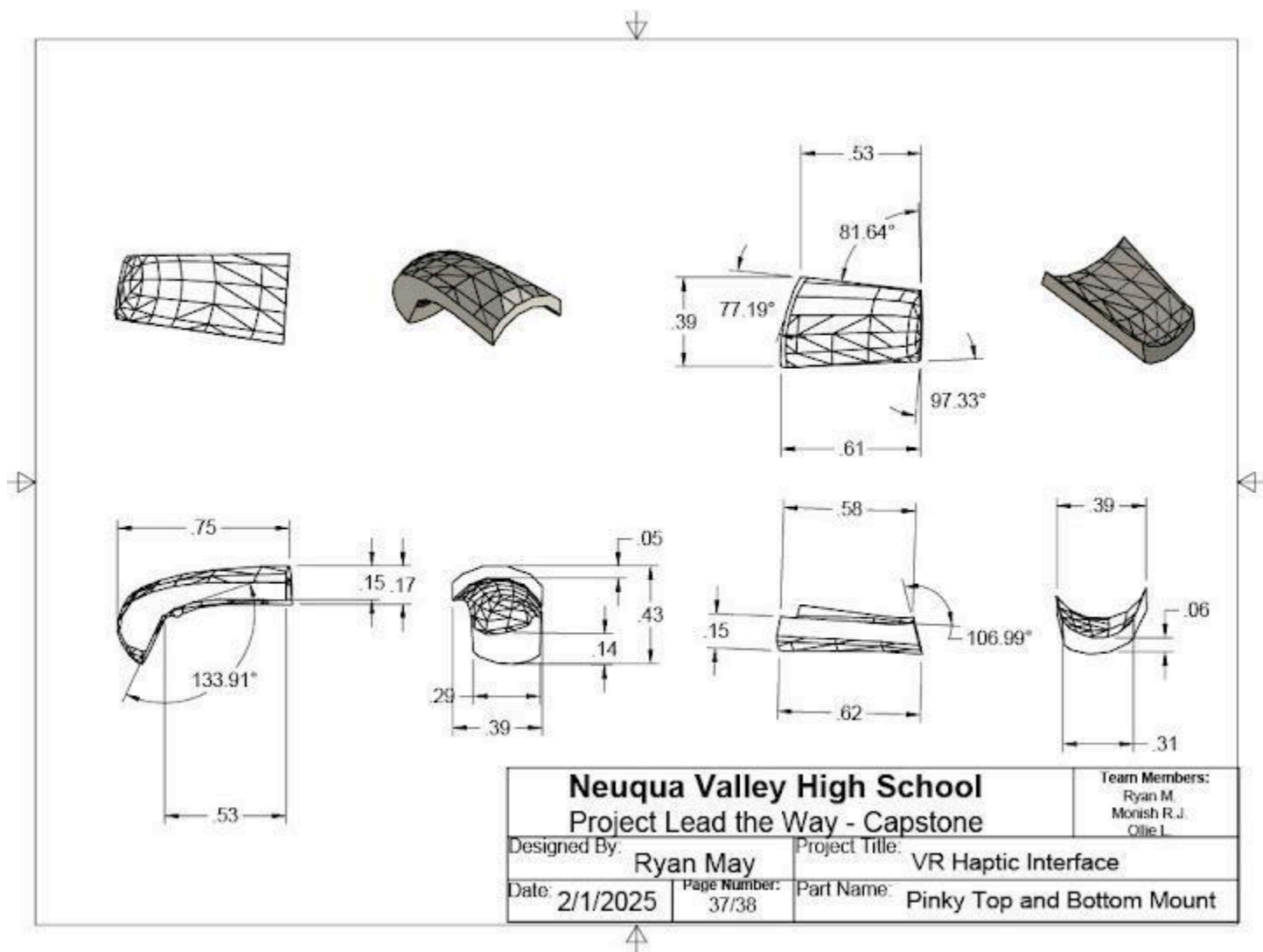


Figure 8.37: Dimensioned Sketch of “Pinky Top” and “Pinky Bottom” used in “String Routing”.

“Pinky Top” and “Pinky Bottom” in Figure 8.37 are the top and bottom parts of the adjustable cap that go on the end of the pinky finger.

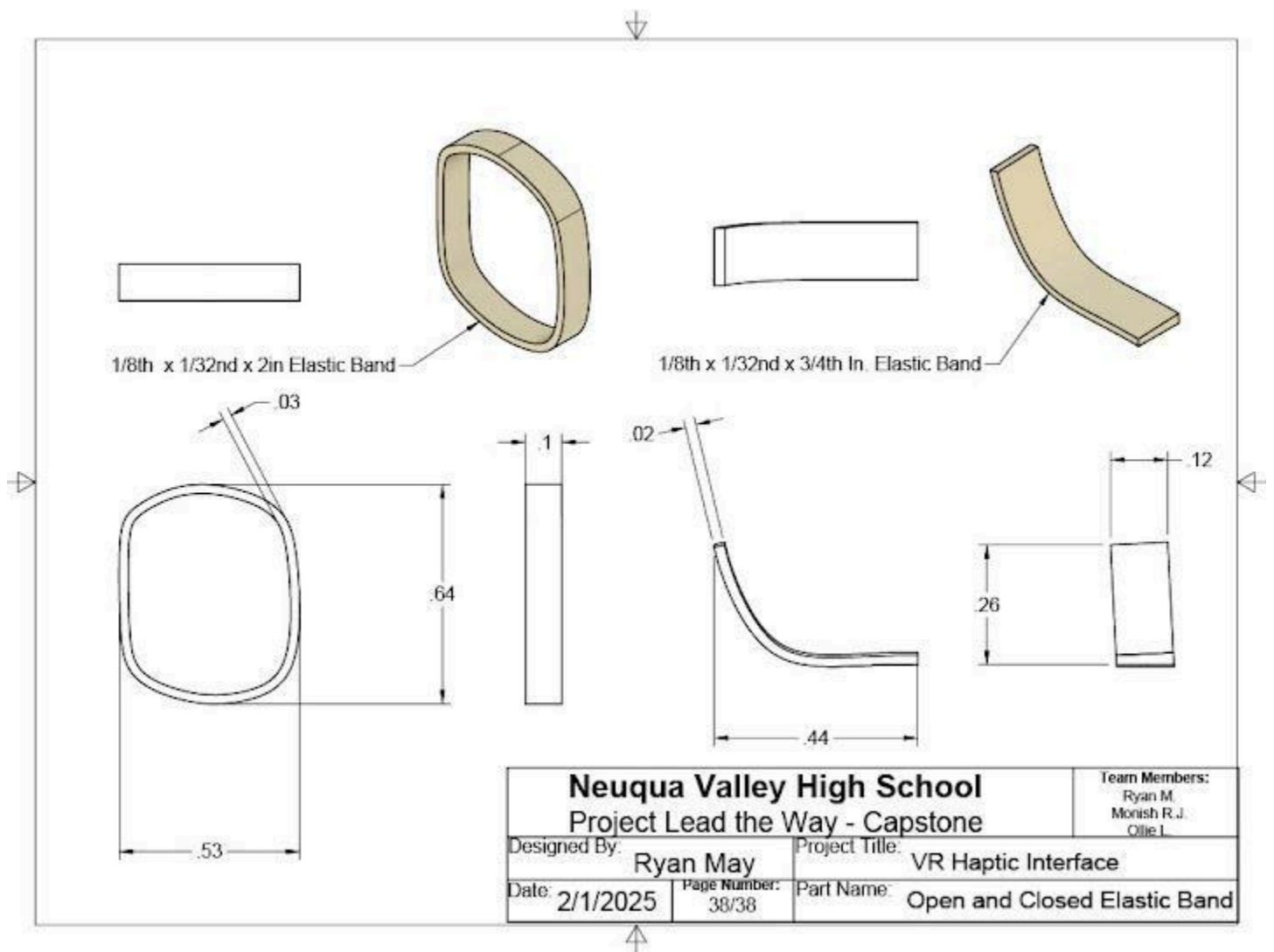


Figure 8.38: Dimensioned Sketch of the elastic bands used in “String Routing”.

Figure 8.38 depicts the elastic bands that connect to the top and bottom parts at the tip of each finger and allow for the caps to change size so they can be used by people with different finger sizes.

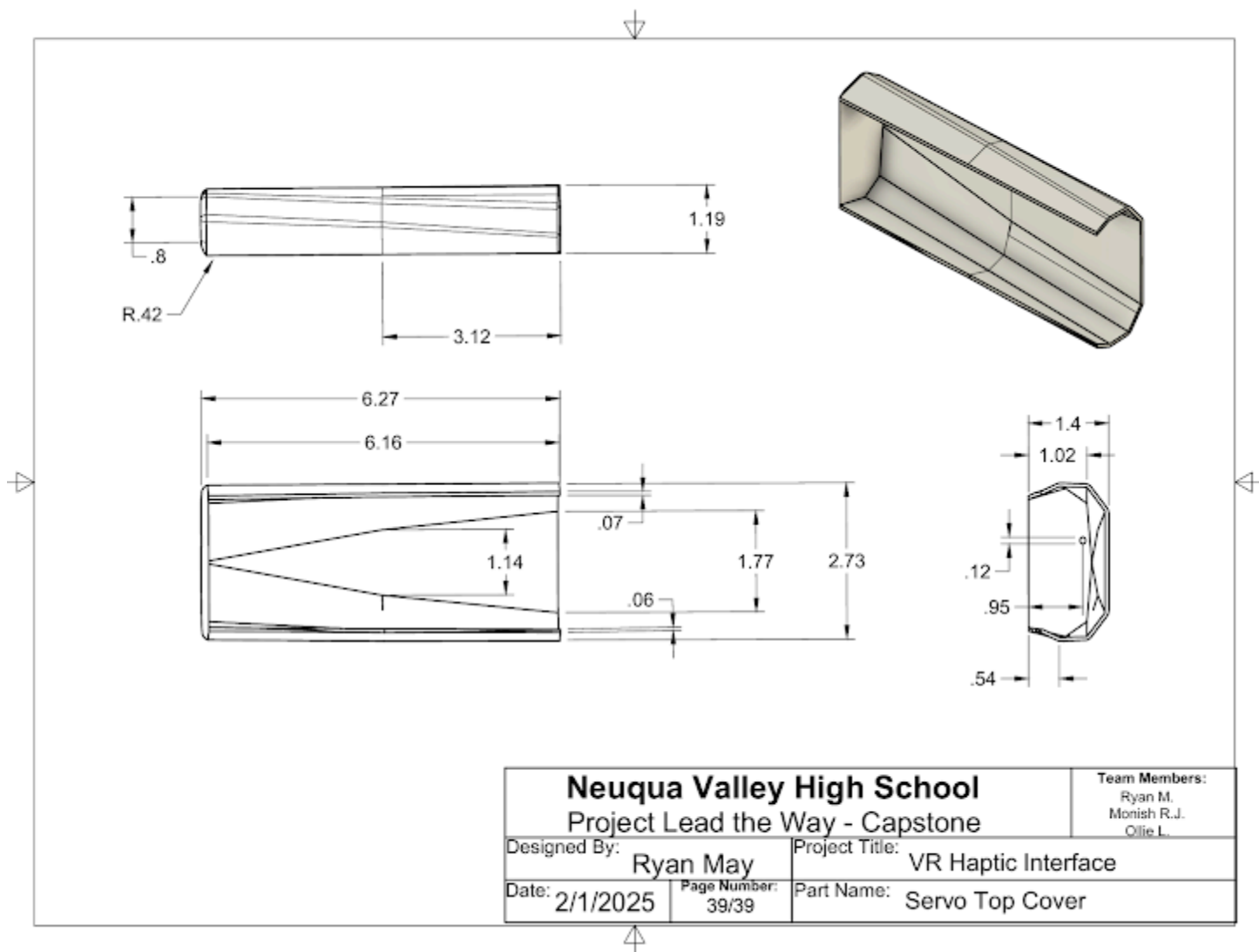


Figure 8.39: Dimensioned Sketch of the Servo Top Cover in “Electronics Mount”.

Figure 8.39 depicts the cover that goes on top of the servos and helps keep dust out of the components. It also acts to hide the intricacies of the glove and make it more visually appealing.

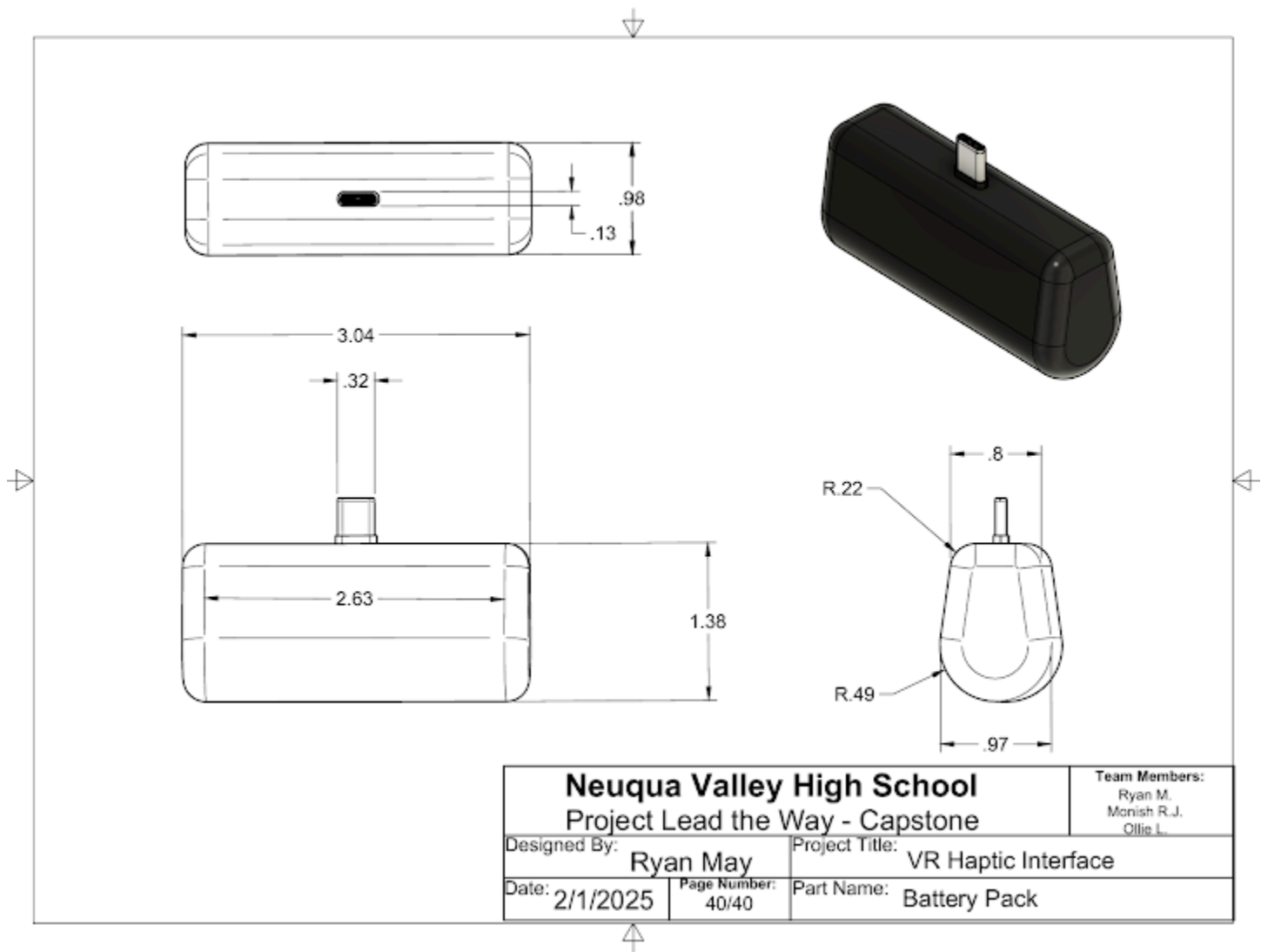


Figure 8.40: Dimensioned Sketch of the Battery Pack.

Figure 8.39 depicts the battery pack which supplies both 5v and ground to the glove. Two of these are required to give the glove sufficient power.